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WESTERN INTERNATIONAL SCHOOL

Physics



Grade 8

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Lesson 1 Speed, Velocity and Acceleration

- Speed
- Velocity
- Acceleration
- Timers
- Practical work: Analysing motion

Speed

If a car travels 300 km from Liverpool to London in five hours, its **average speed** is 300 km/5 h = 60 km/h. The speedometer would certainly not read 60 km/h for the whole journey but might vary considerably from this value. That is why we state the average speed. If a car could travel at a constant speed of 60 km/h for five hours, the distance covered would still be 300 km. It is *always* true that

$$\text{average speed} = \frac{\text{distance moved}}{\text{time taken}}$$

To find the actual speed at any instant we would need to know the distance moved in a very short interval of time. This can be done by multiflash photography. In Figure 2.1 the golfer is photographed while a flashing lamp illuminates him 100 times a second. The speed of the club-head as it hits the ball is about 200 km/h.

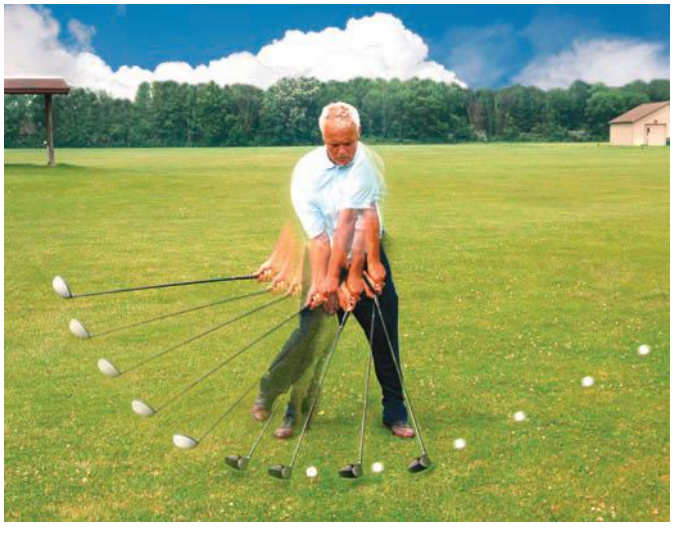


Figure 2.1 Multiflash photograph of a golf swing

Velocity

Speed is the distance travelled in unit time; **velocity is the distance travelled in unit time in a stated direction**. If two trains travel due north at 20 m/s, they have the same speed of 20 m/s and the same velocity of 20 m/s *due north*. If one travels north and the other south, their speeds are the same but not their velocities since their directions of motion are different. Speed is a **scalar** quantity and velocity a **vector** quantity.

$$\text{velocity} = \frac{\text{distance moved in a stated direction}}{\text{time taken}}$$

The velocity of a body is uniform or constant if it moves with a steady speed in a straight line. It is not uniform if it moves in a curved path. Why?

The units of speed and velocity are the same, km/h, m/s.

$$60\,\text{km/h} = \frac{6000\,\text{m}}{3600\,\text{s}} = 17\,\text{m/s}$$

Distance moved in a stated direction is called the **displacement**. It is a vector, unlike distance which is a scalar. Velocity may also be defined as

$$\text{velocity} = \frac{\text{displacement}}{\text{time taken}}$$

Acceleration

When the velocity of a body changes we say the body **accelerates**. If a car starts from rest and moving due north has velocity 2 m/s after 1 second, its velocity has increased by 2 m/s in 1 s and its acceleration is 2 m/s per second due north. We write this as 2 m/s^2.

Acceleration is the change of velocity in unit time, or

$$\text{acceleration} = \frac{\text{change of velocity}}{\text{time taken for change}}$$

For a steady increase of velocity from 20 m/s to 50 m/s in 5 s

$$\text{acceleration} = \frac{(50-20)\,\text{m/s}}{5\,\text{s}} = 6\,\text{m/s}^2$$

Acceleration is also a vector and both its magnitude and direction should be stated. However, at present we will consider only motion in a straight line and so the magnitude of the velocity will equal the speed, and the magnitude of the acceleration will equal the change of speed in unit time.

The speeds of a car accelerating on a straight road are shown below.

Time/s	0	1	2	3	4	5	6
Speed/m/s	0	5	10	15	20	25	30

The speed increases by 5 m/s every second and the acceleration of 5 m/s² is said to be **uniform**.

An acceleration is positive if the velocity increases and negative if it decreases. A negative acceleration is also called a **deceleration** or **retardation**.

Timers

A number of different devices are useful for analysing motion in the laboratory.

a) Motion sensors

Motion sensors use the ultrasonic echo technique (see p. 143) to determine the distance of an object from the sensor. Connection of a datalogger and computer to the motion sensor then enables a distance–time graph to be plotted directly (see Figure 2.6). Further data analysis by the computer allows a velocity–time graph to be obtained, as in Figures 3.1 and 3.2, p. 13.

b) Tickertape timer: tape charts

A **tickertape timer** also enables us to measure speeds and hence accelerations. One type, Figure 2.2, has

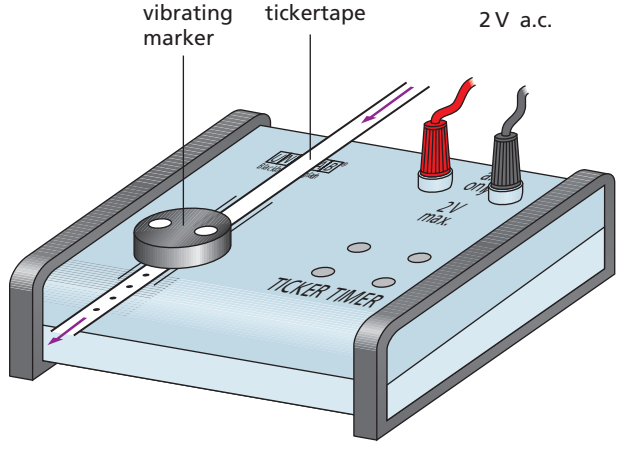


Figure 2.2 Tickertape timer

a marker that vibrates 50 times a second and makes dots at $\frac{1}{50}$ s intervals on the paper tape being pulled through it; $\frac{1}{50}$ s is called a '**tick**'.

The distance between successive dots equals the average speed of whatever is pulling the tape in, say, cm per $\frac{1}{50}$ s, i.e. cm per tick. The '**tentick**' ($\frac{1}{5}$ s) is also used as a unit of time. Since ticks and tenticks are small we drop the 'average' and just refer to the 'speed'.

Tape charts are made by sticking successive strips of tape, usually tentick lengths, side by side. That in Figure 2.3a represents a body moving with **uniform speed** since equal distances have been moved in each tentick interval.

The chart in Figure 2.3b is for **uniform acceleration**: the 'steps' are of equal size showing that the speed increased by the same amount in every tentick ($\frac{1}{5}$ s). The acceleration (average) can be found from the chart as follows.

The speed during the *first* tentick is 2 cm for every $\frac{1}{5}$ s, or 10 cm/s. During the *sixth* tentick it is 12 cm per $\frac{1}{5}$ s or 60 cm/s. And so during this interval of 5 tenticks, i.e. 1 second, the change of speed is (60 – 10) cm/s = 50 cm/s.

$$\text{acceleration} = \frac{\text{change of speed}}{\text{time taken}}$$

$$= \frac{50\,\text{cm/s}}{1\,\text{s}}$$

$$= 50\,\text{cm/s}^2$$

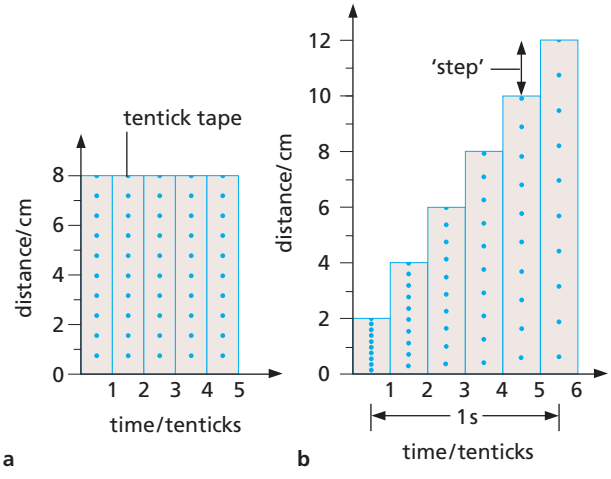


Figure 2.3 Tape charts: **a** uniform speed; **b** uniform acceleration

c) Photogate timer

Photogate timers may be used to record the time taken for a trolley to pass through the gate, Figure 2.4. If the length of the 'interrupt card' on the trolley is measured, the velocity of the trolley can then be calculated. Photogates are most useful in experiments where the velocity at only one or two positions is needed.

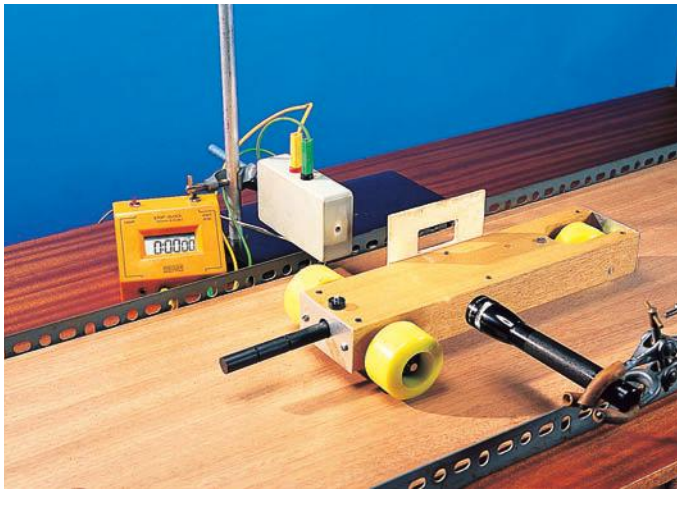


Figure 2.4 Use of a photogate timer

Practical work

Analysing motion

a) Your own motion

Pull a 2 m length of tape through a tickertape timer as you walk away from it quickly, then slowly, then speeding up again and finally stopping.

Cut the tape into tentick lengths and make a tape chart. Write labels on it to show where you speeded up, slowed down, etc.

b) Trolley on a sloping runway

Attach a length of tape to a trolley and release it at the top of a runway (Figure 2.5). The dots will be very crowded at the start – ignore those; but beyond them cut the tape into tentick lengths.

Make a tape chart. Is the acceleration uniform? What is its average value?

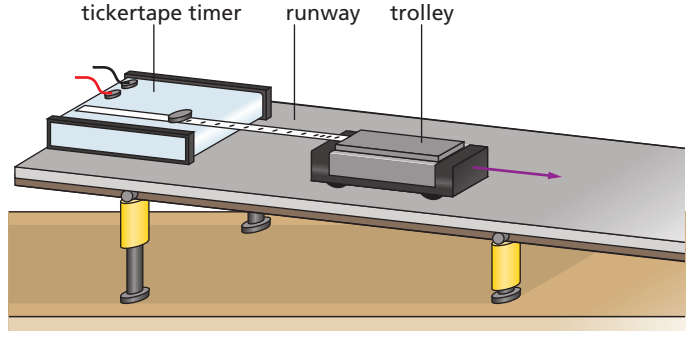


Figure 2.5

c) Datalogging

Replace the tickertape timer with a motion sensor connected to a datalogger and computer (Figure 2.6). Repeat the experiments in **a)** and **b)** and obtain distance–time and velocity–time graphs for each case; identify regions where you think the acceleration changes or remains uniform.

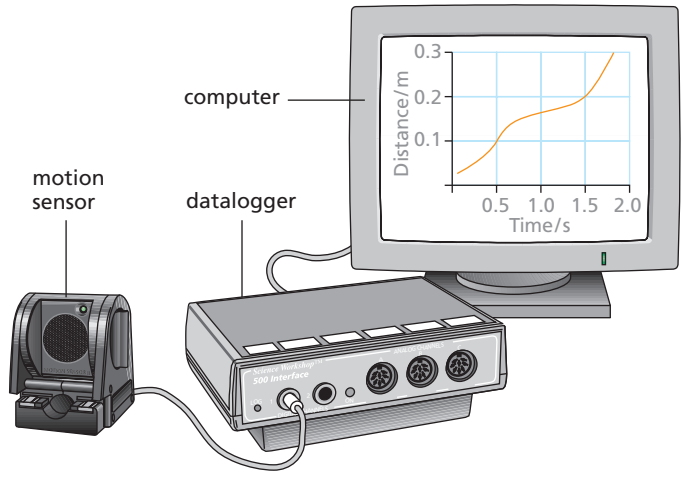


Figure 2.6 Use of a motion sensor

Questions

1 What is the average speed of
- **a** a car that travels 400 m in 20 s,
- **b** an athlete who runs 1500 m in 4 minutes?

2 A train increases its speed steadily from 10 m/s to 20 m/s in 1 minute.
- **a** What is its average speed during this time, in m/s?
- **b** How far does it travel while increasing its speed?

3 A motorcyclist starts from rest and reaches a speed of 6 m/s after travelling with uniform acceleration for 3 s. What is his acceleration?

4 An aircraft travelling at 600 km/h accelerates steadily at 10 km/h per second. Taking the speed of sound as 1100 km/h at the aircraft's altitude, how long will it take to reach the 'sound barrier'?

5 A vehicle moving with a uniform acceleration of 2 m/s^2 has a velocity of 4 m/s at a certain time. What will its velocity be
- **a** 1 s later,
- **b** 5 s later?

6 If a bus travelling at 20 m/s is subject to a steady deceleration of 5 m/s^2, how long will it take to come to rest?

7 The tape in Figure 2.7 was pulled through a timer by a trolley travelling down a runway. It was marked off in tentick lengths.
- **a** What can you say about the trolley's motion?
- **b** Find its acceleration in cm/s^2.

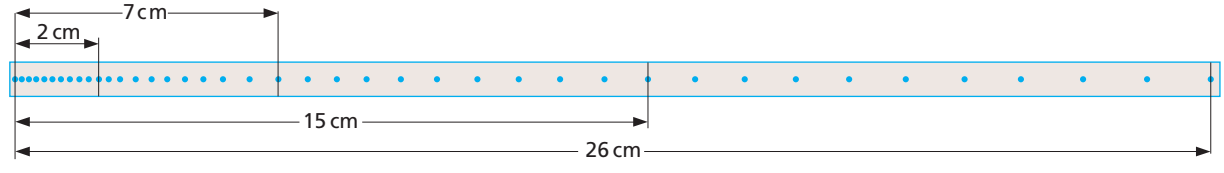


Figure 2.7

8 Each strip in the tape chart of Figure 2.8 is for a time interval of 1 tentick.
- **a** If the timer makes 50 dots per second, what time intervals are represented by OA and AB?
- **b** What is the acceleration between O and A in
  - **(i)** cm/tentick^2,
  - **(ii)** cm/s per tentick,
  - **(iii)** cm/s^2?
- **c** What is the acceleration between A and B?

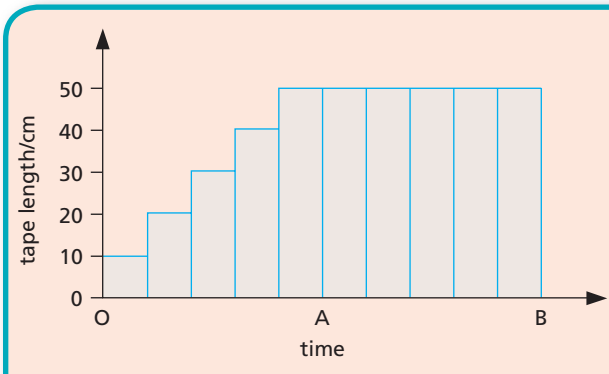


Figure 2.8

9 The speeds of a car travelling on a straight road are given below at successive intervals of 1 second.

Time/s	0	1	2	3	4
Speed/m/s	0	2	4	6	8

The car travels
1. with an average velocity of 4 m/s
2. 16 m in 4 s
3. with a uniform acceleration of 2 m/s^2.

Which statement(s) is (are) correct?
A 1, 2, 3 **B** 1, 2 **C** 2, 3 **D** 1 **E** 3

10 If a train travelling at 10 m/s starts to accelerate at 1 m/s^2 for 15 s on a straight track, its final velocity in m/s is
A 5 **B** 10 **C** 15 **D** 20 **E** 25

Checklist

After studying this chapter you should be able to

- explain the meaning of the terms speed and acceleration,
- distinguish between speed and velocity,
- describe how speed and acceleration may be found using tape charts and motion sensors.

Lesson 2 Graphs of Equations

- Velocity–time graphs
- Distance–time graphs
- Equations for uniform acceleration

1. Velocity–time graphs

If the velocity of a body is plotted against the time, the graph obtained is a **velocity–time graph**. It provides a way of solving motion problems. Tape charts are crude velocity–time graphs that show the velocity changing in jumps rather than smoothly, as occurs in practice. A motion sensor gives a smoother plot.

The area under a velocity–time graph measures the distance travelled.

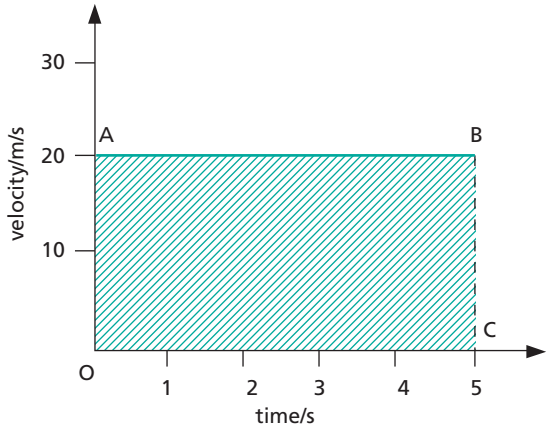


Figure 3.1 Uniform velocity

In Figure 3.1, AB is the velocity–time graph for a body moving with a **uniform velocity** of 20 m/s. Since distance = average velocity × time, after 5 s it will have moved 20 m/s × 5 s = 100 m. This is the shaded area under the graph, i.e. rectangle OABC.

In Figure 3.2a, PQ is the velocity–time graph for a body moving with **uniform acceleration**. At the start of the timing the velocity is 20 m/s but it increases steadily to 40 m/s after 5 s. If the distance covered equals the area under PQ, i.e. the shaded area OPQS, then

$$\begin{aligned}\text{distance} &= \text{area of rectangle OPRS} \\ &\quad + \text{area of triangle PQR} \\ &= \text{OP} \times \text{OS} + \tfrac{1}{2} \times \text{PR} \times \text{QR} \\ &\quad (\text{area of a triangle} = \tfrac{1}{2}\text{base} \times \text{height}) \\ &= 20\,\text{m/s} \times 5\,\text{s} + \tfrac{1}{2} \times 5\,\text{s} \times 20\,\text{m/s} \\ &= 100\,\text{m} + 50\,\text{m} = 150\,\text{m}\end{aligned}$$

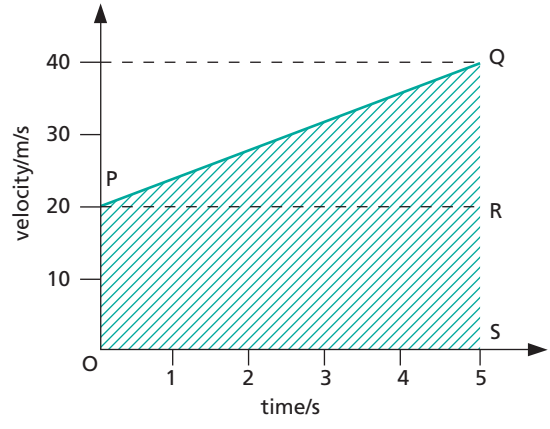


Figure 3.2a Uniform acceleration

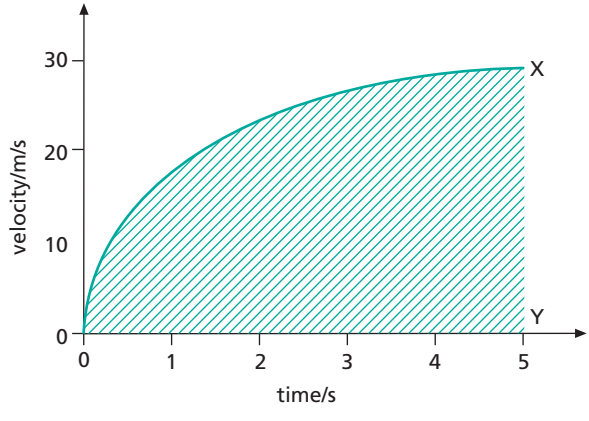


Figure 3.2b Non-uniform acceleration

Notes

1. When calculating the area from the graph, the unit of time must be the same on both axes.
2. This rule for finding distances travelled is true even if the acceleration is not uniform. In Figure 3.2b, the distance travelled equals the shaded area OXY.

The slope or gradient of a velocity–time graph represents the acceleration of the body.

In Figure 3.1, the slope of AB is zero, as is the acceleration. In Figure 3.2a, the slope of PQ is QR/PR = 20/5 = 4: the acceleration is 4 m/s². In Figure 3.2b, when the slope along OX changes, so does the acceleration.

2. Distance–time graphs

A body travelling with uniform velocity covers equal distances in equal times. Its **distance–time graph** is a straight line, like OL in Figure 3.3 for a velocity of 10 m/s. The slope of the graph is LM/OM = 40 m/4 s = 10 m/s, which is the value of the velocity. The following statement is true in general:

The slope or gradient of a distance–time graph represents the velocity of the body.

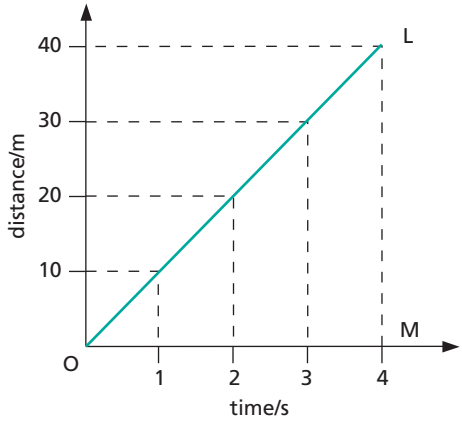


Figure 3.3 Uniform velocity

When the velocity of the body is changing, the slope of the distance–time graph varies, as in Figure 3.4, and at any point equals the slope of the tangent. For example, the slope of the tangent at T is AB/BC = 40 m/2 s = 20 m/s. The velocity at the instant corresponding to T is therefore 20 m/s.

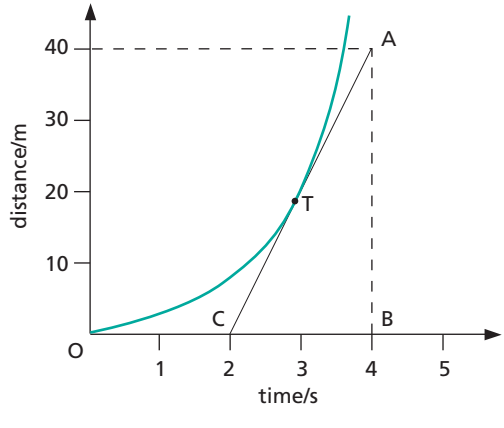


Figure 3.4 Non-uniform velocity

3. Equations for uniform acceleration

Problems involving bodies moving with **uniform acceleration** can often be solved quickly using the **equations of motion**.

First equation

If a body is moving with uniform acceleration a and its velocity increases from u to v in time t, then

$$a = \frac{\text{change of velocity}}{\text{time taken}} = \frac{v-u}{t}$$

$\therefore \quad at = v - u$

or

$$v = u + at \qquad (1)$$

Note that the initial velocity u and the final velocity v refer to the start and the finish of the *timing* and do not necessarily mean the start and finish of the motion.

Second equation

The velocity of a body moving with uniform acceleration increases steadily. Its average velocity therefore equals half the sum of its initial and final velocities, that is,

$$\text{average velocity} = \frac{u+v}{2}$$

If s is the distance moved in time t, then since average velocity = distance/time = s/t,

$$\frac{s}{t} = \frac{u+v}{2}$$

or

$$s = \frac{(u+v)}{2}t \qquad (2)$$

Third equation

From equation (1), $v = u + at$
From equation (2),

$$\frac{s}{t} = \frac{u+v}{2}$$

$$\frac{s}{t} = \frac{u + u + at}{2} = \frac{2u + at}{2}$$
$$= u + \frac{1}{2}at$$

and so

$$s = ut + \frac{1}{2}at^2 \quad (3)$$

Fourth equation

This is obtained by eliminating t from equations (1) and (3). Squaring equation (1) we have

$$v^2 = (u + at)^2$$
$$\therefore \quad v^2 = u^2 + 2uat + a^2t^2$$
$$= u^2 + 2a\left(ut + \frac{1}{2}at^2\right)$$

But $s = ut + \frac{1}{2}at^2$

$$\therefore \quad v^2 = u^2 + 2as$$

If we know any *three* of u, v, a, s and t, the others can be found from the equations.

Worked example

A sprint cyclist starts from rest and accelerates at $1\,m/s^2$ for 20 seconds. He then travels at a constant speed for 1 minute and finally decelerates at $2\,m/s^2$ until he stops. Find his maximum speed in km/h and the total distance covered in metres.

First stage

$u = 0 \quad a = 1\,m/s^2 \quad t = 20\,s$

We have $v = u + at = 0 + 1\,m/s^2 \times 20\,s$

$= 20\,m/s$

$= \frac{20}{1000} \times 60 \times 60 = 72\ km/h$

The distance s moved in the first stage is given by

$$s = ut + \frac{1}{2}at^2 = 0 \times 20\,s + \frac{1}{2} \times 1\,m/s^2 \times 20^2\,s^2$$
$$= \frac{1}{2} \times 1\,m/s^2 \times 400\,s^2 = 200\,m$$

Second stage

$u = 20\,m/s$ (constant) $\quad t = 60\,s$

distance moved = speed × time = $20\,m/s \times 60\,s$

$= 1200\,m$

Third stage

$u = 20\,m/s \quad v = 0 \quad a = -2\,m/s^2$ (a deceleration)

We have

$v^2 = u^2 + 2as$

$$\therefore \quad s = \frac{v^2 - u^2}{2a} = \frac{0 - 20^2\,m^2/s^2}{2 \times (-2)\,m/s^2} = \frac{-400\,m^2/s^2}{-4\,m/s^2}$$
$$= 100\,m$$

Answers

Maximum speed = 72 km/h

Total distance covered = 200 m + 1200 m + 100 m

= 1500 m

Questions

1 The distance–time graph for a girl on a cycle ride is shown in Figure 3.5.

- **a** How far did she travel?
- **b** How long did she take?
- **c** What was her average speed in km/h?
- **d** How many stops did she make?
- **e** How long did she stop for altogether?
- **f** What was her average speed *excluding* stops?
- **g** How can you tell from the shape of the graph when she travelled fastest? Over which stage did this happen?

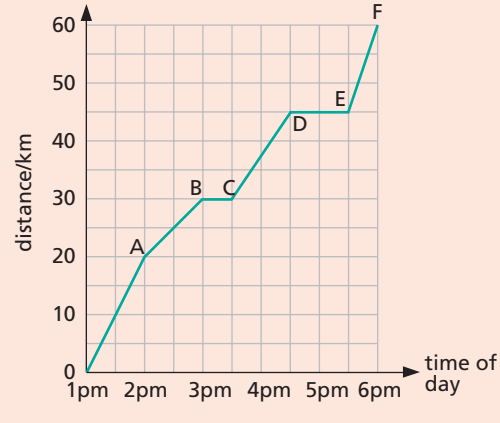


Figure 3.5

2 The graph in Figure 3.6 represents the distance travelled by a car plotted against time.
- **a** How far has the car travelled at the end of 5 seconds?
- **b** What is the speed of the car during the first 5 seconds?
- **c** What has happened to the car after A?
- **d** Draw a graph showing the speed of the car plotted against time during the first 5 seconds.

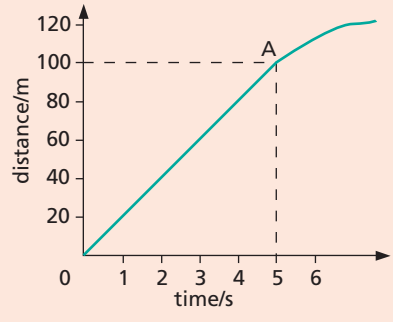


Figure 3.6

3 Figure 3.7 shows an incomplete velocity–time graph for a boy running a distance of 100 m.
- **a** What is his acceleration during the first 4 seconds?
- **b** How far does the boy travel during **(i)** the first 4 seconds, **(ii)** the next 9 seconds?
- **c** Copy and complete the graph showing clearly at what time he has covered the distance of 100 m. Assume his speed remains constant at the value shown by the horizontal portion of the graph.

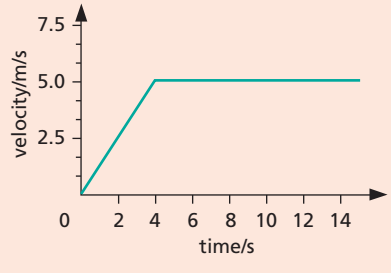


Figure 3.7

4 The approximate velocity–time graph for a car on a 5-hour journey is shown in Figure 3.8. (There is a very quick driver change midway to prevent driving fatigue!)
- **a** State in which of the regions OA, AB, BC, CD, DE the car is **(i)** accelerating, **(ii)** decelerating, **(iii)** travelling with uniform velocity.
- **b** Calculate the value of the acceleration, deceleration or constant velocity in each region.
- **c** What is the distance travelled over each region?
- **d** What is the total distance travelled?
- **e** Calculate the average velocity for the whole journey.

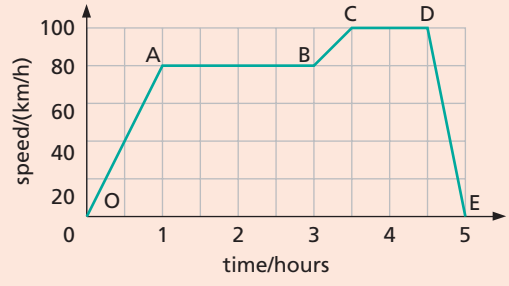


Figure 3.8

5 The distance–time graph for a motorcyclist riding off from rest is shown in Figure 3.9.
- **a** Describe the motion.
- **b** How far does the motorbike move in 30 seconds?
- **c** Calculate the speed.

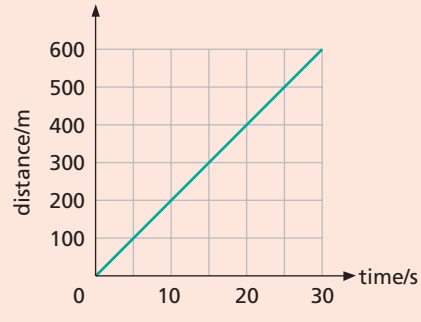


Figure 3.9

Checklist

After studying this chapter you should be able to

- draw, interpret and use velocity–time and distance–time graphs to solve problems.

Lesson 3 Free Falling Bodies

- Acceleration of free fall
- Measuring *g*
- Distance–time graphs
- Projectiles
- Practical work: Motion of a falling body

In air, a coin falls faster than a small piece of paper. In a vacuum they fall at the same rate, as may be shown with the apparatus of Figure 4.1. The difference in air is due to **air resistance** having a greater effect on light bodies than on heavy bodies. The air resistance to a light body is large when compared with the body's weight. With a dense piece of metal the resistance is negligible at low speeds.

There is a story, untrue we now think, that in the 16th century the Italian scientist Galileo dropped a small iron ball and a large cannonball ten times heavier from the top of the Leaning Tower of Pisa (Figure 4.2). And we are told that, to the surprise of onlookers who expected the cannonball to arrive first, they reached the ground almost simultaneously. You will learn more about air resistance in Chapter 8.

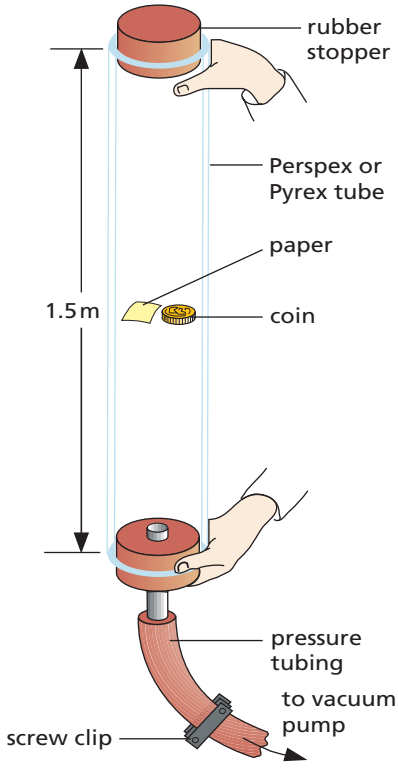


Figure 4.1 A coin and a piece of paper fall at the same rate in a vacuum.



Figure 4.2 The Leaning Tower of Pisa, where Galileo is said to have experimented with falling objects

Practical work

Motion of a falling body

Arrange things as shown in Figure 4.3 and investigate the motion of a 100 g mass falling from a height of about 2 m.

Construct a tape chart using *one-tick* lengths. Choose as dot '0' the first one you can distinguish clearly. What does the tape chart tell you about the motion of the falling mass? Repeat the experiment with a 200 g mass; what do you notice?

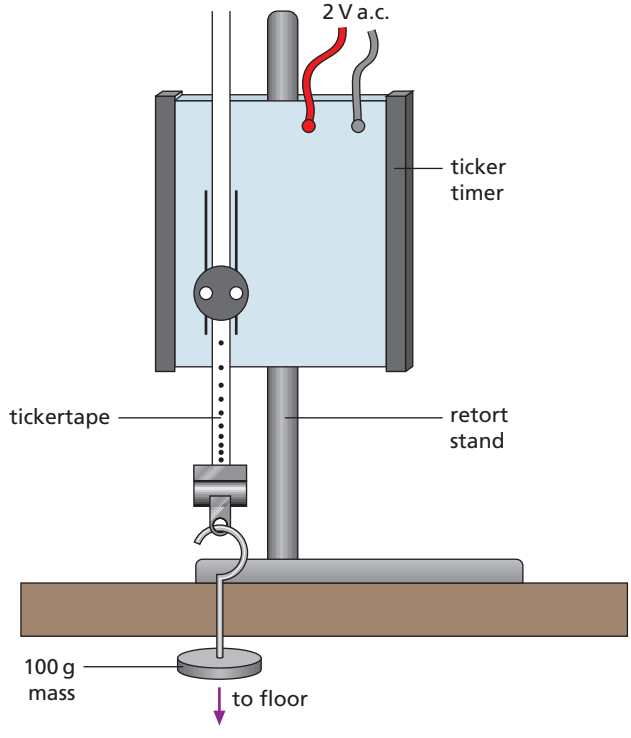


Figure 4.3

1. Acceleration of free fall

All bodies falling freely under the force of gravity do so with uniform acceleration if air resistance is negligible (i.e. the 'steps' in the tape chart from the practical work should all be equal).

This acceleration, called the **acceleration of free fall,** is denoted by the italic letter g. Its value varies slightly over the Earth but is constant in each place; in India for example, it is about 9.8 m/s² or near enough 10 m/s². The velocity of a free-falling body therefore increases by 10 m/s every second. A ball shot straight upwards with a velocity of 30 m/s decelerates by 10 m/s every second and reaches its highest point after 3 s.

In calculations using the equations of motion, g replaces a. It is given a positive sign for falling bodies (i.e. $a = g = +10\,\text{m/s}^2$) and a negative sign for rising bodies since they are decelerating (i.e. $a = -g = -10\,\text{m/s}^2$).

2. Measuring g

Using the arrangement in Figure 4.4 the time for a steel ball-bearing to fall a known distance is measured by an electronic timer.

When the two-way switch is changed to the 'down' position, the electromagnet releases the ball and simultaneously the clock starts. At the end of its fall the ball opens the 'trap-door' on the impact switch and the clock stops.

The result is found from the third equation of motion $s = ut + \frac{1}{2}at^2$, where s is the distance fallen (in m), t is the time taken (in s), $u = 0$ (the ball starts from rest) and $a = g$ (in m/s²). Hence

$$s = \frac{1}{2}gt^2$$

or

$$g = 2s/t^2$$

Air resistance is negligible for a dense object such as a steel ball-bearing falling a short distance.

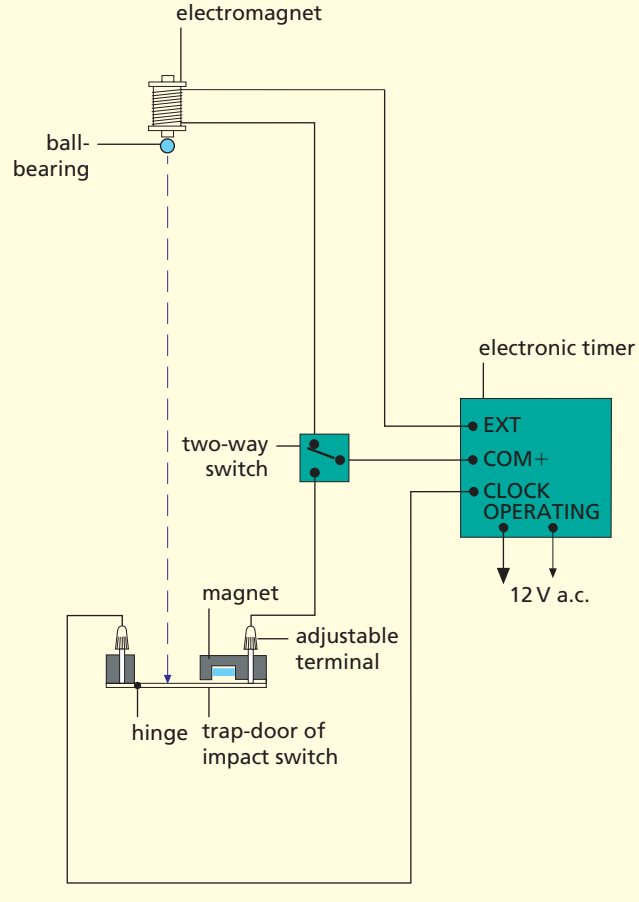


Figure 4.4

Worked example

A ball is projected vertically upwards with an initial velocity of 30 m/s. Find **a** its maximum height and **b** the time taken to return to its starting point. Neglect air resistance and take $g = 10\,\text{m/s}^2$.

a We have $u = 30\,\text{m/s}$, $a = -10\,\text{m/s}^2$ (a deceleration) and $v = 0$ since the ball is momentarily at rest at its highest point. Substituting in $v^2 = u^2 + 2as$,

$$0 = 30^2\,\text{m}^2/\text{s}^2 + 2(-10\,\text{m/s}^2) \times s$$

or

$$-900\,\text{m}^2/\text{s}^2 = -s \times 20\,\text{m/s}^2$$

$$\therefore \quad s = \frac{-900\,\text{m}^2/\text{s}^2}{-20\,\text{m/s}^2} = 45\,\text{m}$$

b If t is the time to reach the highest point, we have, from $v = u + at$,

$$0 = 30\,\text{m/s} + (-10\,\text{m/s}^2) \times t$$

or

$$-30\,\text{m/s} = -t \times 10\,\text{m/s}^2$$

$$\therefore \quad t = \frac{-30\,\text{m/s}}{-10\,\text{m/s}^2} = 3\,\text{s}$$

The downward trip takes exactly the same time as the upward one and so the answer is 6 s.

3. Distance–time graphs

For a body falling freely from rest we have

$$s = \tfrac{1}{2}gt^2$$

A graph of distance s against time t is shown in Figure 4.5a and for s against t^2 in Figure 4.5b. The second graph is a straight line through the origin since $s \propto t^2$ (g being constant at one place).

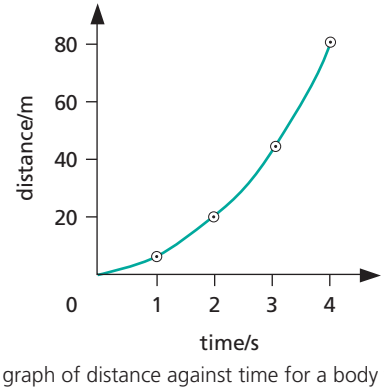


Figure 4.5a A graph of distance against time for a body falling freely from rest

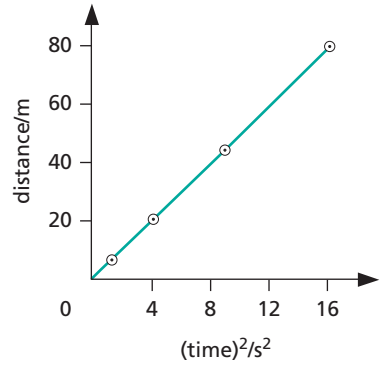


Figure 4.5b A graph of distance against (time)² for a body falling freely from rest

4. Projectiles

The photograph in Figure 4.6 was taken while a lamp emitted regular fl ashes of light. One ball was *dropped from rest* and the other, a '**projectile**', was *thrown sideways* at the same time. Their vertical accelerations (due to gravity) are equal, showing that a projectile falls like a body which is dropped from rest. Its horizontal velocity does not affect its vertical motion.

The horizontal and vertical motions of a body are independent and can be treated separately.

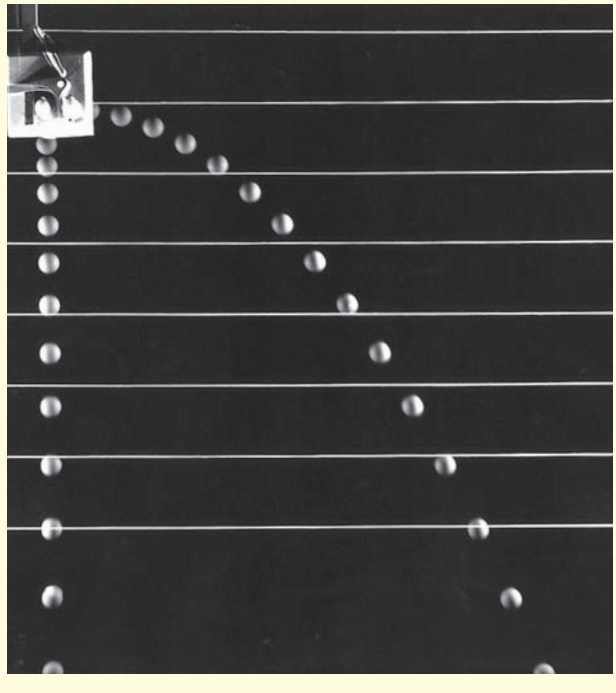


Figure 4.6 Comparing free fall and projectile motion using multiflash photography

For example if a ball is thrown horizontally from the top of a cliff and takes 3 s to reach the beach below, we can calculate the height of the cliff by considering the vertical motion only. We have $u = 0$ (since the ball has no vertical velocity initially), $a = g = +10\,\text{m/s}^2$ and $t = 3\,\text{s}$. The height s of the cliff is given by

$$\begin{aligned} s &= ut + \frac{1}{2}at^2 \\ &= 0 \times 3\,\text{s} + \frac{1}{2}(+10\,\text{m/s}^2)3^2\,\text{s}^2 \\ &= 45\,\text{m} \end{aligned}$$

Projectiles such as cricket balls and explosive shells are projected from near ground level and at an angle. The horizontal distance they travel, i.e. their range, depends on

(i) the speed of projection – the greater this is, the greater the range, and

(ii) the angle of projection – it can be shown that, neglecting air resistance, the range is a maximum when the angle is 45° (Figure 4.7).

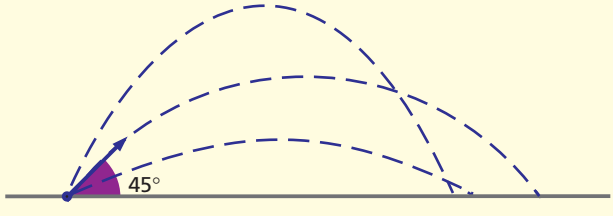


Figure 4.7 The range is greatest for an angle of projection of 45°

Questions

1 A stone falls from rest from the top of a high tower. Ignore air resistance and take $g = 10\,\text{m/s}^2$.

a What is its velocity after
- **(i)** 1 s,
- **(ii)** 2 s,
- **(iii)** 3 s,
- **(iv)** 5 s?

b How far has it fallen after
- **(i)** 1 s,
- **(ii)** 2 s,
- **(iii)** 3 s,
- **(iv)** 5 s?

2 An object falls from a hovering helicopter and hits the ground at a speed of 30 m/s. How long does it take the object to reach the ground and how far does it fall? Sketch a velocity–time graph for the object (ignore air resistance).

Checklist

After studying this chapter you should be able to

- describe the behaviour of falling objects,
- state that the acceleration of free fall for a body near the Earth is constant.

Lesson 4 Weight and Stretching

- Force
- Weight
- The newton
- Hooke's law
- Practical work: Stretching a spring

1. Force

A **force** is a push or a pull. It can cause a body at rest to move, or if the body is already moving it can change its speed or direction of motion. A force can also change a body's shape or size.





Figure 6.1 A weightlifter in action exerts first a pull and then a push.

2. Weight

We all constantly experience the force of **gravity**, in other words. the pull of the Earth. It causes an unsupported body to fall from rest to the ground.

The weight of a body is the force of gravity on it.

For a body above or on the Earth's surface, the nearer it is to the centre of the Earth, the more the Earth attracts it. Since the Earth is not a perfect sphere but is flatter at the poles, the weight of a body varies over the Earth's surface. It is greater at the poles than at the equator.

Gravity is a force that can act through space, i.e. there does not need to be contact between the Earth and the object on which it acts as there does when we push or pull something. Other action-at-a-distance forces which, like gravity, decrease with distance are:

(i) **magnetic** forces between magnets, and
(ii) **electric** forces between electric charges.

3. The newton

The unit of force is the **newton** (N). It will be defined later (Chapter 8); the definition is based on the change of speed a force can produce in a body. Weight is a force and therefore should be measured in newtons.

The weight of a body can be measured by hanging it on a spring balance marked in newtons (Figure 6.2) and letting the pull of gravity stretch the spring in the balance. The greater the pull, the more the spring stretches.

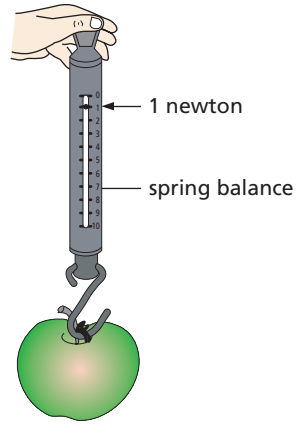


Figure 6.2 The weight of an average-sized apple is about 1 newton.

On most of the Earth's surface:

The weight of a body of mass 1 kg is 9.8 N.

Often this is taken as 10 N. A mass of 2 kg has a weight of 20 N, and so on. The mass of a body is the same wherever it is and, unlike weight, does not depend on the presence of the Earth.

Practical work

Stretching a spring

Arrange a steel spring as in Figure 6.3. Read the scale opposite the bottom of the hanger. Add 100 g loads one at a time (thereby increasing the stretching force by steps of 1 N) and take the readings after each one. Enter the readings in a table for loads up to 500 g.

Note that at the head of columns (or rows) in data tables it is usual to give the name of the quantity or its symbol followed by **/** and the unit.

Stretching force/N	Scale reading/mm	Total extension/mm

Do the results suggest any rule about how the spring behaves when it is stretched?

Sometimes it is easier to discover laws by displaying the results on a graph. Do this on graph paper by plotting stretching force readings along the *x*-axis (horizontal axis) and total extension readings along the *y*-axis (vertical axis). Every pair of readings will give a point; mark them by small crosses and draw a smooth line through them. What is its shape?

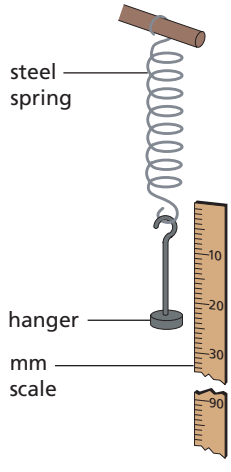


Figure 6.3

4. Hooke's law

Springs were investigated by Robert Hooke nearly 350 years ago. He found that the extension was proportional to the stretching force provided the spring was not permanently stretched. This means that doubling the force doubles the extension, trebling the force trebles the extension, and so on.

Using the sign for proportionality, ∝, we can write **Hooke's law** as

extension ∝ stretching force

It is true only if the **elastic limit** or 'limit of proportionality' of the spring is not exceeded. In other words, the spring returns to its original length when the force is removed.

The graph of Figure 6.4 is for a spring stretched beyond its elastic limit, E. OE is a straight line passing through the origin O and is graphical proof that Hooke's law holds over this range. If the force for point A on the graph is applied to the spring, the proportionality limit is passed and on removing the force some of the extension (OS) remains. Over which part of the graph does a spring balance work?

The **force constant**, k, of a spring is the force needed to cause unit extension, i.e. 1 m. If a force F produces extension x then

$$k = \frac{F}{x}$$

Rearranging the equation gives

$$F = kx$$

This is the usual way of writing Hooke's law in symbols.

Hooke's law also holds when a force is applied to a straight metal wire or an elastic band, provided they are not permanently stretched. Force–extension graphs similar to Figure 6.4 are obtained. You should label each axis of your graph with the name of the quantity or its symbol followed by / and the unit, as shown in Figure 6.4.

For a rubber band, a small force causes a large extension.

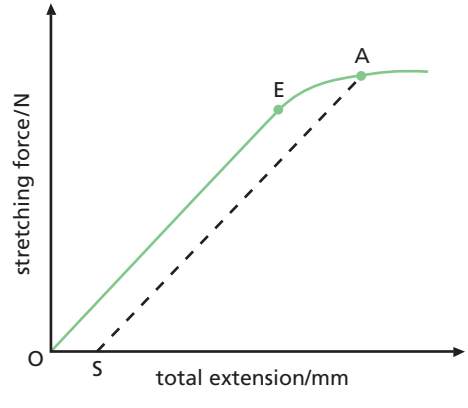


Figure 6.4

Worked example

A spring is stretched 10 mm (0.01 m) by a weight of 2.0 N. Calculate: **a** the force constant k, and **b** the weight W of an object that causes an extension of 80 mm (0.08 m).

a $k = \frac{F}{x} = \frac{2.0\ \text{N}}{0.01\ \text{m}} = 200\ \text{N/m}$

b W = stretching force F
$= k \times x$
$= 200\ \text{N/m} \times 0.08\ \text{m}$
$= 16\ \text{N}$

Checklist

After studying this chapter you should be able to

- recall that a force can cause a change in the motion, size or shape of a body,
- recall that the weight of a body is the force of gravity on it,
- recall the unit of force and how force is measured,
- describe an experiment to study the relation between force and extension for springs,
- draw conclusions from force–extension graphs,
- recall Hooke's law and solve problems using it,
- recognise the significance of the term limit of proportionality.

Practice

1.A body of mass 1 kg has weight 10 N at a certain place. What is the weight of
a. 100 g,
b. 5 kg,
c. 50 g?
2. The force of gravity on the Moon is said to be one-sixth of that on the Earth. What would a mass of 12 kg weigh
a. on the Earth, and
b. on the Moon?
3. What is the force constant of a spring which is stretched
a. 2 mm by a force of 4 N, b, 4 cm by a mass of 200 g?
4. Does Hooke's Law apply to stretch springs, compressed springs or both?

5.a) A load of 45 N is attached to a spring that is hanging vertically. The spring is stretched 0.14 m from its equilibrium position. What is the spring constant?
b) If a 60 N weight is used instead, what would you expect the spring stretch to be instead?

6.a) A slingshot consists of a light leather cup attached between two rubber bands. If it takes a force of 32 N to stretch the bands 1.2cm, what is the equivalent spring constant of the rubber bands?
b) How much force is required to pull the cup of the slingshot 3.0 cm from the equilibrium position?

Hooke's Law

Basic

1. a) What is the equation that links force, spring constant and extension?
   b) What are the units of force, spring constant and extension?
2. Calculate the force on a spring if:
   a) k = 10 N/m, e = 0.20 m.
   b) k = 25 N/m, e = 0.05 m.
   c) k = 150 N/m, e = 0.15 m.
3. If the spring constant is 30 N/m and a spring is stretched by 0.3m, how much force has been applied?
4. If the spring constant is 12.6 N/m and a spring is stretch by 0.25m, how much force has been applied?
5. What force would be needed to extend a spring with a spring constant k = 10 N/m by an extension of 0.3 m?

Medium

6. Re-arrange Hooke's law to give equations for the spring constant k, and the extension e. You will need to use these equations for the rest of the medium questions.
7. Calculate the spring constant if:
   a) F = 150 N, e = 0.075 m.
   b) F = 50 N, e = 0.1 m.
8. Calculate the extension if:
   a) F = 15 N, k = 150 N/m.
   b) F = 45 N, k = 90 N/m.
9. If a 6N weight is hung on a spring, and it extends by 0.2m, what is the spring constant?
10. If the force applied is 4.5 N and the spring constant is 9 N/m, how much will the spring extend by?

Hard

11. A mass of 620 g is hung on a spring of spring constant 31 N/m.
    a) Convert 620 g into kg.
    b) Using F = m × g, what is the force of the mass acting on the spring (g = 10 N/kg)?
    c) Calculate the extension of the spring.

To go from g to kg → ÷ 1000

12. A spring of spring constant 40 N/m starts at a length of 13 cm, and it extends to a length of 21 cm.
    a) What is the extension of the spring (in cm)?
    b) Convert this extension into metres.
    c) What is the force on this spring?

To go from cm to m → ÷ 100

13. A spring has a weight of 200g hanging on it, and has a spring constant of 40 N/m. Calculate the extension of the spring.
14. A spring has a weight of 500g hanging on it, and is stretched from a length of 5cm to a length of 15 cm. What is the spring constant of the spring?
15. A spring has a weight of 750g hanging on it, and is stretched from a length of 2.5cm to a length of 10 cm. What is the spring constant of the spring?

Lesson 5 Adding forces

- Forces and resultants
- Examples of addition of forces
- Vectors and scalars
- Friction
- Practical work: Parallelogram law

Forces and resultants

Force has both magnitude (size) and direction. It is represented in diagrams by a straight line with an arrow to show its direction of action.

Usually more than one force acts on an object. As a simple example, an object resting on a table is pulled downwards by its weight W and pushed upwards by a force R due to the table supporting it (Figure 7.1). Since the object is at rest, the forces must balance, i.e. $R = W$.

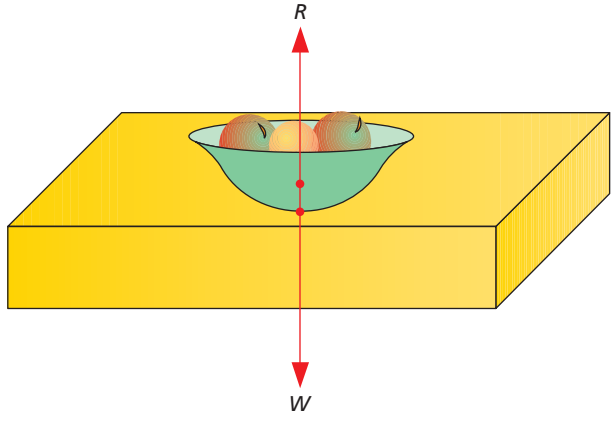


Figure 7.1

In structures such as a giant oil platform (Figure 7.2), two or more forces may act at the same point. It is then often useful for the design engineer to know the value of the single force, i.e. the **resultant**, which has exactly the same effect as these forces. If the forces act in the same straight line, the resultant is found by simple addition or subtraction as shown in Figure 7.3; if they do not they are added by using the **parallelogram law**.



Figure 7.2 The design of an offshore oil platform requires an understanding of the combination of many forces.

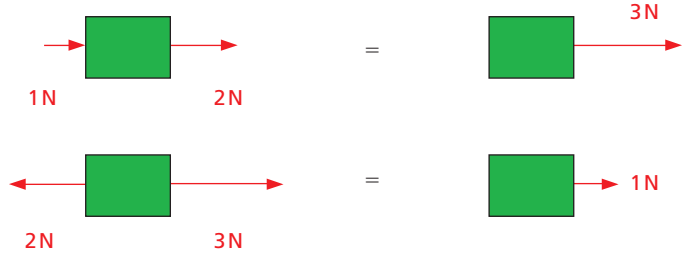


Figure 7.3 The resultant of forces acting in the same straight line is found by addition or subtraction.

Practical work

Parallelogram law

Arrange the apparatus as in Figure 7.4a with a sheet of paper behind it on a vertical board. We have to find the resultant of forces P and Q.

Read the values of P and Q from the spring balances. Mark on the paper the directions of P, Q and W as shown by the strings.

Remove the paper and, using a scale of 1 cm to represent 1 N, draw OA, OB and OD to represent the three forces P, Q and W which act at O, as in Figure 7.4b. (W = weight of the 1 kg mass = 9.8 N; therefore OD = 9.8 cm.)

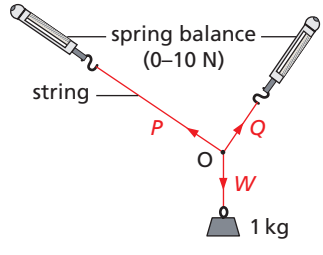


Figure 7.4a

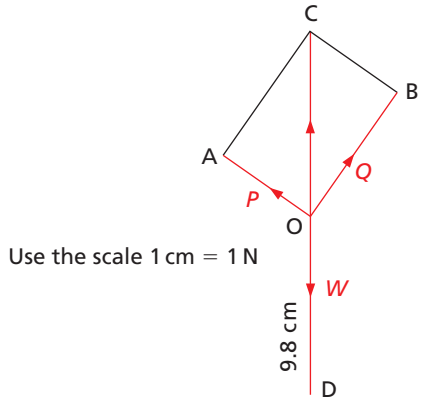


Figure 7.4b Finding a resultant by the parallelogram law

P and *Q* together are balanced by *W* and so their resultant must be a force equal and opposite to *W*.

Complete the parallelogram OACB. Measure the diagonal OC; if it is equal in size (i.e. 9.8 cm) and opposite in direction to *W* then it represents the resultant of *P* and *Q*.

The parallelogram law for adding two forces is:

> If two forces acting at a point are represented in size and direction by the sides of a parallelogram drawn from the point, their resultant is represented in size and direction by the diagonal of the parallelogram drawn from the point.

Worked example

Find the resultant of two forces of 4.0 N and 5.0 N acting at an angle of 45° to each other.

Using a scale of 1.0 cm = 1.0 N, draw parallelogram ABDC with AB = 5.0 cm, AC = 4.0 N and angle CAB = 45° (Figure 7.5). By the parallelogram law, the diagonal AD represents the resultant in magnitude and direction; it measures 8.3 cm, and angle BAD = 20°.

∴ Resultant is a force of 8.3 N acting at an angle of 20° to the force of 5.0 N.

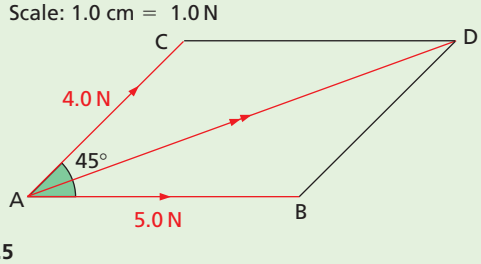
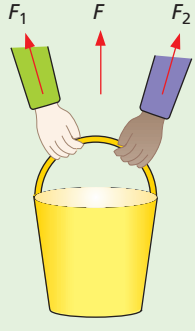


Figure 7.5

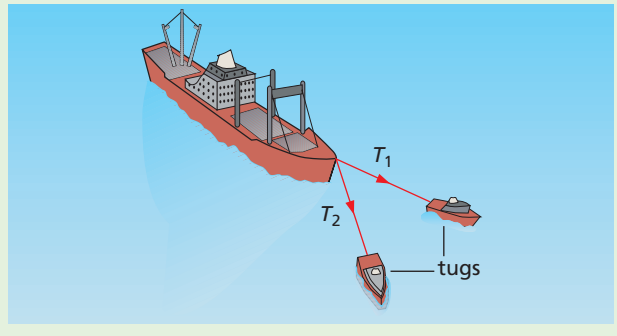
Examples of addition of forces

1 **Two people carrying a heavy bucket.** The weight of the bucket is balanced by the force *F*, the resultant of F_1 and F_2 (Figure 7.6a).

2 **Two tugs pulling a ship.** The resultant of T_1 and T_2 is forwards in direction (Figure 7.6b), and so the ship moves forwards (as long as the resultant is greater than the resistance to motion of the sea and the wind).



a



b

Figure 7.6

Vectors and scalars

A **vector** quantity is one such as force which is described completely only if both its size (magnitude) and direction are stated. It is not enough to say, for example, a force of 10 N, but rather a force of 10 N acting vertically downwards.

A vector can be represented by a straight line whose length represents the magnitude of the quantity and whose direction gives its line of action. An arrow on the line shows which way along the line it acts.

A **scalar** quantity has magnitude only. Mass is a scalar and is completely described when its value is known. Scalars are added by ordinary arithmetic; vectors are added geometrically, taking account of their directions as well as their magnitudes.

Friction

Friction is the force that opposes one surface moving, or trying to move, over another. It can be a help or a hindrance. We could not walk if there was no friction between the soles of our shoes and the ground. Our feet would slip backwards, as they tend to if we walk on ice. On the other hand, engineers try to reduce friction to a minimum in the moving parts of machinery by using lubricating oils and ball-bearings.

When a gradually increasing force P is applied through a spring balance to a block on a table (Figure 7.7), the block does not move at first. This is because an equally increasing but opposing frictional force F acts where the block and table touch. At any instant P and F are equal and opposite.

If P is increased further, the block eventually moves; as it does so F has its maximum value, called **starting** or **static friction**. When the block is moving at a steady speed, the balance reading is slightly less than that for starting friction. **Sliding** or **dynamic friction** is therefore less than starting or static friction.

Placing a mass on the block increases the force pressing the surfaces together and increases friction.

When work is done against friction, the temperatures of the bodies in contact rise (as you can test by rubbing your hands together); mechanical energy is being changed into heat energy (see Chapter 13).

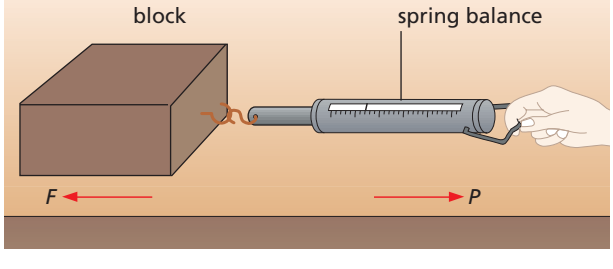


Figure 7.7 Friction opposes motion between surfaces in contact.

Questions

1 Jo, Daniel and Helen are pulling a metal ring. Jo pulls with a force of 100 N in one direction and Daniel with a force of 140 N in the opposite direction. If the ring does not move, what force does Helen exert if she pulls in the same direction as Jo?
2 A boy drags a suitcase along the ground with a force of 100 N. If the frictional force opposing the motion of the suitcase is 50 N, what is the resultant forward force on the suitcase?
3 A picture is supported by two vertical strings; if the weight of the picture is 50 N what is the force exerted by each string?
4 Using a scale of 1 cm to represent 10 N, find the size and direction of the resultant of forces of 30 N and 40 N acting at right angles to each other.
5 Find the size of the resultant of two forces of 5 N and 12 N acting
   a in opposite directions to each other,
   b at 90° to each other.

Checklist

After studying this chapter you should be able to

- combine forces acting along the same straight line to find their resultant,
- add vectors graphically to determine a resultant,
- distinguish between vectors and scalars and give examples of each,
- understand friction as the force between two surfaces that impedes motion and results in heating.

Lesson 6 Force and Acceleration

- Newton's first law
- Mass and inertia
- Newton's second law
- Weight and gravity
- Gravitational field
- Newton's third law
- Air resistance: terminal velocity
- Practical work: Effect of force and mass on acceleration

Newton's first law

Friction and air resistance cause a car to come to rest when the engine is switched off. If these forces were absent we believe that a body, once set in motion, would go on moving forever with a constant speed in a straight line. That is, force is not needed to keep a body moving with uniform velocity provided that no opposing forces act on it.

This idea was proposed by Galileo and is summed up in **Newton's first law of motion**:

> A body stays at rest, or if moving it continues to move with uniform velocity, unless an external force makes it behave differently.

It seems that the question we should ask about a moving body is not 'what keeps it moving' but 'what changes or stops its motion'.

The smaller the external forces opposing a moving body, the smaller is the force needed to keep it moving with uniform velocity. An 'airboard', which is supported by a cushion of air (Figure 8.1), can skim across the ground with little frictional opposition, so that relatively little power is needed to maintain motion.



Figure 8.1 Friction is much reduced for an airboard.

Mass and inertia

Newton's first law is another way of saying that all matter has a built-in opposition to being moved if it is at rest or, if it is moving, to having its motion changed. This property of matter is called **inertia** (from the Latin word for laziness).

Its effect is evident on the occupants of a car that stops suddenly; they lurch forwards in an attempt to continue moving, and this is why seat belts are needed. The reluctance of a stationary object to move can be shown by placing a large coin on a piece of card on your finger (Figure 8.2). If the card is flicked *sharply* the coin stays where it is while the card flies off.

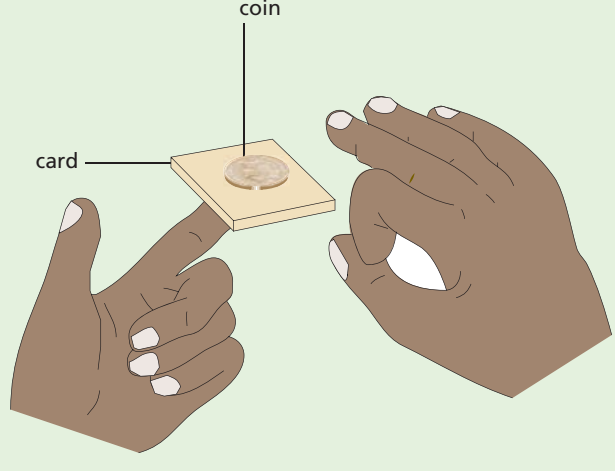


Figure 8.2 Flick the card sharply

The larger the mass of a body, the greater is its inertia, i.e. the more difficult it is to move it when at rest and to stop it when in motion. Because of this we consider that **the mass of a body measures its inertia**. This is a better definition of mass than the one given earlier (Chapter 1) in which it was stated to be the 'amount of matter' in a body.

Practical work

Effect of force and mass on acceleration

The apparatus consists of a trolley to which a force is applied by a stretched length of elastic (Figure 8.3). The velocity of the trolley is found from a tickertape timer or a motion sensor, datalogger and computer (see Figure 2.6, p.11).

First compensate the runway for friction: raise one end until the trolley runs down with uniform velocity when given a push. The dots on the tickertape should be equally spaced, or a horizontal trace obtained on a velocity–time graph. There is now no resultant force on the trolley and any acceleration produced later will be due only to the force caused by the stretched elastic.

a) Force and acceleration (mass constant)

Fix one end of a short length of elastic to the rod at the back of the trolley and stretch it until the other end is level with the front of the trolley. Practise pulling the trolley down the runway, keeping the same stretch on the elastic. After a few trials you should be able to produce a steady accelerating force.

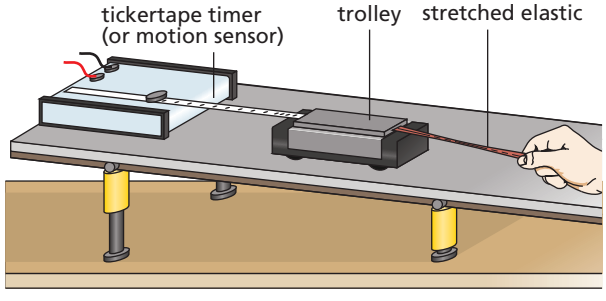


Figure 8.3

Repeat using first two and then three *identical* pieces of elastic, stretched side by side by the same amount, to give two and three units of force.

If you are using tickertape, make a tape chart for each force and use it to find the acceleration produced in cm/tentick² (see Chapter 2). Ignore the start of the tape (where the dots are too close) and the end (where the force may not be steady). If you use a motion sensor and computer to plot a velocity–time graph, the acceleration can be obtained in m/s² from the slope of the graph (Chapter 3).

Does a steady force cause a steady acceleration? Put the results in a table. Do they suggest any relationship between acceleration, *a*, and force *F*?

Force (*F*)/(no. of pieces of elastic)	1	2	3
Acceleration (*a*)/cm/tentick² or m/s²			

b) Mass and acceleration (force constant)

Do the experiment as in **a)** using two pieces of elastic (i.e. constant *F*) to accelerate first one trolley, then two (stacked one above the other) and finally three. Check the friction compensation of the runway each time.

Find the accelerations from the tape charts or computer plots and tabulate the results. Do they suggest any relationship between *a* and *m*?

Mass (*m*)/(no. of trolleys)	1	2	3
Acceleration (*a*)/cm/tentick² or m/s²			

Newton's second law

The previous experiment should show roughly that the acceleration a is

(i) directly proportional to the applied force F for a fixed mass, i.e., $a \propto F$, and
(ii) inversely proportional to the mass m for a fixed force, i.e., $a \propto 1/m$.

Combining the results into one equation, we get

$$a \propto \frac{F}{m} \quad \text{or} \quad F \propto ma$$

Therefore

$$F = kma$$

where k is the constant of proportionality.

One newton is defined as the force which gives a mass of 1 kg an acceleration of 1 m/s², i.e., 1 N = 1 kg m/s².

So if $m = 1$ kg and $a = 1$ m/s², then $F = 1$ N. Substituting in $F = kma$, we get $k = 1$ and so we can write

$$F = ma$$

This is **Newton's second law of motion**. When using it two points should be noted. First, F is the resultant (or unbalanced) force causing the acceleration a. Second, F must be in newtons, m in kilograms and a in metres per second squared, otherwise k is not 1. The law shows that a will be largest when F is large and m small.

You should now appreciate that when the forces acting on a body do not balance there is a net (resultant) force which causes a change of motion, i.e. the body accelerates or decelerates. If the forces balance, there is no change in the motion of the body. However, there may be a change of shape, in which case internal forces in the body (i.e. forces between neighbouring atoms) balance the external forces.

Worked example

A block of mass 2 kg has a constant velocity when it is pushed along a table by a force of 5 N. When the push is increased to 9 N what is

a the resultant force,
b the acceleration?

When the block moves with constant velocity the forces acting on it are balanced. The force of friction opposing its motion must therefore be 5 N.

a When the push is increased to 9 N the resultant (unbalanced) force F on the block is $(9-5)$ N = 4 N (since the frictional force is still 5 N).

b The acceleration a is obtained from $F = ma$ where $F = 4$ N and $m = 2$ kg.

$$\therefore \quad a = \frac{F}{m} = \frac{4\text{ N}}{2\text{ kg}} = \frac{4\text{ kg m/s}^2}{2\text{ kg}} = 2\text{ m/s}^2$$

Weight and gravity

The weight W of a body is the force of gravity acting on it which gives it an acceleration g when it is falling freely near the Earth's surface. If the body has mass m, then W can be calculated from $F = ma$. We put $F = W$ and $a = g$ to give

$$W = mg$$

Taking $g = 9.8$ m/s² and $m = 1$ kg, this gives $W = 9.8$ N, i.e. a body of mass 1 kg has weight 9.8 N, or near enough 10 N. Similarly a body of mass 2 kg has weight of about 20 N, and so on. While the mass of a body is always the same, its weight varies depending on the value of g. On the Moon the acceleration of free fall is only about 1.6 m/s², and so a mass of 1 kg has a weight of just 1.6 N there.

The weight of a body is directly proportional to its mass, which explains why g is the same for all bodies. The greater the mass of a body, the greater is the force of gravity on it but it does not accelerate faster when falling because of its greater inertia (i.e. its greater resistance to acceleration).

Gravitational field

The force of gravity acts through space and can cause a body, not in contact with the Earth, to fall to the ground. It is an invisible, action-at-a-distance force. We try to 'explain' its existence by saying that the Earth is surrounded by a **gravitational field** which exerts a force on any body in the field. Later, magnetic and electric fields will be considered.

The strength of a gravitational field is defined as the force acting on unit mass in the field.

Measurement shows that on the Earth's surface a mass of 1 kg experiences a force of 9.8 N, i.e. its weight is 9.8 N. The strength of the Earth's field is therefore 9.8 N/kg (near enough 10 N/kg). It is denoted by g, the letter also used to denote the acceleration of free fall. Hence

$$g = 9.8\text{ N/kg} = 9.8\text{ m/s}^2$$

We now have two ways of regarding g. When considering bodies *falling freely*, we can think of it as an acceleration of 9.8 m/s². When a body of known mass is *at rest* and we wish to know the force of gravity (in N) acting on it we think of g as the Earth's gravitational field strength of 9.8 N/kg.

Newton's third law

If a body A exerts a force on body B, then body B exerts an equal but opposite force on body A.

This is **Newton's third law of motion** and states that forces never occur singly but always in pairs as a result of the action between two bodies. For example, when you step forwards from rest your foot pushes backwards on the Earth, and the Earth exerts an equal and opposite force forward on you. Two bodies and two forces are involved. The small force you exert on the large mass of the Earth gives no noticeable acceleration to the Earth but the equal force it exerts on your very much smaller mass causes you to accelerate.

Note that the pair of equal and opposite forces **do not act on the same body**; if they did, there could

never be any resultant forces and acceleration would be impossible. For a book resting on a table, the book exerts a downward force on the table and the table exerts an equal and opposite upward force on the book; this pair of forces act on different objects and are represented by the red arrows in Figure 8.4. The weight of the book (blue arrow) does not form a pair with the upward force on the book (although they are equal numerically) as these two forces act on the same body.

An appreciation of the third law and the effect of friction is desirable when stepping from a rowing boat (Figure 8.5). You push backwards on the boat and, although the boat pushes you forwards with an equal force, it is itself now moving backwards (because friction with the water is slight). This reduces your forwards motion by the same amount – so you may fall in!

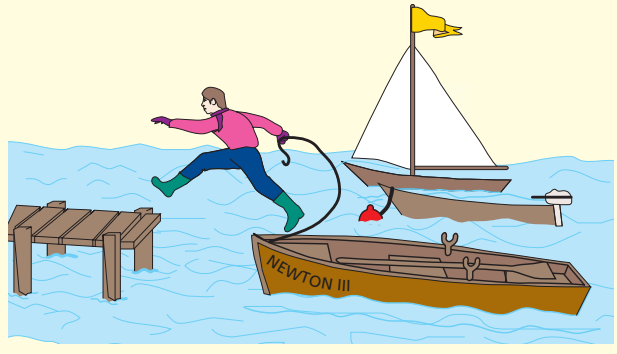


Figure 8.5 The boat moves backwards when you step forwards!

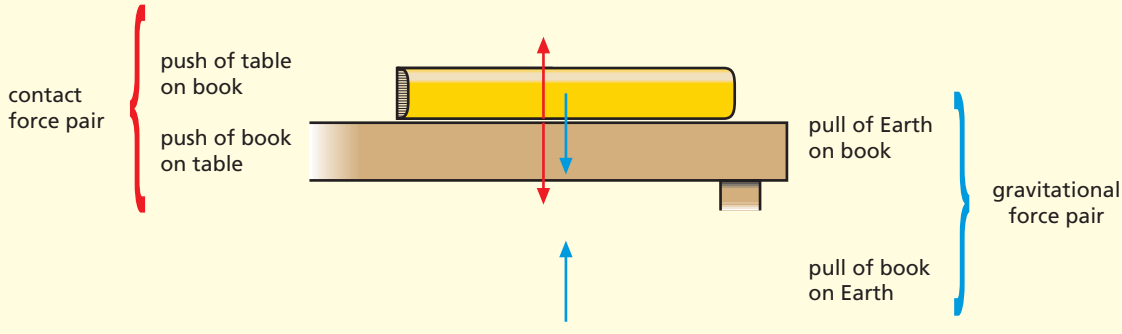


Figure 8.4 Forces between book and table

Questions

1 Which one of the diagrams in Figure 8.7 shows the arrangement of forces that gives the block of mass *M* the greatest acceleration?

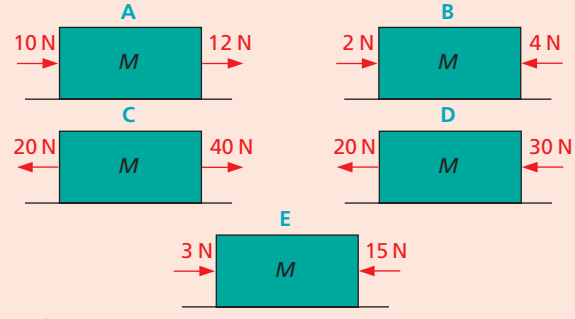


Figure 8.7

2 In Figure 8.8 if *P* is a force of 20 N and the object moves with constant velocity, what is the value of the opposing force *F*?

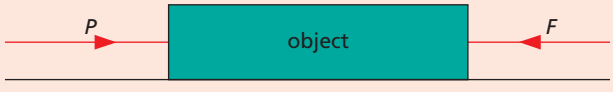


Figure 8.8

3 **a** What resultant force produces an acceleration of 5 m/s² in a car of mass 1000 kg?
b What acceleration is produced in a mass of 2 kg by a resultant force of 30 N?

4 A block of mass 500 g is pulled from rest on a horizontal frictionless bench by a steady force *F* and travels 8 m in 2 s. Find
a the acceleration,
b the value of *F*.

5 Starting from rest on a level road a girl can reach a speed of 5 m/s in 10 s on her bicycle. Find
a the acceleration,
b the average speed during the 10 s,
c the distance she travels in 10 s.
Eventually, even though she is still pedalling as fast as she can, she stops accelerating and her speed reaches a maximum value. Explain in terms of the forces acting why this happens.

6 What does an astronaut of mass 100 kg weigh
a on Earth where the gravitational field strength is 10 N/kg,
b on the Moon where the gravitational field strength is 1.6 N/kg?

7 A rocket has a mass of 500 kg.
a What is its weight on Earth where $g = 10$ N/kg?
b At lift-off the rocket engine exerts an upward force of 25 000 N. What is the resultant force on the rocket? What is its initial acceleration?

8 Figure 8.9 shows the forces acting on a raindrop which is falling to the ground.
a **(i)** *A* is the force which causes the raindrop to fall. What is this force called?
(ii) *B* is the total force opposing the motion of the drop. State *one* possible cause of this force.
b What happens to the drop when force *A* = force *B*?

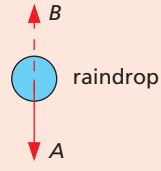


Figure 8.9

9 Explain the following using $F = ma$.
a A racing car has a powerful engine and is made of strong but lightweight material.
b A car with a small engine can still accelerate rapidly.

Checklist

After studying this chapter you should be able to

- describe an experiment to investigate the relationship between force, mass and acceleration,
- state the unit of force,
- state Newton's second law of motion and use it to solve problems,
- recall and use the equation $W = mg$
- define the strength of the Earth's gravitational field,
- describe the motion of an object falling in air.

Lesson 7 : Energy Work and Power

- Forms of energy
- Energy transfers
- Energy measurements
- Energy conservation
- Energy of food
- Combustion of fuels
- Practical work: Measuring power

Energy is a theme that pervades all branches of science. It links a wide range of phenomena and enables us to explain them. It exists in different forms and when something happens, it is likely to be due to energy being transferred from one form to another. Energy transfer is needed to enable people, computers, machines and other devices to work and to enable processes and changes to occur. For example, in Figure 13.1, the water skier can only be pulled along by the boat if there is energy transfer in its engine from the burning petrol to its rotating propeller.

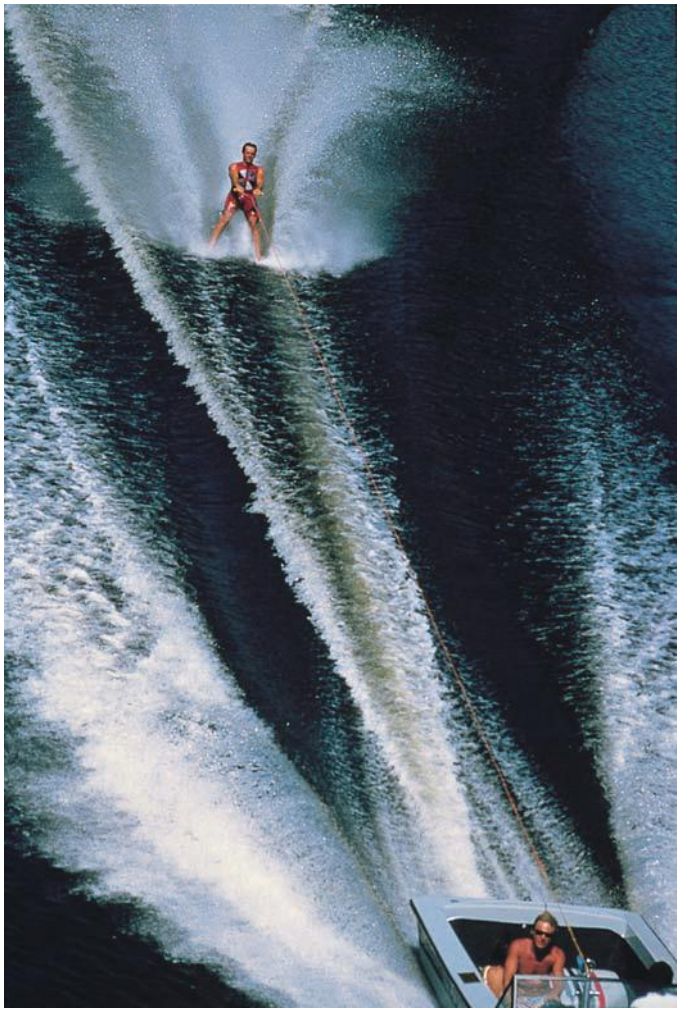


Figure 13.1 Energy transfer in action

Forms of energy

a) Chemical energy

Food and fuels, like oil, gas, coal and wood, are concentrated stores of **chemical energy** (see Chapter 15). The energy of food is released by chemical reactions in our bodies, and during the transfer to other forms we are able to do useful jobs. Fuels cause energy transfers when they are burnt in an engine or a boiler. Batteries are compact sources of chemical energy, which in use is transferred to electrical energy.

b) Potential energy (p.e.)

This is the energy a body has because of its position or condition. A body above the Earth's surface, like water in a mountain reservoir, has **potential energy** (p.e.) stored in the form of **gravitational potential energy**.

Work has to be done to compress or stretch a spring or elastic material and energy is transferred to potential energy; the p.e. is stored in the form of **strain energy** (or **elastic potential energy**). If the catapult in Figure 13.3c were released, the strain energy would be transferred to the projectile.

c) Kinetic energy (k.e.)

Any moving body has **kinetic energy** (k.e.) and the faster it moves, the more k.e. it has. As a hammer drives a nail into a piece of wood, there is a transfer of energy from the k.e. of the moving hammer to other forms of energy.

d) Electrical energy

Electrical energy is produced by energy transfers at power stations and in batteries. It is the commonest form of energy used in homes and industry because of the ease of transmission and transfer to other forms.

e) Heat energy

This is also called **thermal** or **internal energy** and is the final fate of other forms of energy. It is transferred by conduction, convection or radiation.

f) Other forms

These include **light energy** and other forms of **electromagnetic radiation**, **sound** and **nuclear energy**.

Energy transfers

a) Demonstration

The apparatus in Figure 13.2 can be used to show a battery changing **chemical energy** to **electrical energy** which becomes **kinetic energy** in the electric motor. The motor raises a weight, giving it **potential energy**. If the changeover switch is joined to the lamp and the weight allowed to fall, the motor acts as a generator in which there is an energy transfer from **kinetic energy** to **electrical energy**. When this is supplied to the lamp, it produces a transfer to **heat** and **light energy**.

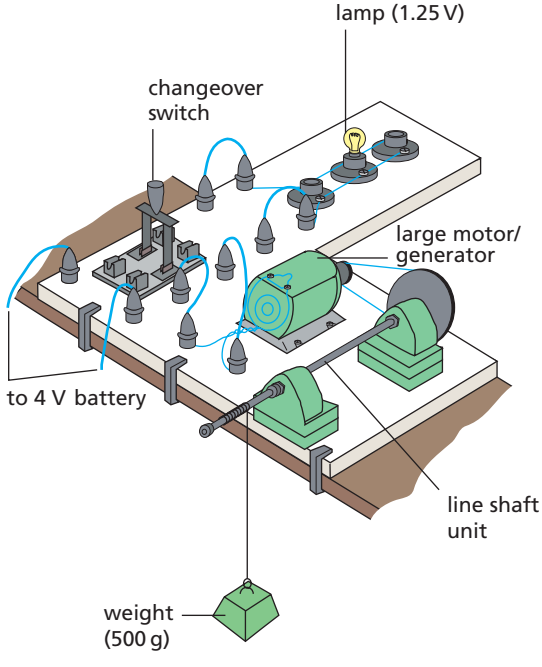


Figure 13.2 Demonstrating energy transfers

b) Other examples

Study the energy transfers shown in Figures 13.3a to d. Some devices have been invented to cause particular energy transfers. For example, a **microphone** changes sound energy into electrical energy; a **loudspeaker** does the reverse. Belts, chains or gears are used to transfer energy between moving parts, such as those in a bicycle.



a Potential energy to kinetic energy



b Electrical energy to heat and light energy



c Chemical energy (from muscles in the arm) to p.e. (strain energy of catapult)



d Potential energy of water to kinetic energy of turbine to electrical energy from generator

Figure 13.3 Some energy transfers

Energy measurements

a) Work

In science the word **work** has a different meaning from its everyday use. **Work is done when a force moves.** No work is done in the scientific sense by someone standing still holding a heavy pile of books: an upward force is exerted, but no motion results.

If a building worker carries ten bricks up to the first floor of a building, he does more work than if he carries only one brick because he has to exert a larger force. Even more work is required if he carries the ten bricks to the second floor. The amount of work done depends on the size of the force applied and the distance it moves. We therefore measure work by

work = force × distance moved in direction of force (1)

The unit of work is the **joule** (J); it is the work done when a force of 1 newton (N) moves through 1 metre (m). For example, if you have to pull with a force of 50 N to move a crate steadily 3 m in the direction of the force (Figure 13.4a), the work done is 50 N × 3 m = 150 N m = 150 J. That is

joules = newtons × metres

If you lift a mass of 3 kg vertically through 2 m (Figure 13.4b), you have to exert a vertically upward force equal to the weight of the body, i.e. 30 N (approximately) and the work done is 30 N × 2 m = 60 N m = 60 J.

Note that we must always take the distance in the direction in which the force acts.

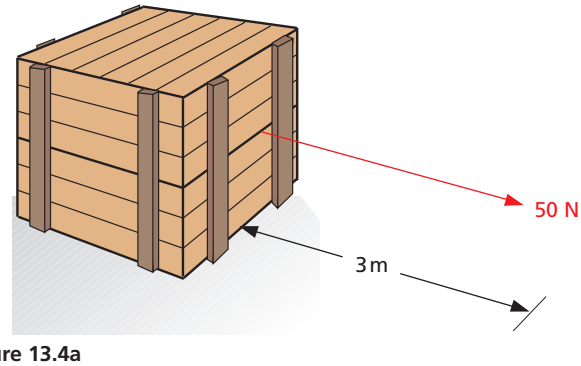


Figure 13.4a

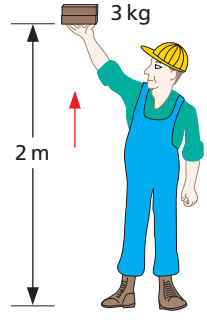


Figure 13.4b

b) Measuring energy transfers

In an energy transfer, work is done. **The work done is a measure of the amount of energy transferred.** For example, if you have to exert an upward force of 10 N to raise a stone steadily through a vertical distance of 1.5 m, the work done is 15 J. This is also the amount of chemical energy transferred from your muscles to potential energy of the stone. All forms of energy, as well as work, are measured in joules.

c) Power

The more powerful a car is, the faster it can accelerate or climb a hill, i.e. the more rapidly it does work. The **power** of a device is the work it does per second, i.e. the rate at which it does work. This is **the same as the rate at which it transfers energy from one form to another.**

$$\text{power} = \frac{\text{work done}}{\text{time taken}} = \frac{\text{energy transfer}}{\text{time taken}} \quad (2)$$

The unit of power is the **watt** (W) and is **a rate of working of 1 joule per second**, i.e. 1 W = 1 J/s. Larger units are the **kilowatt** (kW) and the **megawatt** (MW):

$$1\,\text{kW} = 1000\,\text{W} = 10^3\,\text{W}$$

$$1\,\text{mW} = 1\,000\,000\,\text{W} = 10^6\,\text{W}$$

If a machine does 500 J of work in 10 s, its power is 500 J/10 s = 50 J/s = 50 W. A small car develops a maximum power of about 25 kW.

Practical work

Measuring power

a) Your own power

Get someone with a stopwatch to time you running up a flight of stairs, the more steps the better. Find your weight (in newtons). Calculate the total vertical height (in metres) you have climbed by measuring the height of one step and counting the number of steps.

The work you do (in joules) in lifting your weight to the top of the stairs is (your weight) × (vertical height of stairs). Calculate your power (in watts) from equation (2). About 0.5 kW is good.

b) Electric motor

This experiment is described in Chapter 40.

Energy conservation

a) Principle of conservation of energy

This is one of the basic laws of physics and is stated as follows.

Energy cannot be created or destroyed; it is always conserved.

However, energy is continually being transferred from one form to another. Some forms, such as electrical and chemical energy, are more easily transferred than others, such as heat, for which it is hard to arrange a useful transfer.

Ultimately all energy transfers result in the surroundings being heated (as a result of doing work against friction) and the energy is wasted, i.e. spread out and increasingly more difficult to use. For example, when a brick falls its potential energy becomes kinetic energy; as it hits the ground, its temperature rises and heat and sound are produced. If it seems in a transfer that some energy has disappeared, the 'lost' energy is often converted into non-useful heat. This appears to be the fate of all energy in the Universe and is one reason why new sources of useful energy have to be developed (Chapter 15).

b) Efficiency of energy transfers

The **efficiency** of a device is the percentage of the energy supplied to it that is usefully transferred. It is calculated from the expression:

$$\text{efficiency} = \frac{\text{useful energy output}}{\text{total energy input}} \times 100\%$$

For example, for the lever shown in Figure 10.4 (p. 40)

$$\text{efficiency} = \frac{\text{work done on load}}{\text{work done by effort}} \times 100\%$$

This will be less than 100% if there is friction in the fulcrum.

Table 13.1 lists the efficiencies of some devices and the energy transfers involved.

Table 13.1

Device	% Efficiency	Energy transfer
large electric motor	90	electrical to k.e.
large electric generator	90	k.e. to electrical
domestic gas boiler	75	chemical to heat
compact fluorescent lamp	50	electrical to light
steam turbine	45	heat to k.e.
car engine	25	chemical to k.e.
filament lamp	10	electrical to light

A device is efficient if it transfers energy mainly to useful forms and the 'lost' energy is small.

Energy of food

When food is eaten it reacts with the oxygen we breathe into our lungs and is slowly 'burnt'. As a result chemical energy stored in food becomes thermal energy to warm the body and mechanical energy for muscular movement.

The **energy value** of a food substance is the amount of energy released when 1 kg is completely oxidised.

Energy value is measured in J/kg. The energy values of some foods are given in Figure 13.5 in megajoules per kilogram.

▶▶

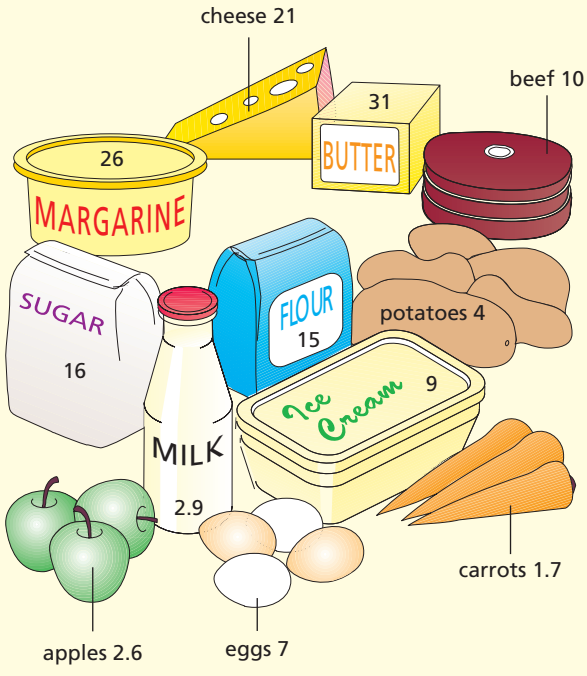


Figure 13.5 Energy values of some foods in MJ/kg

Foods with high values are 'fattening' and if more food is eaten than the body really needs, the extra is stored as fat. The average adult requires about 10 MJ per day.

Our muscles change chemical energy into mechanical energy when we exert a force – to lift a weight, for example. Unfortunately, they are not very good at doing this; of every 100 J of chemical energy they use, they can convert only 25 J into mechanical energy – that is, they are only 25% efficient at changing chemical energy into mechanical energy. The other 75 J becomes thermal energy, much of which the body gets rid of by sweating.

Combustion of fuels

Fuels can be solids such as wood and coal, liquids such as fuel oil and paraffin, or gases such as methane and butane.

Some fuels are better than others for certain jobs. For example, fuels for cooking or keeping us warm should, as well as being cheap, have a high **heating value**. This means that every gram of fuel should produce a large amount of heat energy when burnt.

A fuel for a space rocket (e.g. liquid hydrogen) must also burn very quickly so that the gases created expand rapidly and leave the rocket at high speed.

The heating values of some fuels are given in Table 13.2 in kilojoules per gram.

Table 13.2 Heating values of fuels in kJ/g

Solids	Value	Liquids	Value	Gases	Value
wood	17	fuel oil	45	methane	55
coal	25–33	paraffin	48	butane	50

The thick dark liquid called **petroleum** or **crude oil** is the source of most liquid and gaseous fuels. It is obtained from underground deposits at oil wells in many parts of the world. Natural gas (methane) is often found with it. In an oil refinery different fuels are obtained from petroleum, including fuel oil for industry, diesel oil for lorries, paraffin (kerosene) for jet engines, and petrol for cars, as well as butane (bottled gas).

Questions

1 Name the energy transfers which occur when
   - **a** an electric bell rings,
   - **b** someone speaks into a microphone,
   - **c** a ball is thrown upwards,
   - **d** there is a picture on a television screen,
   - **e** a torch is on.

2 Name the forms of energy represented by the letters A, B, C and D in the following statement.
In a coal-fired power station, the (A) energy of coal becomes (B) energy which changes water into steam. The steam drives a turbine which drives a generator. A generator transfers (C) energy into (D) energy.

3 How much work is done when a mass of 3 kg (weighing 30 N) is lifted vertically through 6 m?

4 A hiker climbs a hill 300 m high. If she has a mass of 50 kg calculate the work she does in lifting her body to the top of the hill.

5 In loading a lorry a man lifts boxes each of weight 100 N through a height of 1.5 m.
   - **a** How much work does he do in lifting one box?
   - **b** How much energy is transferred when one box is lifted?
   - **c** If he lifts four boxes per minute at what power is he working?

6 A boy whose weight is 600 N runs up a flight of stairs 10 m high in 12 s. What is his average power?

7 a When the energy input to a gas-fired power station is 1000 MJ, the electrical energy output is 300 MJ. What is the efficiency of the power station in changing the energy in gas into electrical energy?
b What form does the 700 MJ of 'lost' energy take?
c What is the fate of the 'lost' energy?

8 State what energy transfers occur in
a a hairdryer,
b a refrigerator,
c an audio system.

9 An escalator carries 60 people of average mass 70 kg to a height of 5 m in one minute. Find the power needed to do this.

Checklist

After studying this chapter you should be able to

- recall the different forms of energy,
- describe energy transfers in given examples,
- relate work done to the magnitude of a force and the distance moved,
- use the relation work done = force × distance moved to calculate energy transfer,
- define the unit of work,
- relate power to work done and time taken, and give examples,
- recall that power is the rate of energy transfer, give its unit and solve problems,
- describe an experiment to measure your own power,
- state the principle of conservation of energy,
- understand qualitatively the meaning of efficiency,
- recall and use the equation

$$\text{efficiency} = \frac{\text{useful energy output}}{\text{energy input}} \times 100\%$$

Practice

1. Calculate the work done by a 47 N force pushing a pencil 0.26 m.
2. Calculate the work done by a 47 N force pushing a 0.025 kg pencil 0.25 m against a force of 23 N.
3. Calculate the work done by a 2.4 N force pushing a 400. g sandwich across a table 0.75 m wide.
4. How much work is it to lift a 20. kg sack of potatoes vertically 6.5 m?
5. If a small motor does 520. J of work to move a toy car 260. m, what force does it exert?
6. A 75.0 kg man pushes on a 5.0×10^5 **ton** wall for 250 s but it does not move. How much work does he do on the wall? (2000 lb = 1 ton; 0.454 kg = 1 lb)
7. How much power does it take to lift 30.0 N 10.0 m high in 5.00 s?
8. How much power does it take to lift 30.0 kg 10.0 m high in 5.00 s?
9. You move a 25 N object 5.0 meters. How much work did you do?
10. You carry a 20. N bag of dog food up a 6.0 m flight of stairs. How much work was done?
11. You push down on a 3.0 N box for 10. minutes. How much work was done?
12. You use 35 J of energy to move a 7.0 N object. How far did you move it?
13. You do 45 J of work in 3.0 seconds. How much power do you use?
14. A car uses 2,500 Joules in 25 seconds. Find power.
15. A 60. watt light bulb runs for 5.0 seconds. How much energy does it use?
16. How much power does it take to lift 30.0 N 10.0 m high in 5.00 s?
17. How much power does it take to lift 30.0 kg 10.0 m high in 5.00 s?
18. You move a 25 N object 5.0 meters. How much work did you do?
19. You carry a 20. N bag of dog food up a 6.0 m flight of stairs. How much work was done?
20. You push down on a 3.0 N box for 10. minutes. How much work was done?

21.Transformation of Energy

Directions: Use the following forms of energy to fill in the table below: mechanical, electrical, heat, radiant, chemical, nuclear, and sound. The first one has been done for you.

	ORIGINAL ENERGY FORM	FINAL ENERGY FORM
1. Electric motor	electrical	mechanical

2. A battery that runs a moving toy		
3. A solar panel on the roof of a house		
4. A person lifting a chair		
5. A nuclear power plant		
6. A toaster		
7. A church bell		
8. Gasoline powering a car		
9. A light bulb		
10. Photosynthesis		

Lesson 8 Kinetic and Potential Energy

- Kinetic energy (k.e.)
- Potential energy (p.e.)
- Conservation of energy
- Elastic and inelastic collisions
- Driving and car safety
- Practical work: Change of p.e. to k.e.

Energy and its different forms were discussed earlier (Chapter 13). Here we will consider **kinetic energy** (k.e.) and **potential energy** (p.e.) in more detail.

Kinetic energy (k.e.)

Kinetic energy is the energy a body has because of its motion.

For a body of mass m travelling with velocity v,

$$\text{kinetic energy} = E_k = \frac{1}{2}mv^2$$

If m is in kg and v in m/s, then kinetic energy is in J. For example, a football of mass 0.4 kg (400 g) moving with velocity 20 m/s has

$$\text{k.e.} = \frac{1}{2}mv^2 = \frac{1}{2} \times 0.4\,\text{kg} \times (20)^2\,\text{m}^2/\text{s}^2$$
$$= 0.2 \times 400\,\text{kg m/s}^2 \times \text{m}$$
$$= 80\,\text{N m} = 80\,\text{J}$$

Since k.e. depends on v^2, a high-speed vehicle travelling at 1000 km/h (Figure 14.1), has one hundred times the k.e. it has at 100 km/h.

Potential energy (p.e.)

Potential energy is the energy a body has because of its position or condition.

A body above the Earth's surface is considered to have an amount of gravitational potential energy equal to the work that has been done against gravity by the force used to raise it. To lift a body of mass m through a vertical height h at a place where the Earth's gravitational field strength is g, needs a force equal and opposite to the weight mg of the body. Hence

$$\text{work done by force} = \text{force} \times \text{vertical height}$$
$$= mg \times h$$

$$\therefore \quad \text{potential energy} = E_p = mgh$$

When m is in kg, g in N/kg (or m/s^2) and h in m, the potential energy is in J. For example, if $g = 10$ N/kg, the potential energy gained by a 0.1 kg (100 g) mass raised vertically by 1 m is

$$0.1\,\text{kg} \times 10\,\text{N/kg} \times 1\,\text{m} = 1\,\text{N m} = 1\,\text{J}$$

Note Strictly speaking we are concerned with *changes* in potential energy from that which a body has at the Earth's surface, rather than with actual values. The expression for potential energy is therefore more correctly written

$$\Delta E_p = mgh$$

where Δ (pronounced 'delta') stands for 'change in'.



Figure 14.1 Kinetic energy depends on the square of the velocity.

Practical work

Change of p.e. to k.e.

Friction-compensate a runway and arrange the apparatus as in Figure 14.2 with the bottom of the 0.1 kg (100 g) mass 0.5 m from the floor.

Start the timer and release the trolley. It will accelerate until the falling mass reaches the floor; after that it moves with *constant* velocity *v*.

From your results calculate *v* in m/s (on the tickertape 50 ticks = 1 s). Find the mass of the trolley in kg. Work out:

k.e. gained by trolley and 0.1 kg mass = ___J
p.e. lost by 0.1 kg mass = ___J

Compare and comment on the results.

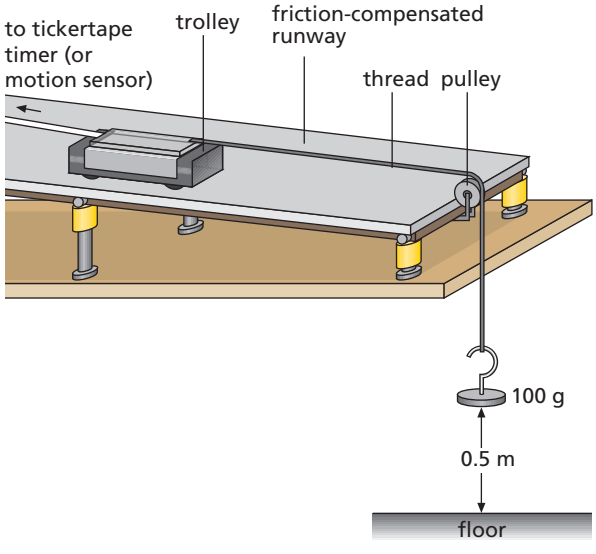


Figure 14.2

Conservation of energy

A mass *m* at height *h* above the ground has potential energy = *mgh* (Figure 14.3). When it falls, its velocity increases and it gains kinetic energy at the expense of its potential energy. If it starts from rest and air resistance is negligible, its kinetic energy on reaching the ground equals the potential energy lost by the mass

$$\frac{1}{2}mv^2 = mgh$$

or

loss of p.e. = gain of k.e.

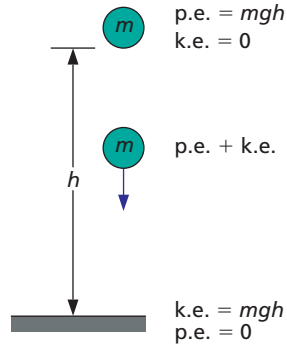


Figure 14.3 Loss of p.e. = gain of k.e.

This is an example of the **principle of conservation of energy** which was discussed in Chapter 13.

In the case of a pendulum (Figure 14.4), kinetic and potential energy are interchanged continually. The energy of the bob is all potential energy at the end of the swing and all kinetic energy as it passes through its central position. In other positions it has both potential and kinetic energy. Eventually all the energy is changed to heat as a result of overcoming air resistance.

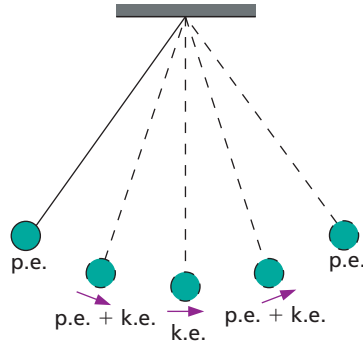


Figure 14.4 Interchange of p.e. and k.e. for a simple pendulum

Elastic and inelastic collisions

In all collisions (where no external force acts) there is normally a loss of kinetic energy, usually to heat energy and to a small extent to sound energy. The greater the proportion of kinetic energy lost, the less **elastic** is the collision, i.e. the more **inelastic** it is. In a perfectly elastic collision, kinetic energy is conserved.

▶▶



Figure 14.5 Newton's cradle is an instructive toy for studying collisions and conservation of energy.

Driving and car safety

a) Braking distance and speed

For a car moving with speed v, the brakes must be applied over a braking distance s to bring the car to rest. The **braking distance is directly proportional to the square of the speed**, i.e. if v is doubled, s is quadrupled. The **thinking distance** (i.e. the distance travelled while the driver is reacting before applying the brakes) has to be added to the **braking distance** to obtain the **overall stopping distance**, in other words

stopping distance = thinking distance + braking distance

Typical values taken from the Highway Code are given in Table 14.1 for different speeds. The greater the speed, the greater the stopping distance for a given braking force. (To stop the car in a given distance a greater braking force is needed for higher speeds.)

Thinking distance depends on the driver's reaction time – this will vary with factors such as the driver's degree of tiredness, use of alcohol or drugs, eyesight and the visibility of the hazard. Braking distance varies with both the road conditions and the state of the car; it is longer when the road is wet or icy, when friction between the tyres and the road is low, than when conditions are dry. Efficient brakes and deep tyre tread help to reduce the braking distance.

Table 14.1

Speed/mph	**20**	**40**	**60**	**80**
Thinking distance/metres	6	12	18	24
Braking distance/metres	6	24	54	96
Total stopping distance/metres	**12**	**36**	**72**	**120**

b) Car design and safety

When a car stops rapidly in a collision, large forces are produced on the car and its passengers, and their kinetic energy has to be dissipated.

Crumple zones at the front and rear collapse in such a way that the kinetic energy is absorbed gradually (Figure 14.6). As we saw in Chapter 12 this extends the collision time and reduces the decelerating force and hence the potential for injury to the passengers.

Extensible seat belts exert a backwards force (of 10 000 N or so) over about 0.5 m, which is roughly the distance between the front seat occupants and the windscreen. In a car travelling at 15 m/s (34 mph), the effect felt by anyone *not* using a seat belt is the same as that produced by jumping off a building 12 m high!

Air bags in some cars inflate and protect the driver from injury by the steering wheel.

Head restraints ensure that if the car is hit from behind, the head goes forwards with the body and not backwards over the top of the seat. This prevents damage to the top of the spine.

All these are **secondary** safety devices which aid **survival** in the event of an accident. **Primary** safety factors help to **prevent** accidents and depend on the car's roadholding, brakes, steering, handling and above all on the driver since most accidents are due to driver error.

The chance of being killed in an accident is about **five times less** if seat belts are worn and head restraints are installed.



Figure 14.6 Cars in an impact test showing the collapse of the front crumple zone

Worked example

A boulder of mass 4 kg rolls over a cliff and reaches the beach below with a velocity of 20 m/s. Find: **a** the kinetic energy of the boulder as it lands; **b** the potential energy of the boulder when it was at the top of the cliff and **c** the height of the cliff.

a Mass of boulder = $m = 4\,\text{kg}$

Velocity of boulder as it lands = $v = 20\,\text{m/s}$

$$\therefore \quad \text{k.e. of boulder as it lands} = E_k = \frac{1}{2}mv^2$$

$$= \frac{1}{2} \times 4\ \text{kg} \times (20)^2\ \text{m}^2/\text{s}^2$$

$$= 800\,\text{kg m/s}^2 \times \text{m}$$

$$= 800\,\text{N m}$$

$$= 800\,\text{J}$$

b Applying the principle of conservation of energy (and neglecting energy lost in overcoming air resistance),

p.e. of boulder on cliff = k.e. as it lands

$$\therefore \quad \Delta E_p = E_k = 800\,\text{J}$$

c If h is the height of the cliff,

$$\Delta E_p = mgh$$

$$\therefore \quad h = \frac{\Delta E_p}{mg} = \frac{800\ \text{J}}{4\ \text{kg} \times 10\ \text{m/s}^2} = \frac{800\ \text{N m}}{40\ \text{kg m/s}^2}$$

$$= \frac{800\ \text{kg m/s}^2 \times \text{m}}{40\ \text{kg m/s}^2} = 20\ \text{m}$$

Questions

1 Calculate the k.e. of
- **a** a 1 kg trolley travelling at 2 m/s,
- **b** a 2 g (0.002 kg) bullet travelling at 400 m/s,
- **c** a 500 kg car travelling at 72 km/h.

2 **a** What is the velocity of an object of mass 1 kg which has 200 J of k.e.?
- **b** Calculate the p.e. of a 5 kg mass when it is **(i)** 3 m, **(ii)** 6 m, above the ground. (g = 10 N/kg)

3 A 100 g steel ball falls from a height of 1.8 m on to a metal plate and rebounds to a height of 1.25 m. Find
- **a** the p.e. of the ball before the fall (g = 10 m/s^2),
- **b** its k.e. as it hits the plate,
- **c** its velocity on hitting the plate,
- **d** its k.e. as it leaves the plate on the rebound,
- **e** its velocity of rebound.

4 It is estimated that 7×10^6 kg of water pours over the Niagara Falls every second. If the falls are 50 m high, and if all the energy of the falling water could be harnessed, what power would be available? (g = 10 N/kg)

Checklist

After studying this chapter you should be able to

- define kinetic energy (k.e.),
- perform calculations using $E_k = \frac{1}{2}mv^2$,
- define potential energy (p.e.),
- calculate changes in p.e. using $\Delta E_p = mgh$,
- apply the principle of conservation of energy to simple mechanical systems, such as a pendulum,
- recall the effect of speed on the braking distance of a vehicle,
- describe secondary safety devices in cars.

I. What can you find in your home that are examples of kinetic and potential energy?

1. Kinetic: _____
2. Kinetic: _____
3. Potential: _____
4. Potential: _____

II. Solve the following word problems using the kinetic and potential energy formulas

PE = KE =

1. Determine the kinetic energy of a 1000-kg roller coaster car that is moving with a speed of 20.0 m/s.
2. If the roller coaster car in the above problem were moving with twice the speed, then what would be its new kinetic energy?
3. What is the gravitational potential energy of a 61.2 kg person standing on the roof of a 10-story building relative to (a) the tenth floor, (b) the sixth floor, (c) the first floor? (Each story is 2.50 m high.)
5. How fast must a 1000. kg car be moving to have a kinetic energy of:
   (a) 2.0×10^3 J, (b) 2.0×10^5 J, (c) 1.0 kWh?
6. A 50.0 kg bicyclist on a 10.0 kg bicycle speeds up from 5.00 m/s to 10.0 m/s.
   (a) What was the total kinetic energy before accelerating?
   (b) What was the total kinetic energy after accelerating?
   (c) How much work was done to increase the kinetic energy of the bicyclist?
   (d) Is it more work to speed up from 0 to 5.00 m/s than from 5.00 to 10.0 m/s?
7. How high would you have to lift a 1000. kg car to give it a potential energy of:
   (a) 2.0×10^3 J, (b) 2.00×10^5 J, (c) 1.00 kWh?
8. A rolling ball has 18 J of kinetic energy and is rolling 3.0 m/s. Find its mass.

Lesson 9 Pressure and Liquid Pressure

- Pressure
- Liquid pressure
- Water supply system
- Hydraulic machines
- Expression for liquid pressure
- Pressure gauges

Pressure

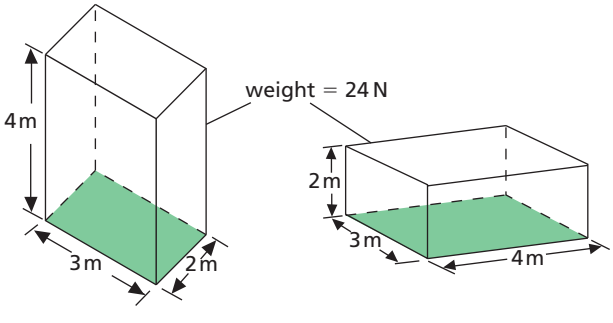
To make sense of some effects in which a force acts on a body we have to consider not only the force but also the area on which it acts. For example, wearing skis prevents you sinking into soft snow because your weight is spread over a greater area. We say the **pressure** is less.

Pressure is the force (or thrust) **acting on unit area** (i.e. 1 m²) and is calculated from

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

The unit of pressure is the **pascal** (Pa); it equals 1 newton per square metre (N/m²) and is quite a small pressure. An apple in your hand exerts about 1000 Pa.

The greater the area over which a force acts, the less the pressure. Figure 16.1 shows the pressure exerted on the floor by the same box standing on end (Figure 16.1a) and lying flat (Figure 16.1b). This is why a tractor with wide wheels can move over soft ground. The pressure is large when the area is small and this is why nails are made with sharp points. Walnuts can be broken in the hand by squeezing two together but not one. Why?



a area = 6 m²
pressure = $\frac{24\text{ N}}{6\text{ m}^2}$ = 4 Pa

b area = 12 m²
pressure = $\frac{24\text{ N}}{12\text{ m}^2}$ = 2 Pa

Figure 16.1

Liquid pressure

1 **Pressure in a liquid increases with depth** because the further down you go, the greater the weight of liquid above. In Figure 16.2a water spurts out fastest and furthest from the lowest hole.

2 **Pressure at one depth acts equally in all directions.** The can of water in Figure 16.2b has similar holes all round it at the same level. Water comes out equally fast and spurts equally far from each hole. Hence the pressure exerted by the water at this depth is the same in all directions.

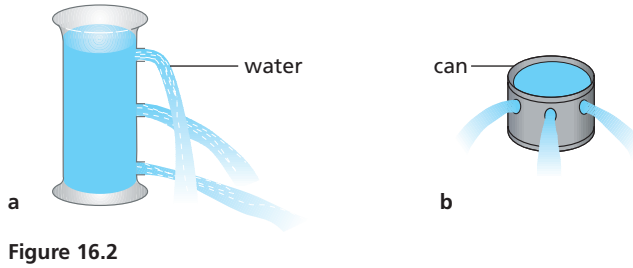


Figure 16.2

3 **A liquid finds its own level.** In the U-tube of Figure 16.3a the liquid pressure at the foot of P is greater than at the foot of Q because the left-hand column is higher than the right-hand one. When the clip is opened the liquid flows from P to Q until the pressure and the levels are the same, i.e. the liquid 'finds its own level'. Although the weight of liquid in Q is now greater than in P, it acts over a greater area because tube Q is wider.
In Figure 16.3b the liquid is at the same level in each tube and confirms that the pressure at the foot of a liquid column depends only on the *vertical* depth of the liquid and not on the tube width or shape.

4 **Pressure depends on the density of the liquid.** The denser the liquid, the greater the pressure at any given depth.

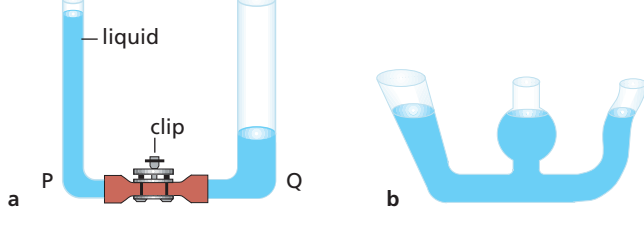


Figure 16.3

Water supply system

A town's water supply often comes from a reservoir on high ground. Water flows from it through pipes to any tap or storage tank that is below the level of water in the reservoir (Figure 16.4). The lower the place supplied, the greater the water pressure. In very tall buildings it may be necessary first to pump the water to a large tank in the roof.

Reservoirs for water supply or for hydroelectric power stations are often made in mountainous regions by building a dam at one end of a valley. The dam must be thicker at the bottom than at the top due to the large water pressure at the bottom.

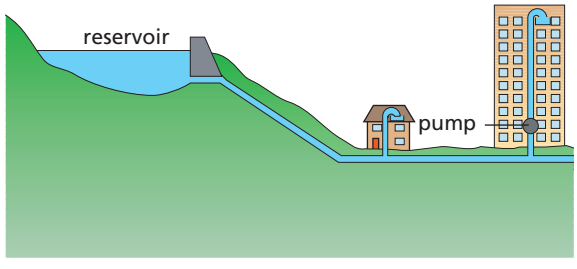


Figure 16.4 Water supply system. Why is the pump needed in the high-rise building?

Hydraulic machines

Liquids are almost incompressible (i.e. their volume cannot be reduced by squeezing) and they 'pass on' any pressure applied to them. Use is made of these facts in **hydraulic** machines. Figure 16.5 shows the principle on which they work.

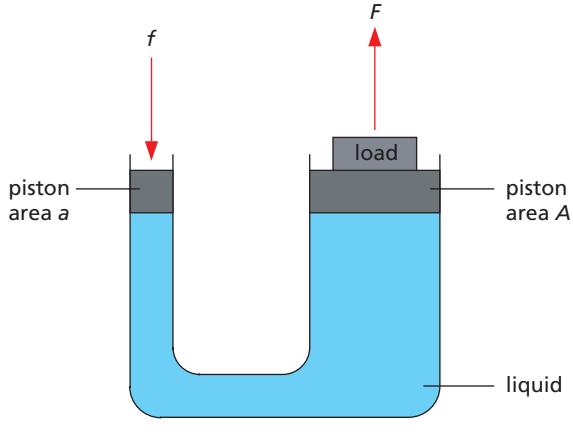


Figure 16.5 The hydraulic principle

Suppose a downward force f acts on a piston of area a. The pressure transmitted through the liquid is

$$\text{pressure} = \frac{\text{force}}{\text{area}} = \frac{f}{A}$$

This pressure acts on a second piston of larger area A, producing an upward force, F = pressure × area:

$$F = \frac{f}{a} \times A$$

or

$$F = f \times \frac{A}{a}$$

Since A is larger than a, F must be larger than f and the hydraulic system is a force multiplier; the **multiplying factor** is A/a.

For example, if f = 1 N, $a = \frac{1}{100}$ m² and A = $\frac{1}{2}$ m² then

$$F = 1\text{ N} \times \frac{\frac{1}{2}\text{ m}^2}{\frac{1}{100}\text{ m}^2}$$

$$= 50\text{ N}$$

A force of 1 N could lift a load of 50 N; the hydraulic system multiplies the force 50 times.

A **hydraulic jack** (Figure 16.6) has a platform on top of piston B and is used in garages to lift cars. Both valves open only to the right and they allow B to be raised a long way when A moves up and down repeatedly. When steel is forged using a hydraulic press there is a fixed plate above piston B and the sheets of steel are placed between B and the plate.

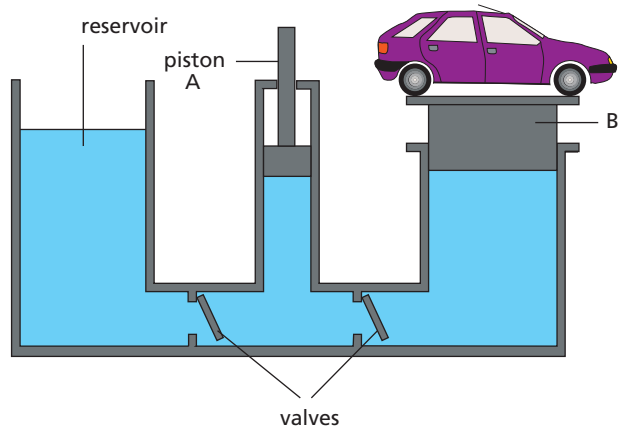


Figure 16.6 A hydraulic jack

Hydraulic fork-lift trucks and similar machines such as loaders (Figure 16.7) work in the same way.



Figure 16.7 A hydraulic machine in action

Hydraulic car brakes are shown in Figure 16.8. When the brake pedal is pushed, the piston in the master cylinder exerts a force on the brake fluid and the resulting pressure is transmitted equally to eight other pistons (four are shown). These force the brake shoes or pads against the wheels and stop the car.

Expression for liquid pressure

In designing a dam an engineer has to calculate the pressure at various depths below the water surface. The pressure increases with depth and density.

An expression for the pressure at a depth h in a liquid of density ρ can be found by considering a horizontal area A (Figure 16.9). The force acting vertically downwards on A equals the weight of a liquid column of height h and cross-sectional area A above it. Then

$$\text{volume of liquid column} = hA$$

Since mass = volume × density we can say

$$\text{mass of liquid column} = hA\rho$$

Taking a mass of 1 kg to have weight 10 N,

$$\text{weight of liquid column} = 10hA\rho$$

$$\therefore \quad \text{force on area A} = 10hA\rho$$

As

$$\text{pressure} = \text{force}/\text{area} = 10hA\rho/A$$

then

$$\text{pressure} = 10h\rho$$

This is usually written as

$$\text{pressure} = \text{depth} \times \text{density} \times g$$

$$= h\rho g$$

where g is the acceleration of free fall (Chapter 4).

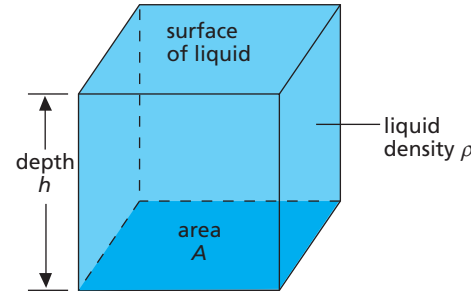


Figure 16.9

This pressure acts equally in all directions at depth h and depends only on h and ρ. Its value will be in Pa if h is in m and ρ in kg/m³.

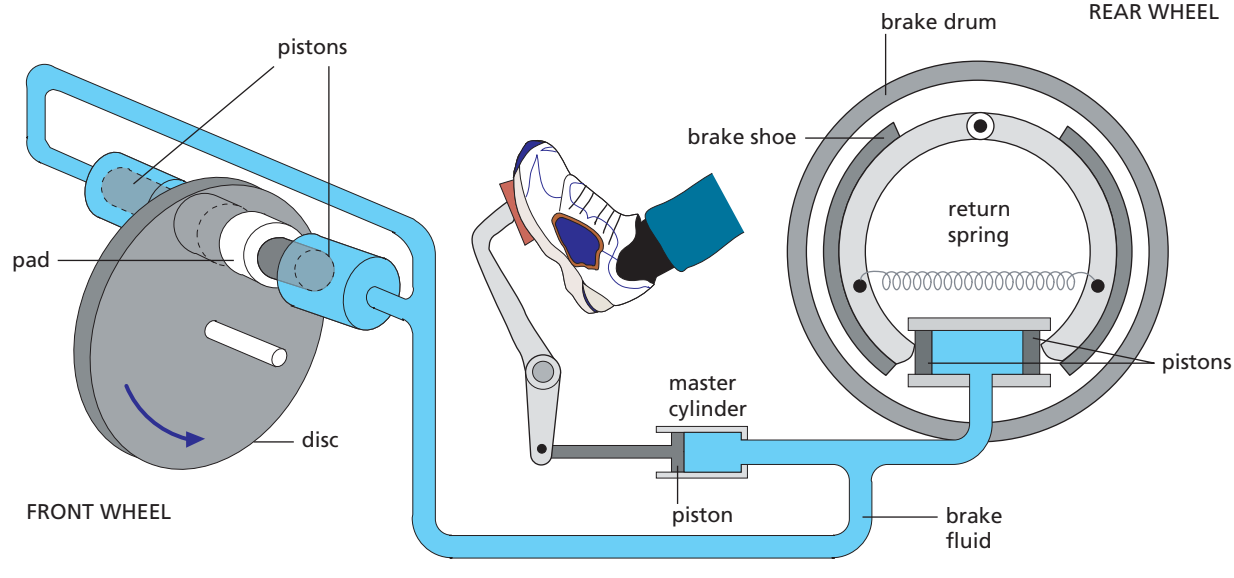


Figure 16.8 Hydraulic car brakes

Pressure gauges

These measure the pressure exerted by a fluid, in other words by a liquid or a gas.

a) Bourdon gauge

This works like the toy in Figure 16.10. The harder you blow into the paper tube, the more it uncurls. In a **Bourdon gauge** (Figure 16.11), when a fluid pressure is applied, the curved metal tube tries to straighten out and rotates a pointer over a scale. Car oil-pressure gauges and the gauges on gas cylinders are of this type.



Figure 16.10 The harder you blow, the greater the pressure and the more it uncurls.

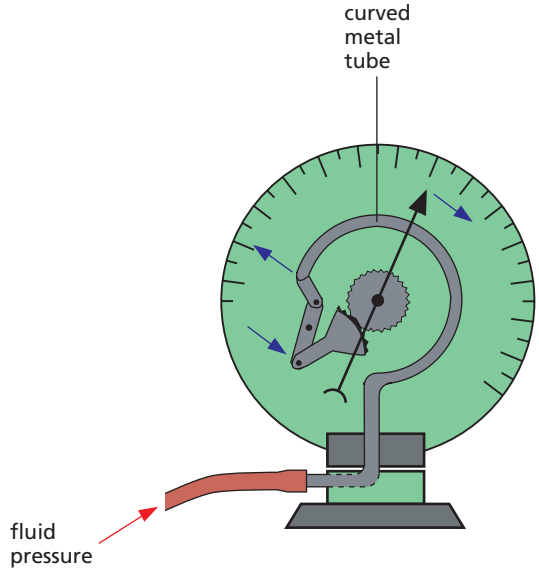


Figure 16.11 A Bourdon gauge

b) U-tube manometer

In Figure 16.12a each surface of the liquid is acted on equally by atmospheric pressure and the levels are the same. If one side is connected to, for example, the gas supply (Figure 16.12b), the gas exerts a pressure on surface A and level B rises until

pressure of gas = atmospheric pressure
+ pressure due to liquid column BC

The pressure of the liquid column BC therefore equals the amount by which the gas pressure *exceeds* atmospheric pressure. It equals $h\rho g$ (in Pa) where h is the vertical height of BC (in m) and ρ is the density of the liquid (in kg/m^3). The height h is called the **head of liquid** and sometimes, instead of stating a pressure in Pa, we say that it is so many cm of water (or mercury for higher pressures).

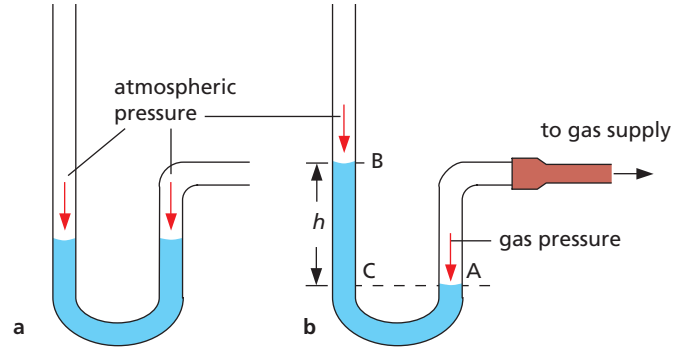


Figure 16.12 A U-tube manometer

c) Mercury barometer

A **barometer** is a manometer which measures atmospheric pressure. A simple barometer is shown in Figure 16.13. The pressure at X due to the weight of the column of mercury XY equals the atmospheric pressure on the surface of the mercury in the bowl. The height XY measures the atmospheric pressure in mm of mercury (mmHg).

The *vertical* height of the column is unchanged if the tube is tilted. Would it be different with a wider tube? The space above the mercury in the tube is a vacuum (except for a little mercury vapour).

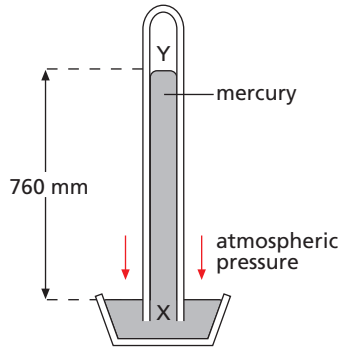


Figure 16.13 Mercury barometer

Questions

1 a What is the pressure on a surface when a force of 50 N acts on an area of
(i) 2.0 m^2,
(ii) 100 m^2,
(iii) 0.50 m^2?
b A pressure of 10 Pa acts on an area of 3.0 m^2. What is the force acting on the area?

2 In a hydraulic press a force of 20 N is applied to a piston of area 0.20 m^2. The area of the other piston is 2.0 m^2. What is
a the pressure transmitted through the liquid,
b the force on the other piston?

3 a Why must a liquid and not a gas be used as the 'fluid' in a hydraulic machine?
b On what other important property of a liquid do hydraulic machines depend?

4 What is the pressure 100 m below the surface of sea water of density 1150 kg/m^3?

5 Figure 16.14 shows a simple barometer.
a What is the region A?
b What keeps the mercury in the tube?
c What is the value of the atmospheric pressure being shown by the barometer?
d What would happen to this reading if the barometer were taken up a high mountain? Give a reason.

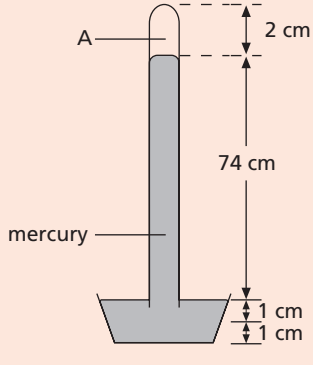


Figure 16.14

6 Which of the following will damage a wood-block floor that can withstand a pressure of 2000 kPa (2000 kN/m^2)?
1 A block weighing 2000 kN standing on an area of 2 m^2.
2 An elephant weighing 200 kN standing on an area of 0.2 m^2.
3 A girl of weight 0.5 kN wearing stiletto-heeled shoes standing on an area of 0.0002 m^2.
Use the answer code:
A 1, 2, 3
B 1, 2
C 2, 3
D 1
E 3

7 The pressure at a point in a liquid
1 increases as the depth increases
2 increases if the density of the liquid increases
3 is greater vertically than horizontally.
Which statement(s) is (are) correct?
A 1, 2, 3
B 1, 2
C 2, 3
D 1
E 3

Checklist

After studying this chapter you should be able to

- relate pressure to force and area and give examples,
- define pressure and recall its unit,
- connect the pressure in a fluid with its depth and density,
- recall that pressure is transmitted through a liquid and use it to explain the hydraulic jack and hydraulic car brakes,
- use pressure = $h\rho g$ to solve problems,
- describe how a U-tube manometer may be used to measure fluid pressure,
- describe and use a simple mercury barometer.

Lesson 10 Conduction and Convection

- Conduction
- Uses of conductors
- Conduction and the kinetic theory
- Convection in liquids
- Convection in air
- Natural convection currents
- Energy losses from buildings
- Ventilation

To keep a building or a house at a comfortable temperature in winter and in summer, if it is to be done economically and efficiently, requires a knowledge of how heat travels.

Conduction

The handle of a metal spoon held in a hot drink soon gets warm. Heat passes along the spoon by **conduction**.

Conduction is the flow of thermal energy (heat) through matter from places of higher temperature to places of lower temperature without movement of the matter as a whole.

A simple demonstration of the different conducting powers of various metals is shown in Figure 23.1. A match is fixed to one end of each rod using a little melted wax. The other ends of the rods are heated by a burner. When the temperatures of the far ends reach the melting point of wax, the matches drop off. The match on copper falls first, showing it is the best **conductor**, followed by aluminium, brass and then iron.

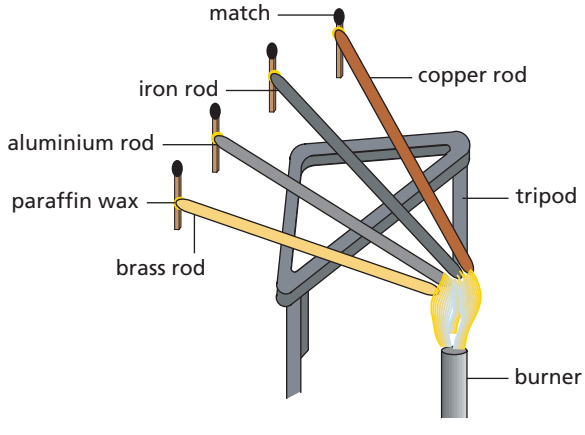


Figure 23.1 Comparing conducting powers

Heat is conducted faster through a rod if it has a large cross-sectional area, is short and has a large temperature difference between its ends.

Most metals are good conductors of heat; materials such as wood, glass, cork, plastics and fabrics are bad conductors. The arrangement in Figure 23.2 can be used to show the difference between brass and wood. If the rod is passed through a flame several times, the paper over the wood scorches but not the paper over the brass. The brass conducts the heat away from the paper quickly, preventing the paper from reaching the temperature at which it burns. The wood conducts the heat away only very slowly.

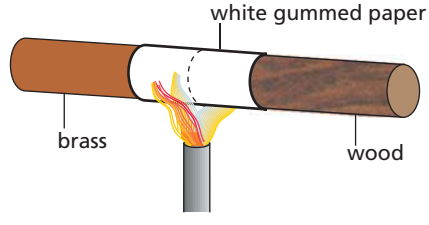


Figure 23.2 The paper over the brass does not burn.

Metal objects below body temperature *feel* colder than those made of bad conductors – even if all the objects are at exactly the same temperature – because they carry heat away faster from the hand.

Liquids and gases also conduct heat but only very slowly. Water is a very poor conductor, as shown in Figure 23.3. The water at the top of the tube can be boiled before the ice at the bottom melts.

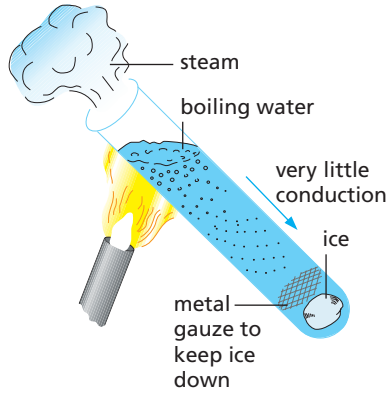


Figure 23.3 Water is a poor conductor of heat.

Uses of conductors

a) Good conductors

These are used whenever heat is required to travel quickly through something. Saucepans, boilers and radiators are made of metals such as aluminium, iron and copper.

b) Bad conductors (insulators)

The handles of some saucepans are made of wood or plastic. Cork is used for table mats.

Air is one of the worst conductors and so one of the best **insulators**. This is why houses with cavity walls (two layers of bricks separated by an air space) and double-glazed windows keep warmer in winter and cooler in summer.

Materials that trap air, such as wool, felt, fur, feathers, polystyrene foam, fibreglass, are also very bad conductors. Some of these materials are used as 'lagging' to insulate water pipes, hot water cylinders, ovens, refrigerators and the walls and roofs of houses (Figures 23.4a and 23.4b). Others are used to make warm winter clothes like 'fleece' jackets (Figure 23.4c).



Figure 23.4a Lagging in a cavity wall provides extra insulation.

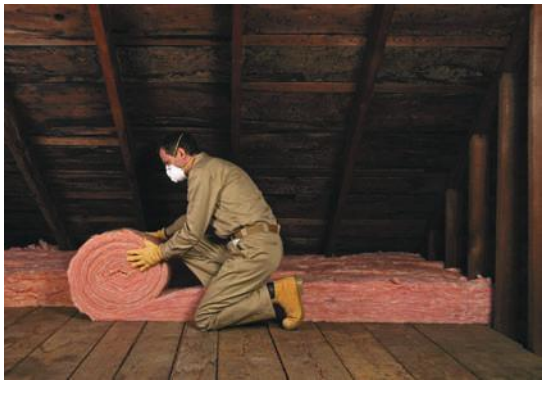


Figure 23.4b Laying lagging in a house loft



Figure 23.4c Fleece jackets help you to retain your body warmth.

'Wet suits' are worn by divers and water skiers to keep them warm. The suit gets wet and a layer of water gathers between the person's body and the suit. The water is warmed by body heat and stays warm because the suit is made of an insulating fabric, such as neoprene, a synthetic rubber.

Conduction and the kinetic theory

Two processes occur in metals. Metals have a large number of 'free' electrons (Chapter 36) which wander about inside them. When one part of a metal is heated, the electrons there move faster (their kinetic energy increases) and further. As a result they 'jostle' atoms in cooler parts, so passing on their energy and raising the temperature of these parts. This process occurs quickly.

The second process is much slower. The atoms themselves at the hot part make 'colder' neighbouring atoms vibrate more vigorously. This is less important in metals but is the only way conduction occurs in non-metals since these do not have 'free' electrons; hence non-metals are poor conductors of heat.

Convection in liquids

Convection is the usual method by which thermal energy (heat) travels through fluids such as liquids and gases. It can be shown in water by dropping a few crystals of potassium permanganate down a tube to the bottom of a beaker or flask of water. When the tube is removed and the beaker heated just below the crystals by a *small* flame (Figure 23.5a), purple streaks of water rise upwards and fan outwards.



Figure 23.5a Convection currents shown by potassium permanganate in water

Streams of warm moving fluids are called **convection currents**. They arise when a fluid is heated because it expands, becomes less dense and is forced upwards by surrounding cooler, denser fluid which moves under it. We say 'hot water (or hot air) rises'. Warm fluid behaves like a cork released under water: being less dense it bobs up. Lava lamps (Figure 23.5b) use this principle.

Convection is the flow of heat through a fluid from places of higher temperature to places of lower temperature by movement of the fluid itself.



Figure 23.5b Lava lamps make use of convection.

Convection in air

Black marks often appear on the wall or ceiling above a lamp or a radiator. They are caused by dust being carried upwards in air convection currents produced by the hot lamp or radiator.

A laboratory demonstration of convection currents in air can be given using the apparatus of Figure 23.6. The direction of the convection current created by the candle is made visible by the smoke from the touch paper (made by soaking brown paper in strong potassium nitrate solution and drying it).

Convection currents set up by electric, gas and oil heaters help to warm our homes. Many so-called 'radiators' are really convector heaters.

Where should the input and extraction ducts for cold/hot air be located in a room?

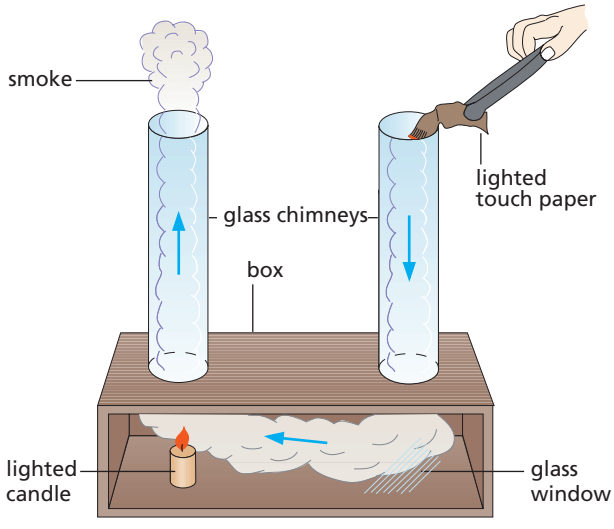


Figure 23.6 Demonstrating convection in air

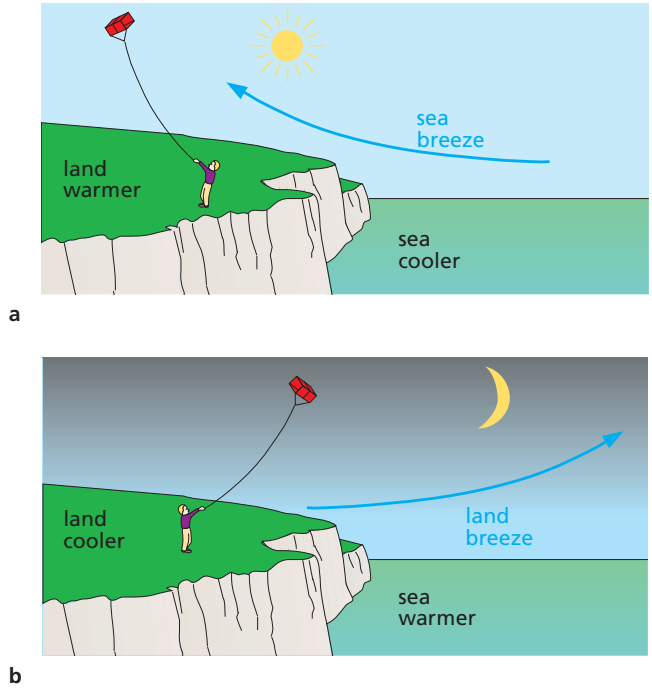


Figure 23.7 Coastal breezes are due to convection: **a** day; **b** night

Natural convection currents

a) Coastal breezes

During the day the temperature of the land increases more quickly than that of the sea (because the specific heat capacity of the land is much smaller; see Chapter 21). The hot air above the land rises and is replaced by colder air from the sea. A breeze from the sea results (Figure 23.7a).

At night the opposite happens. The sea has more heat to lose and cools more slowly. The air above the sea is warmer than that over the land and a breeze blows from the land (Figure 23.7b).

b) Gliding

Gliders, including 'hang-gliders' (Figure 23.8), depend on hot air currents, called **thermals**.



Figure 23.8 Once airborne, a hang-glider pilot can stay aloft for several hours by flying from one thermal to another.

Energy losses from buildings

The inside of a building can only be kept at a steady temperature above that outside by heating it at a rate which equals the rate at which it is losing energy. The loss occurs mainly by conduction through the walls, roof, floors and windows. For a typical house in the UK where no special precautions have been taken, the contribution each of these makes to the total loss is shown in Table 23.1a.

As fuels (and electricity) become more expensive and the burning of fuels becomes of greater environmental concern (Chapter 15), more people are considering it worthwhile to reduce heat losses from their homes. The substantial reduction of this loss which can be achieved, especially by wall and roof insulation, is shown in Table 23.1b.

Table 23.1 Energy losses from a typical house

a

Percentage of total energy loss due to				
walls	**roof**	**floors**	**windows**	**draughts**
35	25	15	10	15

b

Percentage of each loss saved by				
insulating walls	**insulating roof**	**carpets on floors**	**double glazing**	**draught excluders**
65	80	≈30	50	≈60
Percentage of total loss saved = 60				

Ventilation

In addition to supplying heat to compensate for the energy losses from a building, a heating system has also to warm the ventilated cold air, needed for comfort, which comes in to replace stale air.

If the rate of heat loss is, say, 6000 J/s, or 6 kW, and the warming of ventilated air requires 2 kW, then the total power needed to maintain a certain temperature (e.g. 20 °C) in the building is 8 kW. Some of this is supplied by each person's 'body heat', estimated to be roughly equal to a 100 W heater.

d A vacuum is an even better heat insulator than air. Suggest one (scientific) reason why the double glazing should not have a vacuum between the sheets of glass.

e The manufacturers of roof lagging suggest that two layers of fibreglass are more effective than one. Describe how you might set up an experiment in the laboratory to test whether this is true.

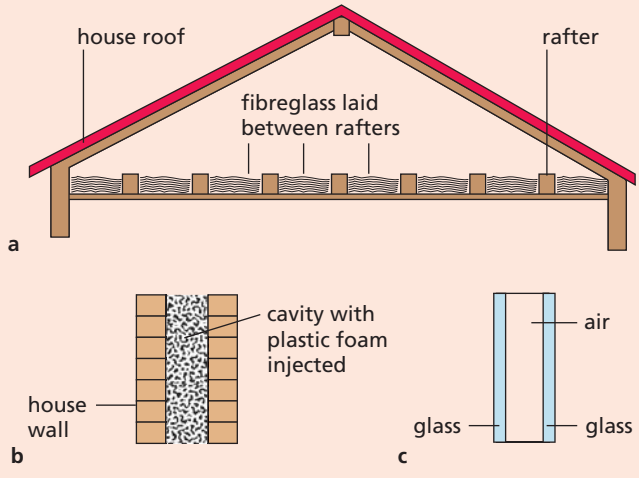


Figure 23.9 a Roof insulation; **b** cavity wall insulation; **c** double glazing

3 What is the advantage of placing an electric immersion heater
- **a** near the top,
- **b** near the bottom,

of a tank of water?

4 Explain why on a cold day the metal handlebars of a bicycle feel colder than the rubber grips.

Questions

1 Explain why
- **a** newspaper wrapping keeps hot things hot, e.g. fish and chips, and cold things cold, e.g. ice cream,
- **b** fur coats would keep their owners warmer if they were worn inside out,
- **c** a string vest helps to keep a person warm even though it is a collection of holes bounded by string.

2 Figure 23.9 illustrates three ways of reducing heat losses from a house.
- **a** As far as you can, explain how each of the three methods reduces heat losses. Draw diagrams where they will help your explanations.
- **b** Why are fibreglass and plastic foam good substances to use?
- **c** Air is one of the worst conductors of heat. What is the point of replacing it by plastic foam as shown in the Figure 23.9b?

Checklist

After studying this chapter you should be able to

- describe experiments to show the different conducting powers of various substances,
- name good and bad conductors and state uses for each,
- explain conduction using the kinetic theory,
- describe experiments to show convection in fluids (liquids and gases),
- relate convection to phenomena such as land and sea breezes,
- explain the importance of insulating a building.

Lesson 11 Radiation

- Good and bad absorbers
- Good and bad emitters
- Vacuum flask
- The greenhouse
- Rate of cooling of an object

Radiation is a third way in which heat can travel, but whereas conduction and convection both need matter to be present, radiation can occur in a vacuum; particles of matter are not involved. Radiation is the way heat reaches us from the Sun.

Radiation has all the properties of electromagnetic waves, such as it travels at the speed of radio waves and gives interference effects. When it falls on an object, it is partly reflected, partly transmitted and partly absorbed; the absorbed part raises the temperature of the object.

Radiation is the flow of heat from one place to another by means of electromagnetic waves.

Radiation is emitted by all bodies above absolute zero and consists mostly of **infrared radiation** but light and **ultraviolet** are also present if the body is very hot (e.g. the Sun).



Figure 1 Why are buildings in hot countries often painted white?

Good and bad absorbers

Some surfaces absorb radiation better than others, as may be shown using the apparatus in Figure 2. The inside surface of one lid is shiny and of the other dull black. The coins are stuck on the outside of each lid with candle wax. If the heater is midway between the lids they each receive the same amount of radiation. After a few minutes the wax on the black lid melts and the coin falls off. The shiny lid stays cool and the wax unmelted.

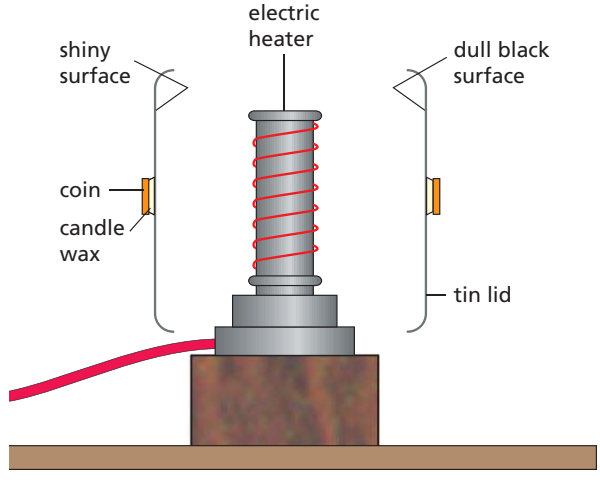


Figure 2 Comparing absorbers of radiation

Dull black surfaces are better absorbers of radiation than white shiny surfaces – the latter are good **reflectors** of radiation. Reflectorsonelectric fir es are made of polished metal because of its good reflectingp roperties.

Good and bad emitters

Some surfaces also emit radiation better than others when they are hot. If you hold the backs of your hands on either side of a hot copper sheet that has one side polished and the other side blackened (Figure 3), it will be found that the **dull black surface is a better emitter of radiation than the shiny one**.

The cooling fi ns on the heat exchangers at the back of a refrigerator are painted black so that they lose heat more quickly. By contrast, saucepans that are polished are poor emitters and keep their heat longer.

In general, surfaces that are good absorbers of radiation are good emitters when hot.

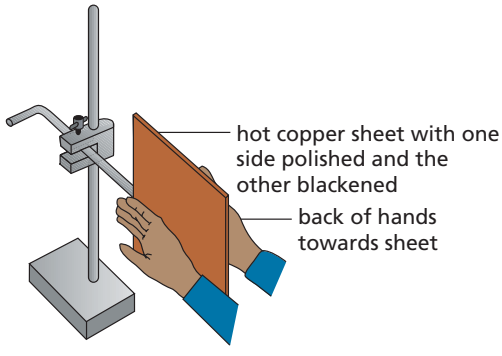


Figure 3 Comparing emitters of radiation

Vacuum flask

A vacuum or Thermos flask keeps hot liquids hot or cold liquids cold. It is very difficult for heat to travel into or out of the flask.

Transfer by conduction and convection is minimised by making the flask a double-walled glass vessel with a vacuum between the walls (Figure 4). Radiation is reduced by silvering both walls on the vacuum side. Then if, for example, a hot liquid is stored, the small amount of radiation from the hot inside wall is reflected back across the vacuum by the silvering on the outer wall. The slight heat loss that does occur is by conduction up the thin glass walls and through the stopper.

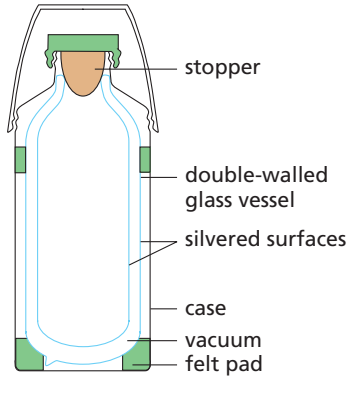


Figure 4 A vacuum flask

The greenhouse

The warmth from the Sun is not cut off by a sheet of glass but the warmth from a red-hot fire can be blocked by glass. The radiation from very hot bodies like the Sun is mostly in the form of light and short-wavelength infrared. The radiation from less hot objects, like a fire, is largely long-wavelength infrared which, unlike light and short-wavelength infrared, cannot pass through glass.

Light and short-wavelength infrared from the Sun penetrate the glass of a greenhouse and are absorbed by the soil, plants, etc., raising their temperature. These in turn emit infrared but, because of their relatively low temperature, this has a long wavelength and is not transmitted by the glass. The greenhouse thus acts as a 'heat-trap' and its temperature rises.

Carbon dioxide and other gases such as methane in the Earth's atmosphere act in a similar way to the glass of a greenhouse in trapping heat; this has serious implications for the global climate.

Rate of cooling of an object

The rate at which an object cools, i.e. at which its temperature falls, can be shown to be proportional to the ratio of its surface area A to its volume V.

For a cube of side l

$$\frac{A_1}{V_1} = 6 \times \frac{l^2}{l^3} = \frac{6}{l}$$

For a cube of side $2l$

$$\frac{A_2}{V_2} = 6 \times \frac{4l^2}{8l^3} = \frac{3}{l}$$

$$= \frac{1}{2} \times \frac{6}{l} = \frac{1}{2}\frac{A_1}{V_1}$$

The larger cube has the smaller A/V ratio and so cools more slowly.

▶▶

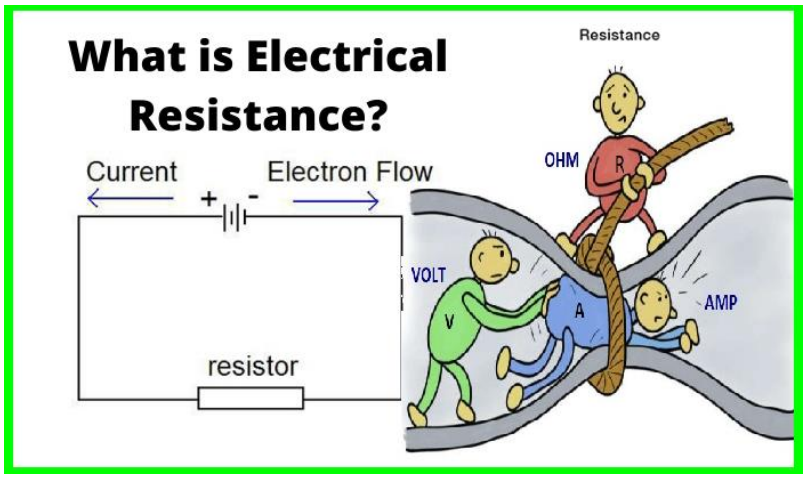
Checklist

After studying this chapter you should be able to

- describe the effect of surface colour and texture on the emission, absorption and reflection of radiation,
- describe experiments to study factors affecting the absorption and emission of radiation,
- recall that good absorbers are also good emitters,
- explain how a knowledge of heat transfer affects the design of a vacuum flask,
- explain how a greenhouse acts as a 'heat-trap'.

Lesson 12

DC Circuits



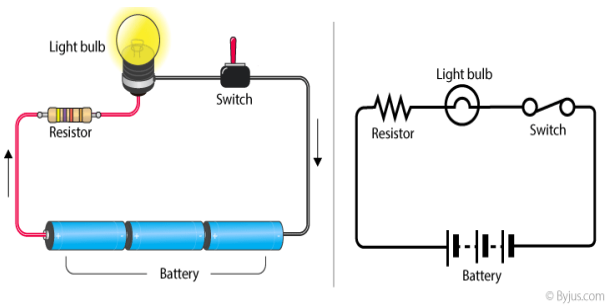
At the end of this chapter you should be able to:

- draw circuit diagrams with power sources (cell or battery), switches, lamps, resistors (fixed and variable), fuses, ammeters and voltmeters:
- recall and apply the formulae for the effective resistance of a number of resistors in series and in parallel to new situations or to solve related problems,
- state that current at every point in a series circuit is the same and apply the principle to new situations or to solve related problems, state that the sum of potential differences in a series circuit is equal to the potential differences across the whole circuit.
- state that the current from the source is the sum of the currents in the separate branches of a parallel circuit
- state that the potential difference across the separate branches of a parallel circuit is the same, and
- recall and apply the relevant relationships, including R=V/I and those for current, potential differences and resistors in series and in parallel circuits, in calculations involving a whole circuit.

1 Putting Them Together

An electric circuit basically consists of

- a power source such as a battery or a cell,
- connecting wires,
- a switch, and
- an electrical device, such as a lamp.



Closed Circuit

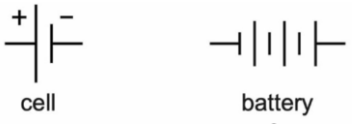
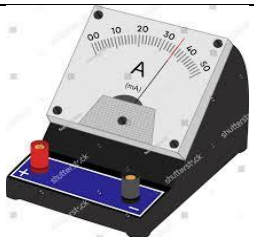
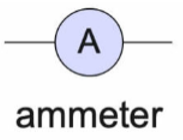
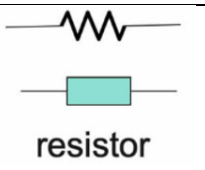
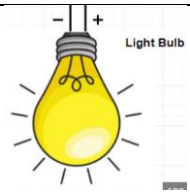
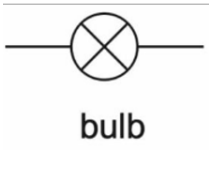
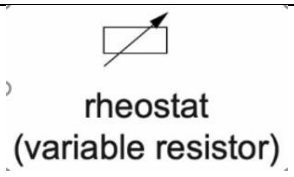
In a closed circuit, charges flow continuously. An electrical device, like a lamp, converts electrical energy to light energy.

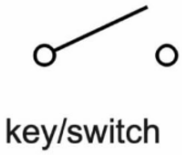
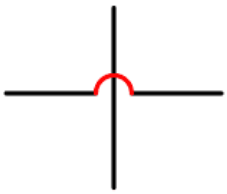
Open Circuit

When there is a break in one part of the circuit, charges stop flowing.

2.Representing the Circuit

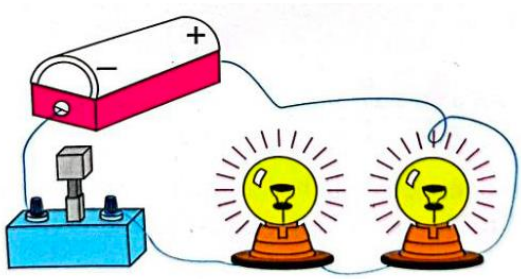
To represent electrical components when we draw an electric circuit, we use circuit symbols, as shown below.

Electrical Components	Symbol	Name
		Cell or batteries Used to supply the electrical energy
		Ammeter Used to measure the current
		Used to measure the voltage
		Resistor Resists the current
		Light Bulb
		Change the value of resistor

		Open/ close the circuit
		Wire Connect the circuit

2 Series Circuit

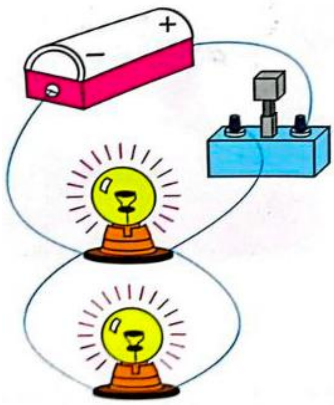
In a television series, you get several episodes, one after the other. A series circuit is similar. You get several components one after the other.
If you follow the circuit diagram from one side of the cell to the other, you should pass through all the different components, one after the other, without any branches.



A circuit with a battery and two lamps connected in series

3 Parallel Circuit

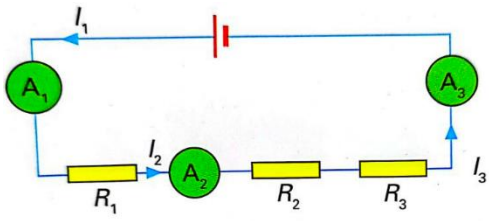
In parallel circuits different components are connected on different branches of the wire. If you follow the circuit diagram from one side of the cell to the other, you can only pass through all the different components if you follow all the branches.



A circuit with a battery and two lamps connected in parallel.

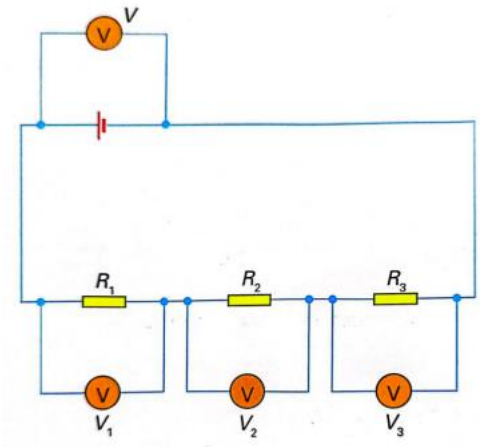
4 Resistors in Series

Resistors in Series



When resistors are connected in series, they result in greater resistance. The effective resistance is the same as joining several short lengths of resistance wires together to form a longer length. For such an arrangement, the same current flows through each resistor.

Ammeter readings show that: $\mathbf{I_1 = I_2 = I_3}$



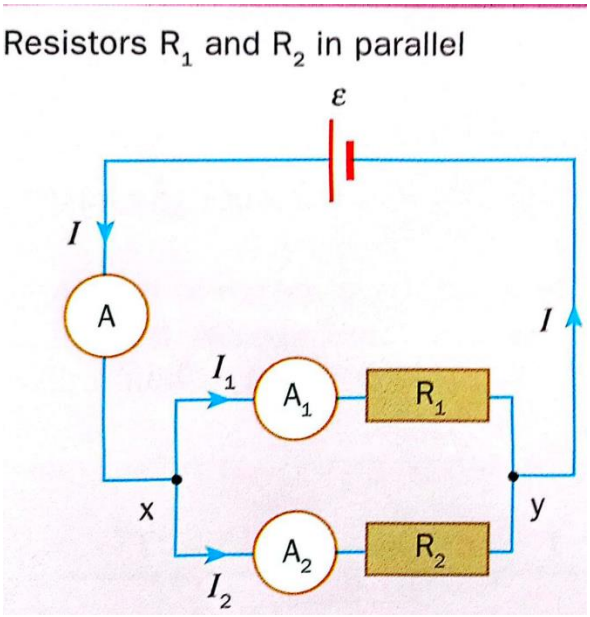
In general, if a number of resistors, for example, R,, R, and R, have an effective resistance R when connected in series, then

$$\mathbf{R} = \boldsymbol{R_1} + \boldsymbol{R_2} + \boldsymbol{R_3}$$

The total potential difference, V across the three resistors $\boldsymbol{R_1}, \boldsymbol{R_2}, \boldsymbol{R_3}$ is the sum of the potential difference across each resistor.

$$\mathbf{V} = \mathbf{V_1} + \mathbf{V_2} + \mathbf{V_3}$$

5 Resistors in Parallel



When resistors are connected in parallel, the effective resistance is less than the resistance of each resistor.

The reciprocal of the effective resistance of resistors in parallel is the sum of the reciprocal of the individual resistors.

In general, a number of resistors, for example, R, and R, connected in parallel. have an effective resistance of R. given by

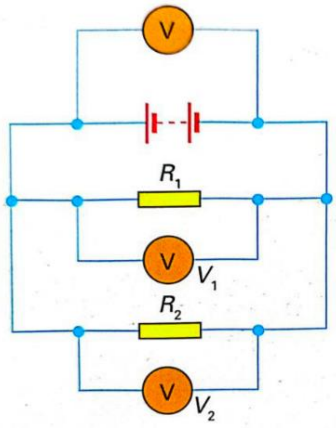
$$\frac{\mathbf{1}}{\mathbf{R}} = \frac{\mathbf{1}}{\mathbf{R_1}} + \frac{\mathbf{1}}{\mathbf{R_2}}$$

The ammeter reading show that The total current is the sum of each current

$$\mathbf{I} = \mathbf{I_1} + \mathbf{I_2}$$

The potential different across each resistor in parallel is the same.

Voltmeter reading show that $\mathbf{V} = \mathbf{V_1} = \mathbf{V_2}$



6 Ohm's Law

In 1826, German physicist Georg Ohm discovered that when physical conditions (such as temperature) are constant, the electric current in a metallic conductor is directly proportional to the potential difference across it. This relationship is known as Ohm's Law.

Ohm's Law states that the current passing through a metallic conductor is directly proportional to the potential difference across it, provided that physical conditions (such as temperature) remain constant.

In symbols, $\mathbf{R} = \frac{V}{I} = constant$

Where, I is the current (in A)

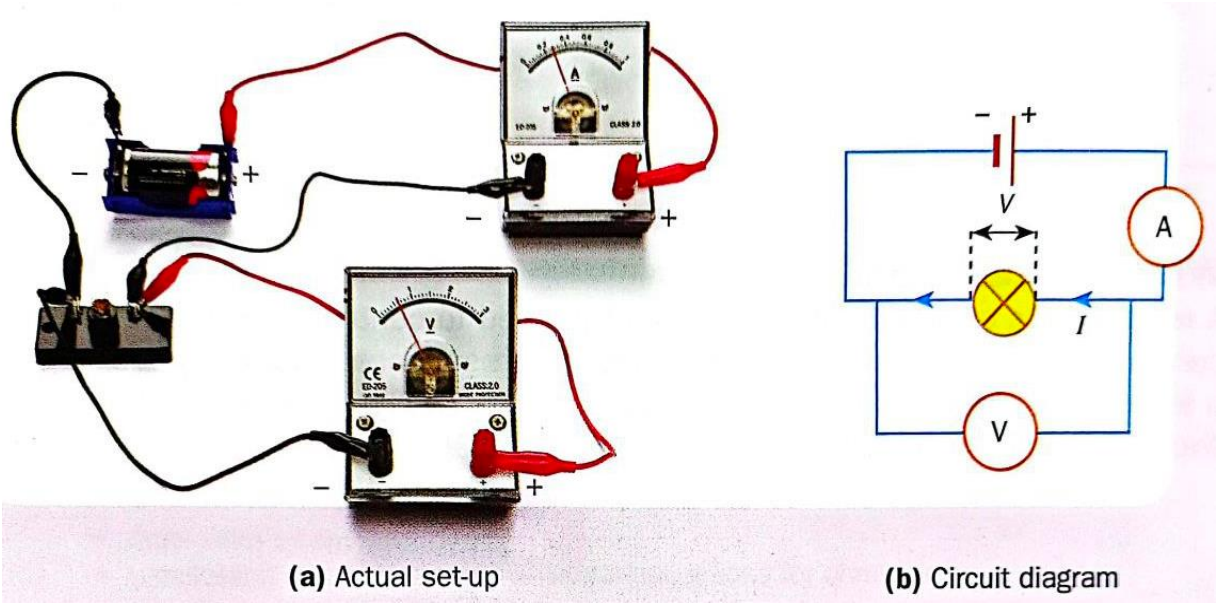
V is the potential difference (in V)

R is the resistance (in Ω)

Thus, according to Ohm's Law, the resistance of metallic conductors remains constant under steady physical conditions.

How do we measure resistance?

Circuit components other than resistors also have resistance. We can measure their resistances by measuring the current flowing through them and the p.d. across them. Figure below shows a current flowing through a lamp and a potential difference V across it. With these two quantities, we can apply the formula R =V/I to calculate the resistance of the lamp.



(a) Actual set-up **(b)** Circuit diagram

SUMMARY

1. A circuit is basically made up of a power source, (cell or battery), a switch and an electrical device that converts electrical energy to other forms of energy.

2. Circuit components can be arranged in series or in parallel.

3. Lamps arranged in series are dependent on one another. If one of them blows, the others will not be able to light up in the same circuit.

4 Light bulbs arranged in parallel are independent of one another. If one of them blows, the others will continue to light up normally.

5 Ammeters must be connected to a circuit in series.

6.Voltmeters must be connected to a circuit in parallel.

7. For a series circuit current at every point in a series circuit is the same, sum of the potential differences in a series circuit is equal to the potential differences across the whole circuit,

$$\mathbf{I_1 = I_2 = I_3}$$

$$\mathbf{V = V_1 + V_2 + V_3}$$

sum of the resistance in a series circuit is equal to the effective resistance for the whole circuit.

$$\mathbf{R = R_1 + R_2 + R_3}$$

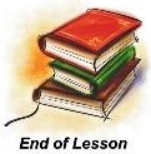
8.For a parallel circuit the current from the source is the sum of the currents in the separate branches of a parallel circuit. effective resistance R, of two resistors R, and R, connected in parallel is given by

$$\mathbf{I = I_1 + I_2}$$

$$\mathbf{\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}}$$

The potential different across each resistor in parallel is the same.

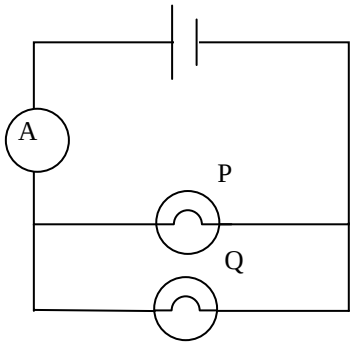
$$\mathbf{V = V_1 = V_2}$$



Practice

QUESTIONSHEET 1

(a) In the circuit below, lamps P and Q are identical. The reading on the ammeter is 3A. The cell shown is of emf. 6V.



Calculate the current that passes through lamp P.

.....

.....

.....

(b) If a voltmeter was connected across lamp P, what would it read ?

.....

(c) Calculate the resistance of lamp P.

.....

.....

.....

QUESTIONSHEET 2

(a) Resistors of 5 ohms and 10 ohms are connected in series with a battery supplying 3 volts.
What is the total resistance ?

.....

.....

(b) Calculate the current which flows in the circuit.

.....

.....

.....

QUESTIONSHEET 3

(a) Resistors of 2 ohms and 4 ohms are connected in parallel across a battery supplying 6 volts. Another resistor of 5 ohms is connected in series with the battery. Draw the circuit.

(b) Calculate the total resistance in the circuit.

.....

.....

.....

.....

.....

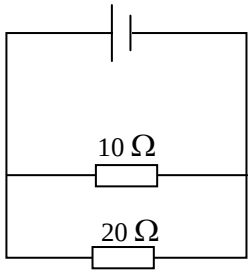
(c) Calculate the current flowing through the 5 ohm resistor.

.....

.....

QUESTIONSHEET 5

In the circuit below, 0.6 A flows through the 10 ohm resistor.



(a) Calculate the current flowing through the 20 ohm resistor.

(b) Calculate the current flowing through the battery.

.....

..... [2]

(c) Calculate the voltage supplied by the battery.

Lesson 13: Electric Power

- Power in electric circuits
- Electric lighting
- Electric heating
- Joulemeter
- House circuits
- Paying for electricity
- Dangers of electricity
- Practical work: Measuring electric power.

Power in electric circuits

In many circuits it is important to know the rate at which electrical energy is transferred into other forms of energy. Earlier (Chapter 13) we said that **energy transfers were measured by the work done** and power was defined by the equation

$$\text{power} = \frac{\text{work done}}{\text{time taken}} = \frac{\text{energy transfer}}{\text{time taken}}$$

In symbols

$$P = \frac{E}{t} \qquad (1)$$

where if E is in joules (J) and t in seconds (s) then P is in J/s or watts (W).

From the definition of p.d. (Chapter 37) we saw that if E is the electrical energy transferred when there is a steady current I (in amperes) for time t (in seconds) in a device (e.g. a lamp) with a p.d. V (in volts) across it, as in Figure 40.1, then

$$E = ItV \qquad (2)$$

Substituting for E in (1) we get

$$P = \frac{E}{t} = \frac{ItV}{t}$$

or

$$P = IV$$

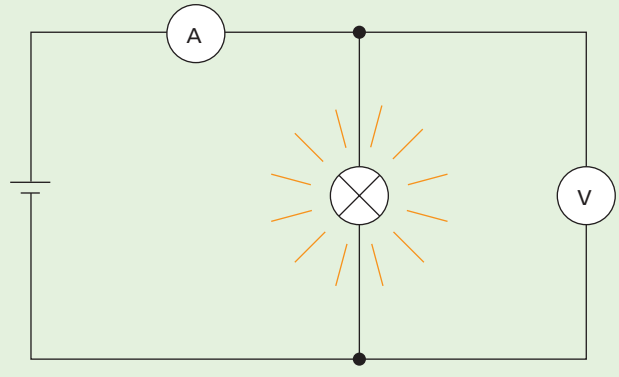


Figure 40.1

Therefore to calculate the power P of an electrical appliance we multiply the current I in it by the p.d. V across it. For example if a lamp on a 240 V supply has a current of 0.25 A in it, its power is $240\,\text{V} \times 0.25\,\text{A} = 60\,\text{W}$. The lamp is transferring 60 J of electrical energy into heat and light each second. Larger units of power are the **kilowatt** (kW) and the **megawatt** (MW) where

$$1\,\text{kW} = 1000\,\text{W} \quad \text{and} \quad 1\,\text{MW} = 1\,000\,000\,\text{W}$$

In units

$$\text{watts} = \text{amperes} \times \text{volts} \qquad (3)$$

It follows from (3) that since

$$\text{volts} = \frac{\text{watts}}{\text{amperes}} \qquad (4)$$

the volt can be defined as a **watt per ampere** and p.d. calculated from (4).

If all the energy is transferred to heat in a resistor of resistance R, then $V = IR$ and the rate of production of heat is given by

$$P = V \times I = IR \times I = I^2R$$

That is, if the current is doubled, four times as much heat is produced per second. Also, $P = V^2/R$.

Practical work

Measuring electric power

a) Lamp

Connect the circuit of Figure 40.2. Note the ammeter and voltmeter readings and work out the electric power supplied to the lamp in watts.

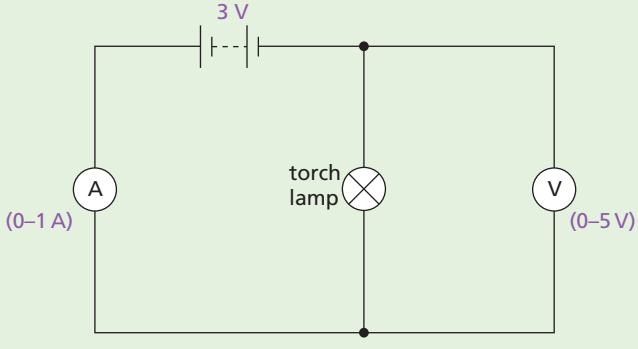


Figure 40.2

b) Motor

Replace the lamp in Figure 40.2 by a small electric motor. Attach a known mass m (in kg) to the axle of the motor with a length of thin string and find the time t (in s) required to raise the mass through a known height h (in m) at a steady speed. Then the power output P_o (in W) of the motor is given by

$$P_o = \frac{\text{work done in raising mass}}{\text{time taken}} = \frac{mgh}{t}$$

If the ammeter and voltmeter readings I and V are noted while the mass is being raised, the power input P_i (in W) can be found from

$$P_i = IV$$

The efficiency of the motor is given by

$$\text{efficiency} = \frac{P_o}{P_i} \times 100\%$$

Also investigate the effect of a greater mass on: (i) the speed, (ii) the power output and (iii) the efficiency of the motor at its rated p.d.

Electric lighting

a) Filament lamps

The **filament** is a small coil of tungsten wire (Figure 40.3) which becomes white hot when there is a current in it. The higher the temperature of the filament, the greater is the proportion of electrical energy transferred to light and for this reason it is made of tungsten, a metal with a high melting point (3400 °C).

Most lamps are gas-filled and contain nitrogen and argon, not air. This reduces evaporation of the tungsten which would otherwise condense on the bulb and blacken it. The coil is coiled compactly so that it is cooled less by convection currents in the gas.

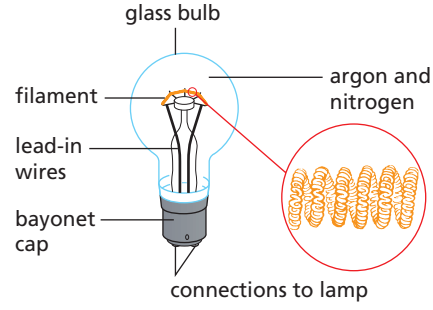


Figure 40.3 A filament lamp

b) Fluorescent strips

A filament lamp transfers only 10% of the electrical energy supplied to light; the other 90% becomes heat. Fluorescent strip lamps (Figure 40.4a) are five times as efficient and may last 3000 hours compared with the 1000-hour life of filament lamps. They cost more to install but running costs are less.

When a fluorescent strip lamp is switched on, the mercury vapour emits invisible ultraviolet radiation which makes the powder on the inside of the tube fluoresce (glow), i.e. visible light is emitted. Different powders give different colours.

c) Compact fluorescent lamps

These energy-saving fluorescent lamps (Figure 40.4b) are available to fit straight into normal light sockets, either bayonet or screw-in. They last up to eight times longer (typically 8000 hours) and use about five times less energy than filament lamps for the same light output. For example, a 20 W compact fluorescent is equivalent to a 100 W filament lamp.

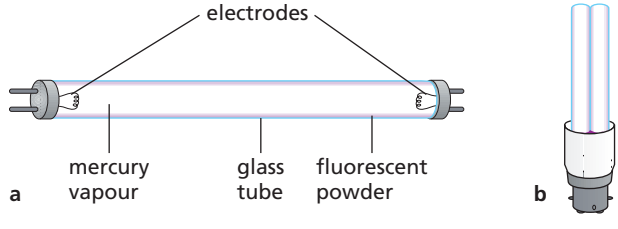


Figure 40.4 Fluorescent lamps

Electric heating

a) Heating elements

In domestic appliances such as electric fires, cookers, kettles and irons the 'elements' (Figure 40.5) are made from Nichrome wire. This is an alloy of nickel and chromium which does not oxidise (and so become brittle) when the current makes it red hot.

The elements in **radiant** electric fires are at red heat (about 900 °C) and the radiation they emit is directed into the room by polished reflectors. In **convector** types the element is below red heat (about 450 °C) and is designed to warm air which is drawn through the heater by natural or forced convection. In **storage** heaters the elements heat fire-clay bricks during the night using 'off-peak' electricity. On the following day these cool down, giving off the stored heat to warm the room.

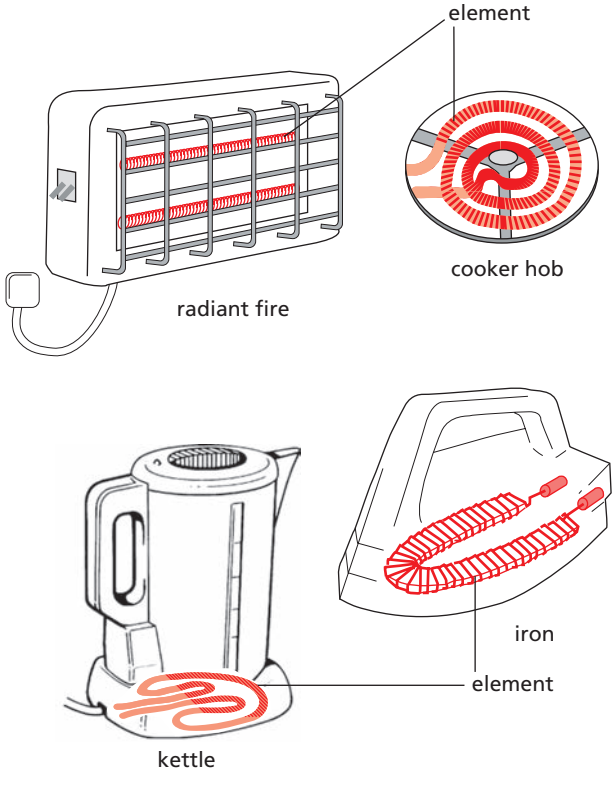


Figure 40.5 Heating elements

b) Three-heat switch

This is sometimes used to control heating appliances. It has three settings and uses two identical elements. On 'high', the elements are in parallel across the supply voltage (Figure 40.6a); on 'medium', there is only current in one (Figure 40.6b); on 'low', they are in series (Figure 40.6c).

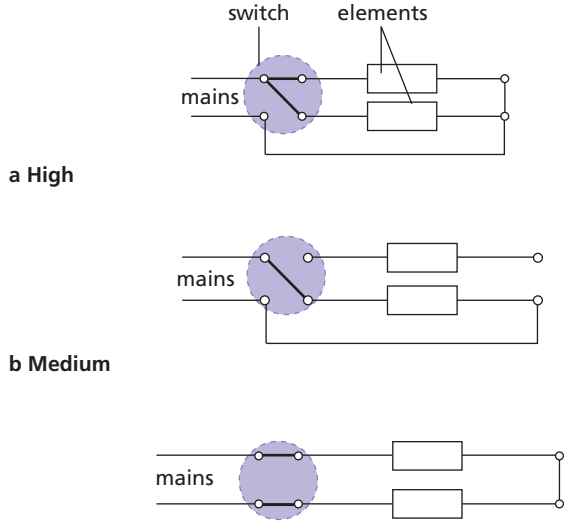


Figure 40.6 Three-heat switch

c) Fuses

A **fuse** protects a circuit. It is a short length of wire of material with a low melting point, often 'tinned copper', which melts and breaks the circuit when the current in it exceeds a certain value. Two reasons for excessive currents are 'short circuits' due to worn insulation on connecting wires and overloaded circuits. Without a fuse the wiring would become hot in these cases and could cause a fire. **A fuse should ensure that the current-carrying capacity of the wiring is not exceeded.** In general the thicker a cable is, the more current it can carry, but each size has a limit.

Two types of fuse are shown in Figure 40.7a. **Always switch off before replacing a fuse**, and always replace with one of the same value as recommended by the manufacturer of the appliance.

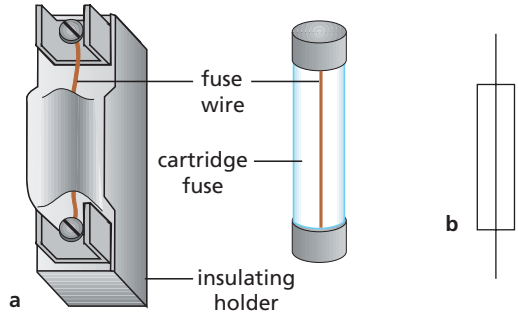


Figure 40.7a Two types of fuse; **b** the circuit symbol for a fuse

Joulemeter

Instead of using an ammeter and a voltmeter to measure the electrical energy transferred by an appliance, a **joulemeter** can be used to measure it directly in joules. The circuit connections are shown in Figure 40.8. A household electricity meter (Figure 40.12) is a joulemeter.

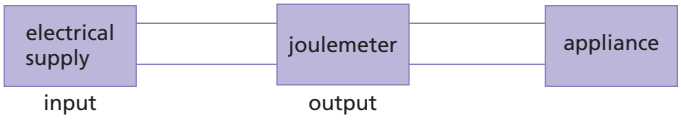


Figure 40.8 Connections to a joulemeter

House circuits

Electricity usually comes to our homes by an underground cable containing two wires, the **live** (L) and the **neutral** (N). The neutral is earthed at the local sub-station and so there is no p.d. between it and earth. The supply is a.c. (Chapter 36) and the live wire is alternately positive and negative. Study the typical house circuits shown in Figure 40.9.

a) Circuits in parallel

Every circuit is connected in parallel with the supply, i.e. across the live and neutral, and receives the full mains p.d. of 230 V (in the UK). The advantages of having appliances connected in parallel, rather than in series, can be seen by studying the lighting circuit in Figure 40.9.

(i) The p.d. across each lamp is fixed (at the mains p.d.), so the lamp shines with the same brightness irrespective of how many other lamps are switched on.

(ii) Each lamp can be turned on and off independently; if one lamp fails, the others can still be operated.

b) Switches and fuses

These are always in the live wire. If they were in the neutral, light switches and power sockets would be 'live' when switches were 'off' or fuses 'blown'. A fatal shock could then be obtained by, for example, touching the element of an electric fire when it was switched off.

c) Staircase circuit

The light is controlled from two places by the two two-way switches.

d) Ring main circuit

The live and neutral wires each run in two complete rings round the house and the power sockets, each rated at 13 A, are tapped off from them. Thinner wires can be used since the current to each socket flows by two paths, i.e. from both directions in the ring. The ring has a 30 A fuse and if it has, say, ten sockets, then all can be used so long as the total current does not exceed 30 A, otherwise the wires overheat. A house may have several ring circuits, each serving a different area.

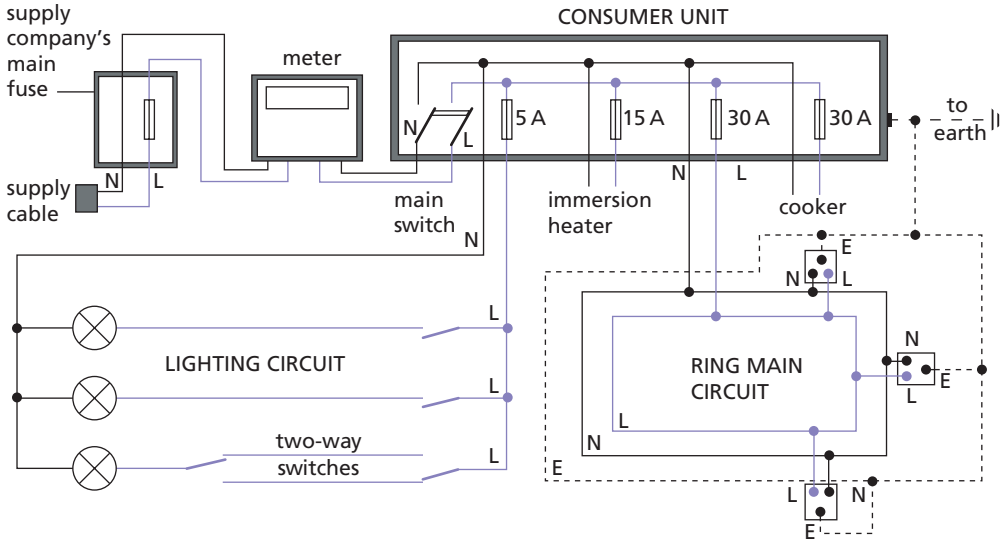


Figure 40.9 Electric circuits in a house

e) Fused plug

Only one type of plug is used in a UK ring main circuit. It is wired as in Figure 40.10a. Note the colours of the wire coverings: L – brown, N – blue, E – green and yellow. It has its own cartridge fuse, 3A (red) for appliances with powers up to 720W, or 13A (brown) for those between 720W and 3kW.

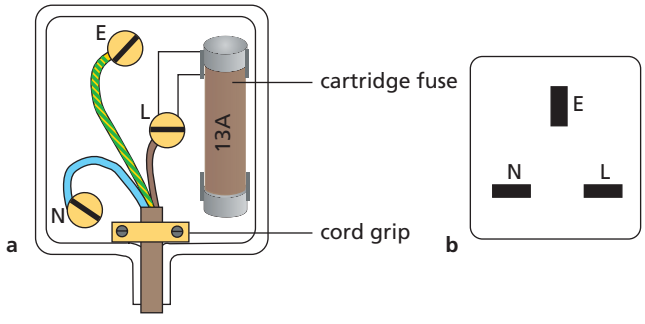


Figure 40.10 a Wiring of a plug; **b** socket

Typical power ratings for various appliances are shown in Table 40.1, p. 182. Calculation of a current in a device allows the correct size of fuse to be chosen.

In some countries the fuse is placed in the appliance rather than in the plug.

f) Safety in electrical circuits

Earthing

A ring main has a third wire which goes to the top sockets on all power points (Figure 40.9) and is earthed by being connected either to a **metal** water pipe entering the house or to an earth connection on the supply cable. This third wire is a safety precaution to prevent electric shock should an appliance develop a fault.

The earth pin on a three-pin plug is connected to the metal case of the appliance which is thus joined to earth by a path of almost zero resistance. If then, for example, the element of an electric fire breaks or sags and touches the case, a large current flows to earth and 'blows' the fuse. Otherwise the case would become 'live' and anyone touching it would receive a shock which might be fatal, especially if they were 'earthed' by, say, standing in a damp environment, such as on a wet concrete floor.

Circuit breakers



Figure 40.11 Circuit breakers

Circuit breakers (Figure 40.11) are now used instead of fuses in consumer units. They contain an electromagnet (Chapter 45) which, when the current exceeds the rated value of the circuit breaker, becomes strong enough to separate a pair of contacts and breaks the circuit. They operate much faster than fuses and have the advantage that they can be reset by pressing a button.

The **residual current circuit breaker** (RCCB), also called a **residual current device** (RCD), is an adapted circuit breaker which is used when the resistance of the earth path between the consumer and the substation is not small enough for a fault-current to blow the fuse or trip the circuit breaker. It works by detecting any difference between the currents in the live and neutral wires; when these become unequal due to an earth fault (i.e. some of the current returns to the substation via the case of the appliance and earth) it breaks the circuit before there is any danger. They have high sensitivity and a quick response.

An RCD should be plugged into a socket supplying power to a portable appliance such as an electric lawnmower or hedge trimmer. In these cases the risk of electrocution is greater because the user is generally making a good earth connection through the feet.

Double insulation

Appliances such as vacuum cleaners, hairdryers and food mixers are usually **double insulated**.

Connection to the supply is by a two-core insulated cable, with no earth wire, and the appliance is enclosed in an insulating plastic case. Any metal attachments that the user might touch are fitted into this case so that they do not make a direct connection with the internal electrical parts, such as a motor. There is then no risk of a shock should a fault develop.

Paying for electricity

Electricity supply companies charge for the **electrical energy** they supply. A joule is a very small amount of energy and a larger unit, the **kilowatt-hour** (kWh), is used.

A kilowatt-hour is the electrical energy used by a 1 kW appliance in 1 hour.

$$1\,\text{kWh} = 1000\,\text{J/s} \times 3600\,\text{s}$$
$$= 3\,600\,000\,\text{J} = 3.6\,\text{MJ}$$

A 3 kW electric fire working for 2 hours uses 6 kWh of electrical energy – usually called 6 'units'. Electricity meters, which are joulemeters, are marked in kWh: the latest have digital readouts like the one in Figure 40.12. At present a 'unit' costs about 8p in the UK.

Typical powers of some appliances are given in Table 40.1.

Table 40.1 Power of some appliances

DVD player	20 W	iron	1 kW
laptop computer	50 W	fire	1, 2, 3 kW
light bulbs	60, 100 W	kettle	2 kW
television	100 W	immersion heater	3 kW
fridge	150 W	cooker	6.4 kW

Note that the current required by a 6.4 kW cooker is given by

$$I = \frac{P}{V} = \frac{6400\,\text{W}}{230\,\text{V}} = 28\,\text{A}$$

This is too large a current to draw from the ring main and so a separate circuit must be used.



Figure 40.12 Electricity meter with digital display

Dangers of electricity

a) Electric shock

Electric shock occurs if current flows from an electric circuit through a person's body to earth. This can happen if there is **damaged insulation** or **faulty wiring**. The typical resistance of dry skin is about 10 000 Ω, so if a person touches a wire carrying electricity at 240 V, an estimate of the current flowing through them to earth would be $I = V/R = 240/10\,000 = 0.024\,\text{A} = 24\,\text{mA}$. For wet skin, the resistance is lowered to about 1000 Ω (since water is a good conductor of electricity) so the current would increase to around 240 mA.

It is the **size of the current** (not the voltage) and the **length of time** for which it acts which determine the strength of an electric shock. The path the current takes influences the effect of the shock; some parts of the body are more vulnerable than others. A current of 100 mA through the heart is likely to be fatal.

Damp conditions increase the severity of an electric shock because water lowers the resistance

of the path to earth; wearing shoes with insulating rubber soles or standing on a dry insulating floor increases the resistance between a person and earth and will reduce the severity of an electric shock.

To avoid the risk of getting an electric shock:

- Switch off the electrical supply to an appliance before starting repairs.
- Use plugs that have an earth pin and a cord grip; a rubber or plastic case is preferred.
- Do not allow appliances or cables to come into contact with water. For example holding a hairdryer with wet hands in a bathroom can be dangerous. Keep electrical appliances well away from baths and swimming pools!
- Do not have long cables trailing across a room, under a carpet that is walked over regularly or in other situations where the insulation can become damaged. Take particular care when using electrical cutting devices (such as hedge cutters) not to cut the supply cable.

In case of an electric shock, take the following action:

1 **Switch off the supply** if the shocked person is still touching the equipment.
2 **Send for qualified medical assistance.**
3 **If breathing or heartbeat has stopped, commence CPR** (cardiopulmonary resuscitation) by applying chest compressions at the rate of about 100 a minute until there are signs of chest movement or medical assistance arrives.

b) Fire risks

If flammable material is placed too close to a hot appliance such as an electric heater, it may catch fire. Similarly if the electrical wiring in the walls of a house becomes overheated, a fire may start. Wires become hot when they carry electrical currents – the larger the current carried, the hotter a particular wire will become, since the rate of production of heat equals I^2R (see p. 177).

To reduce the risk of fire through **overheated cables**, the maximum current in a circuit should be limited by taking these precautions:

- Use plugs that have the correct fuse.
- Do not attach too many appliances to a circuit.
- Don't overload circuits by using too many adapters.
- Appliances such as heaters use large amounts of power (and hence current), so do not connect them to a lighting circuit designed for low current use. (Thick wires have a lower resistance than thin wires so are used in circuits expected to carry high currents.)

Damaged insulation or faulty wiring which leads to a large current flowing to earth through flammable material can also start a fire.

The factors leading to fire or electric shock can be summarised as follows:

damaged insulation	→ electric shock and fire risk
overheated cables	→ fire risk
damp conditions	→ increased severity of electric shocks

Questions

1 How much electrical energy in joules does a 100 watt lamp transfer in
  a 1 second,
  b 5 seconds,
  c 1 minute?
2 **a** What is the power of a lamp rated at 12 V 2 A?
  b How many joules of electrical energy are transferred per second by a 6 V 0.5 A lamp?
3 The largest number of 100 W lamps connected in parallel which can safely be run from a 230 V supply with a 5 A fuse is
  A 2 **B** 5 **C** 11 **D** 12 **E** 0
4 What is the maximum power in kilowatts of the appliance(s) that can be connected safely to a 13 A 230 V mains socket?
5 The circuits of Figures 40.13a and b show 'short circuits' between the live (L) and neutral (N) wires. In both, the fuse has blown but whereas circuit a is now safe, b is still dangerous even though the lamp is out which suggests the circuit is safe. Explain.

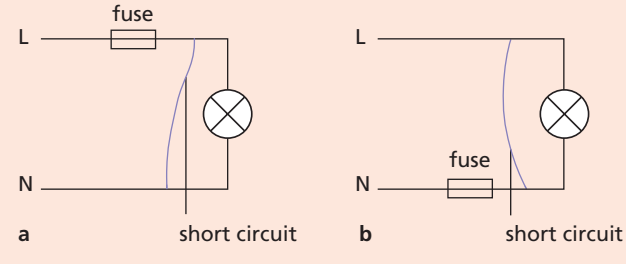


Figure 40.13

6 What steps should be taken before replacing a blown fuse in a plug?
7 What size fuse (3 A or 13 A) should be used in a plug connected to
  a a 150 W television,
  b a 900 W iron,
  c a 2 kW kettle,
  if the supply is 230 V?

▶▶

Checklist

After studying this chapter you should be able to

- recall the relations $E = ItV$ and $P = IV$ and use them to solve simple problems on energy transfers,
- describe experiments to measure electric power,
- describe electric lamps, heating elements and fuses,
- recall that a joulemeter measures electrical energy,
- describe with the aid of diagrams a house wiring system and explain the functions and positions of switches, fuses, circuit breakers and earth,
- state the advantages of connecting lamps in parallel in a lighting circuit,
- wire a mains plug and recall the international insulation colour code,
- perform calculations of the cost of electrical energy in joules and kilowatt-hours,
- recall the hazards of damaged insulation, damp conditions and overheating of cables and the associated risks.

8 What is the cost of heating a tank of water with a 3000 W immersion heater for 80 minutes if electricity costs 10p per kWh?

9 a Below is a list of wattages of various appliances. State which is most likely to be the correct one for each of the appliances named.

60 W 250 W 850 W 2 kW 3.5 kW

(i) kettle
(ii) table lamp
(iii) iron

b What will be the current in a 920 W appliance if the supply voltage is 230 V?

Practice 1

E = P x t

BASIC

1. **Re-arrange** the equations in order to find:
    a) time t
    b) power P
2. **Calculate** the energy transferred by a 3,000 W hairdryer in 60 seconds.

MEDIUM

1. An Xbox uses a power of 125 W and is left on for a time of 2500s. **Calculate** the amount of energy used.
2. A kettle uses a power of 1800 W, and uses 36000 J of energy. **Calculate** the time that the kettle was left on for.
3. A laptop uses a power of 65 W, and uses 1300 J of energy. **Calculate** how long the laptop was used for.
4. **Convert** into Watts:
    a) 3.5 kW.
    b) 0.7 kW.
5. **Convert** into kiloWatts:
    a) 7000 W.
    b) 19000 W.

To go from kW to W → × 1000

To go from W to kW → ÷ 1000

HARD

1. **Calculate** the energy transferred by a 2 kW electric radiator in 5 minutes.
    You need to convert kW into W
    P = 2 kW = _____ W
    You need to convert minutes into seconds
    t = 5 minutes = _____ s
2. **Calculate** the power for an electrical appliance that transfers 5 kJ of energy in 15 minutes.
3. **Calculate** the time it takes a 60 kW appliance to transfer 0.02 kJ of electrical energy.
4. **Describe** what happens to the energy transferred by an appliance if:
    a) the power of the appliance increases (and the time is kept the same)
    b) the time decreases (and the power of the appliance is the same)

Practice 2

1. The current flowing through a heating element is 5 A when a p.d. of 35 V is applied across it. Find the resistance of the element.

2. A 60 W electric light bulb is connected to a 240 V supply. Determine (a) the current flowing in the bulb and (b) the resistance of the bulb.

3. Determine the p.d. which must be applied to a 5 kΩ resistor such that a current of 6 mA may flow.

4. A 20 V source of e.m.f. is connected across a circuit having a resistance of 400 Ω. Calculate the current flowing.

5. The hot resistance of a 250 V filament lamp is 625 Ω. Determine the current taken by the lamp and its power rating.

6. Determine the resistance of a coil connected to a 150 V supply when acurrent of (a) 75 mA (b) 300 μA flows through it.

7. Determine the resistance of an electric fire which takes a current of 12 A from a 240 V supply. Find also the power rating of the fire and the energy used in 20 h.

8. Determine the power dissipated when a current of 10 mA flows through an appliance having a resistance of 8 kΩ.

9. 85.5 J of energy are converted into heat in 9 s. What power is dissipated?

10. A current of 4 A flows through a conductor and 10 W is dissipated. What p.d. exists across the ends of the conductor?

10. Find the power dissipated when:

(a) a current of 5 mA flows through a resistance of 20 kΩ

(b) a voltage of 400 V is applied across a 120 kΩ resistor

(c) a voltage applied to a resistor is 10 kV and the current flow is 4 mA

11. A battery of e.m.f 15 V supplies a current of 2 A for 5 min. How much energy is supplied in this time?

12. A d.c. electric motor consumes 72 MJ when connected to 400 V supply for 2 h 30 min. Find the power rating of the motor and the current taken from the supply.

13. A p.d. of 500 V is applied across the winding of an electric motor and the resistance of the winding is 50 Ω. Determine the power dissipated by the coil.

14. In a household during a particular week three 2 kW fires are used on average 25 h each and eight 100 W light bulbs are used on average 35 h each. Determine the cost of electricity for the

week if 1 unit of electricity costs 15p

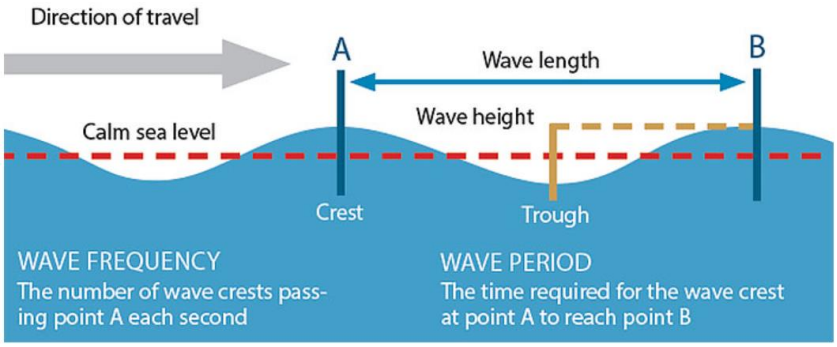
15. Calculate the power dissipated by the element of an electric fire of resistance 30 Ω when a current of 10 A flows in it. If the fire is on for 30 hours in a week determine the energy used. Determine also the weekly cost of energy if electricity costs 13.50p per unit.



End of Lesson

Lesson 14

General Properties of Waves



Have you ever ridden on the waves in the seas?

Sea waves travel on the surface of the water. You can see and feel them. As you swim through the water, you can even create your own waves! These are waves we can see. How about waves that are invisible to our eyes?

At the end of this chapter, you will be able to:

- describe what is meant by wave motion as illustrated by vibration in ropes and springs and by waves in a ripple tank;
- compare transverse and longitudinal wave and give suitable examples of each;
- define speed, frequency, wave length, period and amplitude;
- recall and apply the relationship velocity = frequency wavelength to new situations or to solve related problems;
- state what is meant by the term wave front;
- show understanding that waves transfer energy without transferring matter.

1 What is a Wave?

A wave is a periodic disturbance in a medium. When a stone is thrown into a pool of calm water, the water medium is disturbed, causing water waves to be produced.

Waves are one way in which energy can be transmitted from one place to another.

The waves that you see at the beach are the result of the kinetic energy of water particles passing through the water. Light from a lamp carries light energy to your eye in a form called electromagnetic waves.



Fig.1 Water Wave on the sea

2 Types Wave

Waves come in many shapes and forms, and with different names. It is common to categorize waves based on certain common features.

One way to categorize waves is the direction of movement of the individual particles of the medium relative to the direction in which the waves travel.

Transverse Waves

Particles in such waves vibrate perpendicularly to the direction in which the waves travel.

A ripple on a pond and a wave on a rope are easily visualized transverse waves. Transverse waves can be created when a rope or a slinky is made to swing up and down while its wave is travelling away from the source of vibration. See Figure 2 for an example of a transverse wave in a rope.

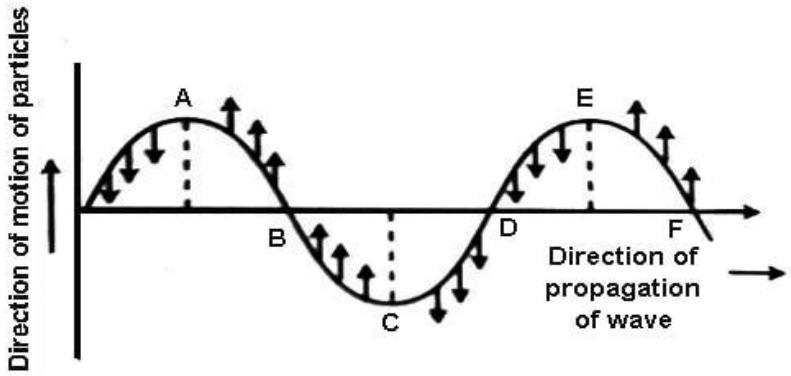


Fig. 2 Example of a transverse wave

As the waves travel outwards from left to right as shown in Figure 2, the particles of the medium vibrate up and down, perpendicular to the direction of wave motion, in a series of crests and troughs.

Other transverse waves include electromagnetic waves (for example, light waves and radio waves) and water waves.

Longitudinal Waves

Particles in such waves vibrate to and fro, along the direction of travel of the waves. Such waves can be created by pulling and pushing a slinky back and forth as shown in Figure 3. Particles in slinky oscillate to and fro, along the left-right direction wave travels from left to right.

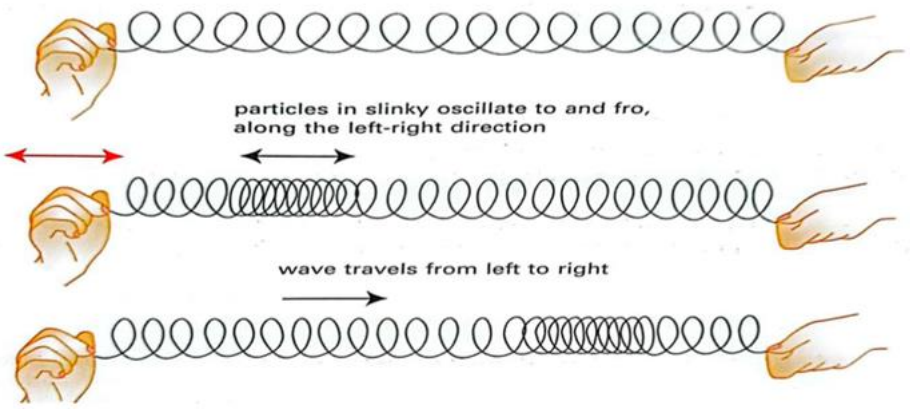


Fig. 3 Example of longitudinal wave

As the wave passes along the spring, the slinky initially at rest, is set into motion, so there is a transfer of energy from one end of the spring to the other.

Such movement of particles causes certain regions to be crowded, such that particles are moving towards each other. This is called compression. Another region where particles are pulling away from one another is called rarefaction.

3 Comparison Between Transverse and Longitudinal Wave Motions

Transverse wave	Longitudinal wave
Example: water waves, light waves. 1. The particle vibration is perpendicular to the direction of wave motion.	Example: sound waves 1. The particle vibration is along the direction of the wave motion.
2. It travels in the form of crests and troughs.	2. It travels in the form of compressions and rarefactions.
3. It can take place in solids, the surface of liquids and gases. Some transverse waves like electromagnetic waves can even travel through vacuum.	3. It takes place in solids, liquids or gases, but not in vacuum.

4. Term used to describe a wave

Part of wave motion	Definition
crest	Refers to the highest point of a particle displacement.
trough	Refers to the lowest point of a particle displacement.
wavelength (λ)	The distance travelled by a complete wave is called the wavelength. It can be determined by measuring the distance between two successive corresponding points, like the distance from one crest to the next successive crest in a transverse wave, or the distance from one compression to the next successive compression in a longitudinal wave.
amplitude (A)	This is the maximum displacement of the wave from its equilibrium or undisturbed or rest position. The value of the amplitude indicates the level of energy the wave. A wave carrying a large amount of energy is represented by a large amplitude.

Period (T)	Time taken to produce a complete wave.
Frequency (F)	This refers to the number of complete waves leaving the source of disturbance every one second. It is measured in Hz or hertz.
Wave speed (v)	The speed at which the wave moves through the medium.
Table 2 Description of parts of a wave	

The Wave Equation

Every wave has three important quantities which differentiate it from others.

They are:

(a) frequency (f);

(b) wavelength (λ); and

(c) wave speed (v);

These quantities are related to one another by the wave equation:

$$\mathbf{v = \lambda \times f}$$

Where **v** is the wave speed in m/s)

λ is the wave length in (m)

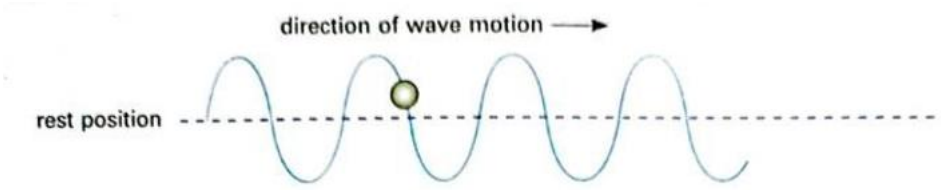
F is the wave frequency in (Hz

5. Wave Transfer Energy Without Transferring Materials

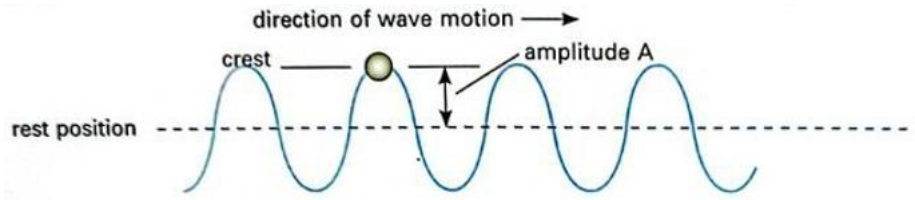
Imagine a styrofoam ball being placed in a ripple tank containing some water. Before the water surface is disturbed, the styrofoam ball will remain at rest.



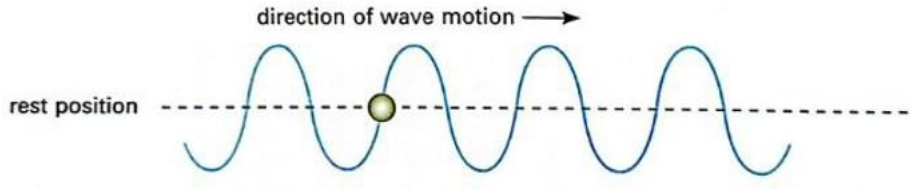
a) Styrofoam ball at rest before water wave passes through direction of wave motion-rest position



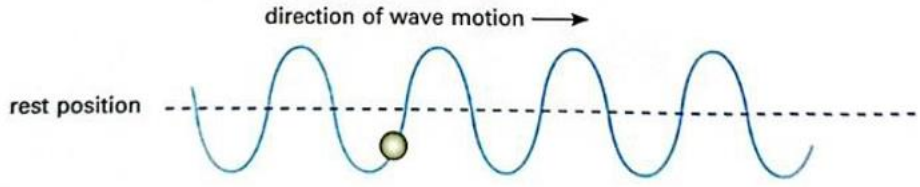
(b) Styrofoam ball is displaced vertically upwards, above its rest position direction of wave motion-rest position



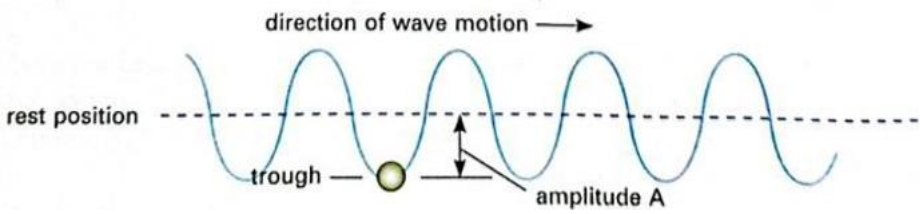
(c) Styrofoam ball is at maximum displacement, above its rest position direction of wave motion-rest position



(d) Styrofoam ball is back to its rest position direction of wave motion rest position



(e) Styrofoam ball is now displaced vertically below the water level



(f) Styrofoam ball is at maximum displacement, below its rest position

Hence waves transfer energy from one point to another without transferring.

SUMMARY

1. Waves are disturbances that allow energy transfer to occur.
2 Waves are described by these terms: amplitude, wavelength, frequency, period and wave speed
3. Wave speed can be calculated using the formula, $v = f \times \lambda$
4. There are two main categories of waves, namely, transverse and longitudinal waves.
5. In a transverse wave, the particles vibrate perpendicular to the direction of wave motion.
6. In a longitudinal wave, the particles vibrate parallel to the direction of wave motion.

Practice

I. Multiple Choice

1. What is a wave?

a. A wave is a force that propagates from the place where it was created.
b. A wave is a disturbance that propagates from the place where it was created.
c. A wave is matter that provides volume to an object.
d. A wave is matter that provides mass to an object.

2. Do all waves require a medium to travel? Explain.

a. No, electromagnetic waves do not require any medium to propagate.
b. No, mechanical waves do not require any medium to propagate.
c. Yes, both mechanical and electromagnetic waves require a medium to propagate.
d. Yes, all transverse waves require a medium to travel.

3. The frequency of a wave is 10 Hz. What is its period?

a. The period of the wave is 100 s.
b. The period of the wave is 10 s.
c. The period of the wave is 0.01 s.
d. The period of the wave is 0.1 s.

4. What is the velocity of a wave whose wavelength is 2 m and whose frequency is 5 Hz?

a. 20 m/s
b. 2.5 m/s
c. 0.4 m/s
d. 10 m/s

5. What is the amplitude of a wave?

a. A quarter of the total height of the wave
b. Half of the total height of the wave
c. Two times the total height of the wave
d. Four times the total height of the wave

II. Exercises

1. If the speed is 335 m/s and its frequency is 67 Hz, what is its wavelength?

2. A wave with a frequency of 60.0 Hz travels through steel with a wavelength of 85.5 m. What is the speed of this wave?

3. A ship anchored at sea is rocked by waves that have crests 14 m apart. The waves travel at 7.0 m/s. What is the frequency of the wave? Show your work.



End of Lesson

Glossary

Acceleration: Rate of change of velocity.
Acceleration of gravity: Rate of change of velocity due to gravitational attraction of the earth.
Ammeter: Electric device used to measure current.
Ampere: Unit for the rate of flow of charged particles. One ampere equals a flow of one coulomb of charge per second.
Amplitude: Maximum displacement from zero of any periodic phenomenon.
Aneroid barometer: Device using a partially evacuated box to measure atmospheric pressure.
Angle of incidence: Angle between a light ray and a line perpendicular to the surface that the ray is striking.
Angle of refraction: Angle between a refracted ray and a line perpendicular to the surface that the ray is leaving.
Antenna: Device used to receive or radiate electromagnetic waves.
Archimedes' principle: An object immerses in a fluid is buoyed up by a force equal to the weight of the fluid displaced by the object.
Armature: Coil of wire that produces current in a generator and rotation in an electric motor.
Barometer: Device for measuring the pressure of the atmosphere.
Battery or galvanic cell: Two dissimilar conductors and an electrolyte that produces a potential difference.
Bernoulli's principle: The pressure exerted by a fluid decreases as its velocity increases.
Black hole: Collapsed astronomical object of sufficient mass to prevent the escape of light.
Components of a vector: Two or more vectors (usually perpendicular) that when added together, produce the original vector.
Concurrent forces: Forces acting on the same point.
Conductor: Material through which charged particles move readily.
Control rods: Devices in a nuclear reactor used to regulate the rate of the nuclear reaction.
Convection current: Current caused by motion of a body of fluid due to differences in density resulting from thermal expansion.
Converging lens: Lens, thick in the middle and thin at the edge, that causes parallel rays to converge.
Converging mirror: Concave mirror capable of causing parallel rays to converge.
Coulomb: Unit of quantity of electric charge equal to the charge found on 6.25×10^{18} electrons.
Candela: Unit of luminous intensity.
Center of curvature: Center of the sphere from which a spherical mirror is taken.
Centripetal acceleration: Acceleration always at right angles to the velocity of a particle.
Centripetal force: Force directed toward the center of a circle, keeps particles moving in uniform circular motion.
Critical angle: Minimum angle of incidence that produces total internal reflection.
Deceleration: Negative acceleration.
Diffraction: Bending of light waves around an object in its path.
Diffuse reflection: Reflected light scattered in many directions.

Displacement: Vector quantity representing the change in position of an object.
Distance: Scalar quantity equal to the sum of the magnitude of the displacements.
Diverging lens: Lens thin in the middle and thick at the edge, that causes parallel rays to diverge.
Diverging mirror: Convex mirror capable of causing parallel light rays to diverge.
Dynamics: Study of the motion of particles acted upon by forces.
Effective resistance: Resistance of a single resistor that could replace a combination of resistors.
Efficiency: Ratio of output work to input work.
Elasticity: Ability of object to return to its original form after removal of deforming forces.
Electric circuit: Continuous path that can be followed by charged particles.
Electric current: Flow of charged particles.
Electric field: The property of space around a charged object that causes forces on other charged objects.
Electric field intensity: Ratio of the force exerted by a field on a charged particle and the charge on the particles.
Electric field lines: Lines representing the direction of the electric field.
Electric force: Force between two objects due to their charges.
Electromagnet: Device in which a magnetic field is generated by an electric current.
Electromagnetic force: One of the fundamental forces. The force that exists between electric charges.
Electromagnetic induction: Generation of an electric current by having a wire cut (or cut by) magnetic flux lines.
Electromagnetic wave: Wave consisting of electric and magnetic fields that move at the speed of light in space.
Electromagnetism: Interrelationship of magnetic fields and electric currents.
Electromotive force (EMF): Potential difference generated by electromagnetic induction.
Energy: Capacity to do work.
Equilibrant: Force equal in magnitude to a resultant, but opposite in direction.
Equilibrium: Condition in which a net force on an object is zero.
Focal length: Distance from the focal point to the vertex of a mirror or lens.
Focal point: Point of convergence, real or apparent (virtual), of rays reflected by a mirror or refracted by a lens.
Force: Action that results in accelerating or deforming an object.
Frequency: Number of occurrences in a unit of time.
Friction: Force opposing motion between two objects that are in contact.
Generator: Device using mechanical energy to produce electric energy.
Gravitational field: Distortion of space due to the presence of a mass. See field.
Gravitational force: Attraction between objects due to their masses.
Grounding: Connecting a charged object to the earth to remove the object's charge.
Heat: Quantity of thermal energy transferred from one object to another object because of a difference in temperature.
Hertz: Unit of frequency equal to one event (cycle) per second.

Index of refraction: $n = \frac{\sin i}{\sin r}$, where n=index of refraction, i=angle of incidence, and r=angle of refraction.
Insulator: Material through which the flow of charged particles is greatly restricted.
Joule: Unit of work or energy equal to a Newton-meter.
Kilowatt hour: Amount of energy equal to $3.6\times10^{6} J$.
Kinematics: Study of the motion of the particles.
Kinetic energy: Energy of an object due to its motion.
Law of conservation of energy: In non nuclear changes, energy can be neither created nor destroyed.
Length: Distance between two points.
Lenz's law: The magnetic field generated by an induced current always opposes the field generating the current.
Light: Electromagnetic radiation of 4×10^{-7} meter to 7×10^{-7} meter wavelength.
Liquid: State of matter in which particles are in close proximity but may readily change their relative positions.
Longitudinal wave: Wave in which the disturbance is in the sane direction as the direction of travel of the wave.
Magnetic field: Space around a magnet in which magnetic force can be detected.
Magnetic force: Force between two object due to the magnetic flux of one or both objects.
Mechanical advantage (MA): Ratio of resistance force to effort force in a machine.
Mechanical wave: Disturbance traveling through a medium.
Normal force: Force perpendicular to a surface.

Ohm: Unit of electric resistance.
Ohm's law: Current flowing in a conductor varies directly with the potential difference and inversely with the resistance.
Opaque material: Material that does not transmit light.
Parallel circuit: Circuit in which there are two or more paths for the charged particles to follow as they complete the circuit.
Pascal: Unit of pressure equal to a Newton per square meter.
Period: Time duration of a phenomenon or event.
Potential energy: Energy of an object due to its position.
Power: Rate of doing work.
Pressure: Force per unit area.
Principle axis: Radius connecting the center of curvature of a spherical mirror with its geometric vertex.
Real image: Image formed by rays that actually recombine to form the image.
Receiver: Device used to detect electromagnetic waves.
Refraction: Change in direction of a wave front as it passes from one medium to another.
Resistance: Opposition to flow of electric current.
Resolution: To find the component of a vector in a given direction.
Resultant: Single force that has the same effect as two or more concurrent vectors.
Satellite: Object in orbit around a planet.
Scalar quantity: Quantity having magnitude (size) only.
Schematic diagram: Diagram of an electric circuit using symbols.

Series circuit: Circuit in which the charged particles must flow through each component of the circuit, one after the other.
SI: An internationally agreed-upon consistent method of using the metric system of measurement.
Significant digits: The reliable digits reported in a measurement.
Simple harmonic motion: Motion in which a particle repeats the same path periodically.
Simple machine: Lever, pulley, wheel and axle, inclined plane, wedge, or screw.
Snell's law: The ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant for two specific substances in contact where a light passes. See index of refraction.
Specific heat: Energy required to change the temperature of 1 kg of a substance I kelvin.
Spectrum: Array of the various wavelengths composing light.
Speed: Ratio of distance traveled to the time interval.
Speed of light: $3.00 \times 10^8 m/s$ (in vacuum).
Spherical concave mirror: A converging mirror that is formed of a spherical segment of one base.
Temperature: Measure of the average kinetic energy of molecules.
Terminal velocity: Velocity of a falling object when the flow of air resistance is equal to the weight.
Thermal energy: Sum of potential energy and kinetic energy or random motion of particles in an object.
Thermal excitation: Exciting an atom by heating it.
Thermal expansion: Moving apart of particles as their temperature rises.
Thermometer: Device used to measure temperature.
Transformer: Device used to transfer energy from one circuit to another circuit by mutual inductance across two coils.
Vector quantity: Quantity having both magnitude (size) and direction.
Velocity: Rate of change of position.
Virtual image: Image whose rays appear to emanate from a point without actually doing so.
Volts: Unit of potential difference.
Voltmeter: Electric device used to measure potential difference.
Watt: Unit of power equal to 1 J/s.
Wave: Traveling disturbance in a field or medium.
Wavelength: Distance between corresponding points on two successive waves.
Work: Product of force and displacement.
Work function: Energy needed to remove an electron from metal as in the photoelectric effect.
X rays: Electromagnetic waves of very short wavelength.

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